

Kandidatuppsats

Statistiska Institutionen

Nr 2026:x

Code

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1 Pre-processing

```
# Loading libraries  
library(dplyr)
```

Warning: package 'dplyr' was built under R version 4.4.3

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
library(tidyr)  
library(kableExtra)
```

Warning: package 'kableExtra' was built under R version 4.4.3

Attaching package: 'kableExtra'

The following object is masked from 'package:dplyr':

group_rows

```
library(corrplot)
```

Warning: package 'corrplot' was built under R version 4.4.3

corrplot 0.95 loaded

```
library(JMbayes2)
```

Warning: package 'JMbayes2' was built under R version 4.4.3

Loading required package: survival

Warning: package 'survival' was built under R version 4.4.3

Loading required package: nlme

Warning: package 'nlme' was built under R version 4.4.3

Attaching package: 'nlme'

The following object is masked from 'package:dplyr':

collapse

Loading required package: GLMMadaptive

Warning: package 'GLMMadaptive' was built under R version 4.4.3

Loading required package: splines

Attaching package: 'JMbayes2'

The following object is masked from 'package:GLMMadaptive':

mixed_model

The following object is masked from 'package:nlme':

lme

The following object is masked from 'package:survival':

coxph

```
library(JM)
```

Warning: package 'JM' was built under R version 4.4.3

Loading required package: MASS

Warning: package 'MASS' was built under R version 4.4.3

Attaching package: 'MASS'

The following object is masked from 'package:JMbayes2':

area

The following object is masked from 'package:GLMMadaptive':

negative.binomial

The following object is masked from 'package:dplyr':

select

```
library(nlme)
```

```
library(lme4)
```

Warning: package 'lme4' was built under R version 4.4.3

Loading required package: Matrix

Warning: package 'Matrix' was built under R version 4.4.3

Attaching package: 'Matrix'

The following objects are masked from 'package:tidyr':

expand, pack, unpack

Attaching package: 'lme4'

The following object is masked from 'package:GLMMadaptive':

negative.binomial

The following object is masked from 'package:nlme':

lmList

```
library(survival)
library(ggplot2)
```

Warning: package 'ggplot2' was built under R version 4.4.3

```
library(glmmTMB)
```

Warning: package 'glmmTMB' was built under R version 4.4.3

```
library(DHARMA)
```

Warning: package 'DHARMA' was built under R version 4.4.3

This is DHARMA 0.4.7. For overview type '?DHARMA'. For recent changes, type news(package = 'DHARMA')

```
library(lcmm)
```

Warning: package 'lcmm' was built under R version 4.4.3

```
library(timeROC)
```

Warning: package 'timeROC' was built under R version 4.4.3

```
library(pec)
```

Warning: package 'pec' was built under R version 4.4.3

Loading required package: prodlim

Warning: package 'prodlim' was built under R version 4.4.3

Attaching package: 'pec'

The following object is masked from 'package:JM':

ibs

```
# Set seed for reproducibility  
set.seed(123)
```

1.1 Loading the data

The data was loaded from the ADNIMERGE2 package accessed via the Alzheimer's Disease Neuroimaging Initiative (ADNI). The data sets include longitudinal ptau-181 and NFL measurements, diagnosis, demographic and genetic data.

```
# Loading data  
# Milou  
#NFL data  
load("ADNIMERGE2_unpacked/ADNIMERGE2/data/BLENNOWPLASMANFLLONG.rda")  
nfl_data <- BLENNOWPLASMANFLLONG  
# P-tau181 data  
load("ADNIMERGE2_unpacked/ADNIMERGE2/data/UGOTPTAU181.rda")  
ptau_data <- UGOTPTAU181  
# Diagnosis data  
load("ADNIMERGE2_unpacked/ADNIMERGE2/data/DXSUM.rda")  
diagnosis_data <- DXSUM  
# Demographics data  
load("ADNIMERGE2_unpacked/ADNIMERGE2/data/PTDEMOG.rda")  
demographics <- PTDEMOG  
# # Apoe data  
load("ADNIMERGE2_unpacked/ADNIMERGE2/data/APOERES.rda")  
apoe <- APOERES
```

```
# Charley
#library(ADNIMERGE2)
#nfl_data = BLENNOWPLASMANFLLONG
#ptau_data = UGOTPTAU181
#diagnosis_data = DXSUM
#demographics = PTDEMOG
#apoe = APOERES
```

```
# Checking which ADNI protocols the data are from
table(nfl_data$ORIGPROT)
```

```
ADNI1  ADNI2  ADNIGO
  934    2317    511
```

```
table(ptau_data$ORIGPROT)
```

```
ADNI1  ADNI2  ADNIGO
  934    2315    509
```

```
table(diagnosis_data$ORIGPROT)
```

```
ADNI1  ADNI2  ADNI3  ADNI4  ADNIGO
  5205    5232    3047    1502     895
```

```
# checking whether there are any observations that should be removed as per the ADNI leaders
grep("381_S_10###", diagnosis_data$PTID)
```

```
integer(0)
```

```
grep("381_S_10###", apoe$PTID)
```

```
integer(0)
```

1.2 Selecting required variables

Variables for the unique identifier for each individual, visit code, exam date, biomarker measurements, diagnosis and genetic data were extracted.

```
# We select the required nfl variables
nfl_data <- nfl_data %>%
  dplyr::select(RID, VISCODE2, EXAMDATE, PLASMA_NFL)

# We select the required ptau variables
ptau_data <- ptau_data %>%
  dplyr::select(RID, VISCODE2, EXAMDATE, PLASMAPTAU181)

# We select the required genotype variables
apoe <- apoe %>%
  dplyr::select(RID, GENOTYPE)

# We select the required diagnosis variables
diagnosis_data <- diagnosis_data %>%
  dplyr::select(ORIGPROT, RID, VISCODE2, EXAMDATE, DIAGNOSIS)

# We add _dx on the diagnosis variables
diagnosis_data <- diagnosis_data %>%
  rename_with(~paste0(.x, "_dx"), -c(RID, VISCODE2))
```

1.3 Checking RID with more than one measurement recorded on the same exam date

```
## Checking which obs have double VISCODE2 codes
# for ptau data
ptau_data %>% group_by(RID, VISCODE2) %>% filter(n() > 1)
```

```
# A tibble: 106 x 4
# Groups:   RID, VISCODE2 [53]
  RID VISCODE2 EXAMDATE PLASMAPTAU181
  <dbl> <chr> <date> <dbl>
1 31 m60 2010-12-02 18.6
2 31 m60 2010-12-02 21.3
3 89 m48 2010-09-28 20.0
4 89 m48 2010-09-28 10.6
```



```

5   125 m60      2011-05-26      25.8
6   125 m60      2011-05-26      18.3
7   138 m96      2013-11-19      21.0
8   138 m96      2013-11-19      20.5
9   225 m96      2014-03-07      20.2
10  225 m96      2014-03-07      24.5
# i 96 more rows

```

```

# for nfl data
nfl_data %>% group_by(RID, VISCODE2) %>% filter(n() > 1)

```

```

# A tibble: 106 x 4
# Groups:   RID, VISCODE2 [53]
   RID VISCODE2 EXAMDATE   PLASMA_NFL
  <dbl> <chr>    <date>     <dbl>
1    31 m60     2010-12-02    50.7
2    31 m60     2010-12-02    56.2
3    89 m48     2010-09-28    29.1
4    89 m48     2010-09-28    29.6
5   125 m60     2011-05-26    49.6
6   125 m60     2011-05-26     47
7   138 m96     2013-11-19    69.5
8   138 m96     2013-11-19    81.3
9   225 m96     2014-03-07    61.7
10  225 m96     2014-03-07    59.9
# i 96 more rows

```

```

# changing VISCODE2 for RID 1097 to m60_1 and m60_2
ptau_data[ptau_data$RID == 1097, "VISCODE2"] <- c("m48", "m60_2", "m60_1")

nfl_data[nfl_data$RID == 1097, "VISCODE2"] <- c("m48", "m60_1", "m60_2")

```

```

# Replacing visits with double measurements with visits with mean-value of the measurements
# Naming the mean as the original variable to do analysis first with this method and then ser

# Adding columns for first, last and averaged ptau measurement
ptau_data <- ptau_data %>%
  group_by(RID, EXAMDATE) %>%
  arrange(RID, EXAMDATE) %>%
  mutate(
    first_ptau = first(PLASMAPTAU181),

```

```

    last_ptau = last(PLASMAPTAU181),
    mean_ptau = mean(PLASMAPTAU181)
  ) %>%
  ungroup()

# Adding columns for first, last and averaged nfl measurement
nfl_data <- nfl_data %>%
  group_by(RID, EXAMDATE) %>%
  arrange(RID, EXAMDATE) %>%
  mutate(
    first_nfl = first(PLASMA_NFL),
    last_nfl = last(PLASMA_NFL),
    mean_nfl = mean(PLASMA_NFL)
  ) %>%
  ungroup()

```

```

# checking the difference in method for those with 2 obs on the same examdate
# for ptau
head(ptau_data %>%
  filter(PLASMAPTAU181 != first_ptau) %>%
  dplyr::select(mean_ptau, first_ptau, last_ptau)
)

```

```

# A tibble: 6 x 3
  mean_ptau first_ptau last_ptau
    <dbl>      <dbl>    <dbl>
1    20.0      18.6     21.3
2    15.3      20.0     10.6
3    22.1      25.8     18.3
4    20.8      21.0     20.5
5    22.4      20.2     24.5
6    29.2      23.2     35.1

```

```

# for nfl
head(nfl_data %>%
  filter(PLASMA_NFL != first_nfl) %>%
  dplyr::select(mean_nfl, first_nfl, last_nfl)
)

```

```

# A tibble: 6 x 3
  mean_nfl first_nfl last_nfl

```

	<dbl>	<dbl>	<dbl>
1	53.4	50.7	56.2
2	29.4	29.1	29.6
3	48.3	49.6	47
4	75.4	69.5	81.3
5	60.8	61.7	59.9
6	254.	451	57.8

1.3.1 Replacing visits with double measurements with visits with mean-value of the measurements

```
#Defining plasma ptau as the mean and collapsing to unique RID and EXAMDATE
ptau_data <- ptau_data %>%
  group_by(RID, EXAMDATE) %>%
  mutate(
    PLASMAPTAU181 = mean(PLASMAPTAU181)
  ) %>%
  ungroup() %>%
  distinct()

# Defining plasma nfl as the mean and collapsing to unique RID and EXAMDATE
nfl_data <- nfl_data %>%
  group_by(RID, EXAMDATE) %>%
  mutate(
    PLASMA_NFL = mean(PLASMA_NFL)
  ) %>%
  ungroup() %>%
  distinct()
```

1.3.1.1 Sensitivity analysis: Replacing visits with double measurements with visits with the first measurement

```
#ptau_data <- ptau_data %>%
#   group_by(RID, EXAMDATE) %>%
#   arrange(RID, EXAMDATE) %>%
#   mutate(
#     PLASMAPTAU181 = first_ptau
#   ) %>%
#   ungroup() %>%
#   distinct()
```

```
#nfl_data <- nfl_data %>%
#   group_by(RID, EXAMDATE) %>%
#   arrange(RID, EXAMDATE) %>%
#   mutate(
#     PLASMA_NFL = first_nfl
#   ) %>%
#   ungroup() %>%
#   distinct()
```

1.3.1.2 Sensitivity analysis: Replacing visits with double measurements with visits with the last measurement

```
# ptau_data <- ptau_data %>%
#   group_by(RID, EXAMDATE) %>%
#   arrange(RID, EXAMDATE) %>%
#   mutate(
#     PLASMAPTAU181 = last_ptau,
#   ) %>%
#   ungroup() %>%
#   distinct()
# nfl_data <- nfl_data %>%
#   group_by(RID, EXAMDATE) %>%
#   arrange(RID, EXAMDATE) %>%
#   mutate(
#     PLASMA_NFL = last_nfl
#   ) %>%
#   ungroup() %>%
#   distinct()
```

1.4 Merging Nfl and ptau-181 data for the longitudinal data set

```
# We merge NFL and ptau data for longitudinal data by RID and VISCODE2
long_data <- nfl_data %>%
  left_join(ptau_data, by = c("RID", "VISCODE2"), suffix = c("_nfl", "_ptau"))
```

1.5 Creating a common EXAMDATE and EXAMDATE_baseline variable

This was performed before preparation of the other variables for the survival and longitudinal models in order to use EXAMDATE_baseline to remove any observations from the diagnosis

data set which occur long before the first biomarker measurements for that individual.

```
# checking how many observations where the EXAMDATE is different for nfl and ptau
diff <- long_data %>% mutate(diff = as.numeric(EXAMDATE_nfl != EXAMDATE_ptau))
table(diff$diff)
```

```
  0    1
3615  82
```

```
# filtering out the observations with different EXAMDATE
diff <- diff %>% filter(diff == TRUE)

# checking how many of these are baseline visits
table(diff$VISCODE2)
```

```
bl m108  m12  m24  m36  m48  m60  m72  m96
31   1    3   27   5   10   1    2    2
```

```
# checking which observations have the same examdate for nfl and ptau
common <- which(long_data$EXAMDATE_nfl %in% long_data$EXAMDATE_ptau)

# checking the observations with different exam dates
long_data[-common, ]
```

```
# A tibble: 10 x 12
```

	RID	VISCODE2	EXAMDATE_nfl	PLASMA_NFL	first_nfl	last_nfl	mean_nfl
	<dbl>	<chr>	<date>	<dbl>	<dbl>	<dbl>	<dbl>
1	384	m108	2015-04-04	36.2	36.2	36.2	36.2
2	896	m24	2008-11-13	69.9	69.9	69.9	69.9
3	2106	bl	2010-10-25	67.3	67.3	67.3	67.3
4	2180	bl	2010-12-28	NA	NA	NA	NA
5	2336	m48	2015-04-09	45.6	45.6	45.6	45.6
6	4309	m48	2015-11-10	42.1	42.1	42.1	42.1
7	4395	m48	2016-02-04	36	36	36	36
8	4521	m48	2016-03-24	35.3	35.3	35.3	35.3
9	4565	m24	2014-04-05	44.8	44.8	44.8	44.8
10	4809	m24	2014-06-19	57.8	57.8	57.8	57.8

```
# i 5 more variables: EXAMDATE_ptau <date>, PLASMAPTAU181 <dbl>,
# first_ptau <dbl>, last_ptau <dbl>, mean_ptau <dbl>
```

```
# calculating the difference in time between EXAMDATE_nfl and EXAMDATE_ptau
diff <- diff %>% mutate(time_diff = EXAMDATE_nfl - EXAMDATE_ptau)

# fetching the time differences
diff$time_diff
```

Time differences in days

```
[1] 7 4 48 -3 231 10 3 1 -17 1 3 4 5 3 6 3 -
2 49 -1
[20] 15 19 -1 6 10 7 7 21 1 31 6 6 4 115 10 6 10 3 -
1
[39] 8 8 18 13 3 1 -29 3 2 6 3 -10 -8 2 6 1 1 29 3
[58] 11 8 8 -10 6 3 8 9 2 9 -7 5 -1 12 4 17 4 16 2
[77] 1 3 19 4 7 -5
```

```
# checking how many observations have a time difference over 30 days (1 month)
diff %>% filter(abs(time_diff) > 30) %>% dplyr::select(time_diff)
```

```
# A tibble: 5 x 1
```

```
time_diff
<drtn>
1 48 days
2 231 days
3 49 days
4 31 days
5 115 days
```

1.5.1 EXAMDATE is EXAMDATE_nfl or EXAMDATE_nfl +- time difference

```
# creating a common EXAMDATE from two cases
# 1: EXAMDATE_nfl == EXAMDATE_ptau -> EXAMDATE == EXAMDATE_nfl
# 2: EXAMDATE_nfl != EXAMDATE_ptau -> EXAMDATE == EXAMDATE_nfl + the difference between the
# then defining the baseline date from the earliest EXAMDATE
long_data <- long_data %>%
  group_by(RID) %>%
  mutate(
    EXAMDATE = EXAMDATE_ptau + 0.5 * (EXAMDATE_nfl - EXAMDATE_ptau),
    EXAMDATE_baseline = min(EXAMDATE),
    time_years = as.numeric(EXAMDATE - EXAMDATE_baseline) / 365.25) %>%
  ungroup()
```

1.5.1.1 Sensitivity analysis: baseline date is the earliest of EXAMDATE_nfl and EXAMDATE_ptau

```
# dates aren't too different, and only one observation at baseline, but to be sure create the
#long_data <- long_data %>%
# group_by(RID) %>%
# mutate(
#   # first biomarker exam across NFL and pTau
#   EXAMDATE = pmin(EXAMDATE_nfl, EXAMDATE_ptau, na.rm = TRUE),
#
#   EXAMDATE_baseline = min(EXAMDATE, na.rm = TRUE),
#
#   time_years = as.numeric(EXAMDATE - EXAMDATE_baseline) / 365.25
# ) %>%
# ungroup()
```

1.6 Removing early diagnosis observations

```
# filtering the observations out of the diagnosis data set where EXAMDATE is much before baseline
# creating a baseline_dx variable, so we can set time_to_event = 0 if baseline_dx == DEMENTIA
diagnosis_data_clean <- diagnosis_data %>%
  merge(
    long_data %>% dplyr::select(RID, EXAMDATE_baseline), by = "RID"
  ) %>%
  filter(EXAMDATE_dx - EXAMDATE_baseline >= -50) %>%
  distinct() %>%
  group_by(RID) %>%
  arrange(EXAMDATE_dx) %>%
  mutate(
    baseline_dx = first(DIAGNOSIS_dx),
    last_dx = last(DIAGNOSIS_dx)
  ) %>%
  dplyr::select(!EXAMDATE_baseline)
```

1.7 Merging longitudinal and diagnosis data for survival data set

```
# We merge NFL, ptau and diagnosis data for survival data, only by RID
surv_data <- long_data %>% left_join(diagnosis_data_clean, by = c("RID"))
```

```
Warning in left_join(., diagnosis_data_clean, by = c("RID")): Detected an unexpected many-
to-many relationship between `x` and `y`.
i Row 1 of `x` matches multiple rows in `y`.
i Row 99 of `y` matches multiple rows in `x`.
i If a many-to-many relationship is expected, set `relationship =
  "many-to-many"` to silence this warning.
```

1.8 Checking missing values

```
# We count missing values per column in long_data: all these are variables we will use in the
  model, so we count them
apply(long_data, function(x) sum(is.na(x)))
```

RID	VISCODE2	EXAMDATE_nfl	PLASMA_NFL
0	0	0	1
first_nfl	last_nfl	mean_nfl	EXAMDATE_ptau
1	1	1	13
PLASMAPTAU181	first_ptau	last_ptau	mean_ptau
13	13	13	13
EXAMDATE	EXAMDATE_baseline	time_years	
13	47	47	

```
# We count missing values per column in surv_data: VISCODE2.y will not be used in the model,
  so we count them
apply(surv_data, function(x) sum(is.na(x)))
```

RID	VISCODE2.x	EXAMDATE_nfl	PLASMA_NFL
0	0	0	1
first_nfl	last_nfl	mean_nfl	EXAMDATE_ptau
1	1	1	13
PLASMAPTAU181	first_ptau	last_ptau	mean_ptau
13	13	13	13
EXAMDATE	EXAMDATE_baseline	time_years	ORIGPROT_dx
13	47	47	49
VISCODE2.y	EXAMDATE_dx	DIAGNOSIS_dx	baseline_dx
49	49	49	49
last_dx			
49			

```
# There are no rows with missing VISCODE2.y with full data for the model variables, so we can
  safely filter them out
surv_data %>% filter(is.na(VISCODE2.y))
```



```
# A tibble: 49 x 21
  RID VISCODE2.x EXAMDATE_nfl PLASMA_NFL first_nfl last_nfl mean_nfl
  <dbl> <chr>      <date>          <dbl>      <dbl>      <dbl>      <dbl>
1   896 m24      2008-11-13      69.9      69.9      69.9      69.9
2   896 m48      2010-11-12      34.2      34.2      34.2      34.2
3   896 m60      2011-12-01      42.3      42.3      42.3      42.3
4   896 m72      2012-11-28      44.3      44.3      44.3      44.3
5  1122 m48      2011-02-25      45.4      45.4      45.4      45.4
6  1122 m72      2013-01-18      38.1      38.1      38.1      38.1
7  1122 m84      2014-01-27      31.1      31.1      31.1      31.1
8  1122 m96      2015-01-14      35.6      35.6      35.6      35.6
9  1203 m48      2011-03-23      54.5      54.5      54.5      54.5
10 1256 m48      2011-04-21      27.1      27.1      27.1      27.1
# i 39 more rows
# i 14 more variables: EXAMDATE_ptau <date>, PLASMAPTAU181 <dbl>,
# first_ptau <dbl>, last_ptau <dbl>, mean_ptau <dbl>, EXAMDATE <date>,
# EXAMDATE_baseline <date>, time_years <dbl>, ORIGPROT_dx <chr>,
# VISCODE2.y <chr>, EXAMDATE_dx <date>, DIAGNOSIS_dx <chr>,
# baseline_dx <chr>, last_dx <chr>
```

```
# Checking for ptau-181 and NfL values = -4, this is missing data code in ADNI documentation
nrow(long_data %>% filter(PLASMA_NFL == -4, PLASMAPTAU181 == -4))
```

```
[1] 0
```

```
# We remove missing values
long_data <- na.omit(long_data)
surv_data <- na.omit(surv_data)

# We re-count missing values
sapply(long_data, function(x) sum(is.na(x)))
```

RID	VISCODE2	EXAMDATE_nfl	PLASMA_NFL
0	0	0	0
first_nfl	last_nfl	mean_nfl	EXAMDATE_ptau
0	0	0	0
PLASMAPTAU181	first_ptau	last_ptau	mean_ptau
0	0	0	0
EXAMDATE	EXAMDATE_baseline	time_years	
0	0	0	

```
sapply(surv_data, function(x) sum(is.na(x)))
```

RID	VISCODE2.x	EXAMDATE_nfl	PLASMA_NFL
0	0	0	0
first_nfl	last_nfl	mean_nfl	EXAMDATE_ptau
0	0	0	0
PLASMAPTAU181	first_ptau	last_ptau	mean_ptau
0	0	0	0
EXAMDATE	EXAMDATE_baseline	time_years	ORIGPROT_dx
0	0	0	0
VISCODE2.y	EXAMDATE_dx	DIAGNOSIS_dx	baseline_dx
0	0	0	0
last_dx			
0			

1.9 Merging selected demographic variables (age and gender) and APOE4 status

```
# We only keep the demographics variables we need and one row per person
demographics_small <- demographics %>%
  dplyr::select(RID, PTDOBY, PTGENDER) %>%
  distinct(RID, .keep_all = TRUE)
```

```
# We define APOE4 status based on number of E4 alleles
apoe <- apoe %>%
```

```
  mutate(APOE4_status = as.factor(
    case_when(
      GENOTYPE == "4/4" ~ "2 alleles",
      GENOTYPE %in% c("2/4", "3/4") ~ "1 allele",
      TRUE ~ "0 alleles")),
    genotype = as.factor(GENOTYPE)
  )
```

```
apoe$APOE4_status <- factor(apoe$APOE4_status,
  levels = c("0 alleles", "1 allele", "2 alleles"))
```

```
# We merge gender with long_data and surv_data
```

```
long_data <- long_data %>%
  left_join(demographics_small, by = "RID")
surv_data <- surv_data %>%
  left_join(demographics_small, by = "RID")
```

```

# We merge apoe with long_data and surv_data
long_data <- merge(long_data, apoe[, c("RID", "APOE4_status", "genotype")], by="RID")

# We make sure that gender is a factor
long_data$PTGENDER <- as.factor(long_data$PTGENDER)
surv_data$PTGENDER <- as.factor(surv_data$PTGENDER)

# We create age_at_visit
long_data$age_at_visit <- as.numeric(format(long_data$EXAMDATE_ptau, "%Y")) - long_data$PTDOI
surv_data$age_at_visit <- as.numeric(format(surv_data$EXAMDATE_ptau, "%Y")) - surv_data$PTDOI

```

1.10 Selecting participants with at least 3 measurements

```

# We filter individuals with at least 3 measurements of ptau_data
keep <- unique((ptau_data %>% group_by(RID) %>% filter(n() > 2))$RID)
long_data <- long_data %>% filter(RID %in% keep)
surv_data <- surv_data %>% filter(RID %in% keep)
# We check number of individuals with at least 3 measurements
length(keep)

```

[1] 782

1.10.1 Sensitivity analysis: Selecting participants with at least 1 measurement

```

# We filter individuals with at least 1 measurement of ptau_data
#keep <- unique((ptau_data %>% group_by(RID) %>% filter(n() > 0))$RID)
#long_data <- long_data %>% filter(RID %in% keep)
#surv_data <- surv_data %>% filter(RID %in% keep)
# We check number of individuals with at least 1 measurement
#length(keep)

```

1.11 Preparing for Cox PH model

1.11.1 Constructing global (common) baseline, age at baseline, event and time-to-event variables

```
surv_data <- surv_data %>%
  group_by(RID) %>%
  summarise(
    #carrying these forward to check events and event times are coded correctly
    baseline_dx = unique(baseline_dx),
    last_dx = unique(last_dx),

#    baseline EXAMDATE defined as midpoint between ptau and nfl exam dates if different
    EXAMDATE_baseline = min(EXAMDATE_baseline),

    #age at baseline visit
    age_at_baseline = age_at_visit[
      EXAMDATE_ptau == min(EXAMDATE_ptau[!is.na(PLASMAPTAU181)]) |
      EXAMDATE_nfl == min(EXAMDATE_nfl[!is.na(PLASMA_NFL)])
    ][1],

    #last dx/nfl/ptau exam
    last_followup = max(c(EXAMDATE_ptau, EXAMDATE_nfl, EXAMDATE_dx), na.rm = TRUE),

    #1 = dementia, 0 = otherwise
    event = as.integer(any(DIAGNOSIS_dx == "Dementia")),

    #0 if dementia dx at baseline, otherwise time to dementia dx or date of censoring
    time_to_event = case_when(
      event == 1 & unique(baseline_dx) == "Dementia" ~ 0,
      event == 1 & unique(baseline_dx) != "Dementia" ~
        as.numeric(min(EXAMDATE_dx[DIAGNOSIS_dx == "Dementia"]) - EXAMDATE_baseline)/365,
      TRUE ~
        as.numeric(last_followup - EXAMDATE_baseline)/365),

    #Sensitivity: event time = midpoint between last clean exam and dementia dx exam
    time_to_event_midpoint = case_when(
      event == 1 & unique(baseline_dx) == "Dementia" ~ 0,
      event == 1 & unique(baseline_dx) != "Dementia" ~ {
        dx_date <- min(EXAMDATE_dx[DIAGNOSIS_dx == "Dementia"])
        prev_date <- max(EXAMDATE_dx[DIAGNOSIS_dx != "Dementia" &
```

```

                                EXAMDATE_dx < dx_date],
                                na.rm = TRUE)
  as.numeric((prev_date + (dx_date - prev_date) / 2) - EXAMDATE_baseline) / 365
},
TRUE ~
  as.numeric(last_followup - EXAMDATE_baseline) / 365
)
)

```

Warning: There were 1120 warnings in `summarise()`.
 The first warning was:
 i In argument: `time\_to\_event = case\_when(...)`.
 i In group 1: `RID = 2`.
 Caused by warning in `min.default()`:
 ! no non-missing arguments to min; returning Inf
 i Run `dplyr::last\_dplyr\_warnings()` to see the 1119 remaining warnings.

```

# extracting number of events with time_to_event != 0
table((surv_data %>% filter(time_to_event > 0))$event)

```

```

  0    1
527 177

```

```

table((surv_data %>% filter(time_to_event == 0))$event)

```

```

  1
66

```

1.12 Removing individuals with dementia at start

```

# Removing individuals who had a dementia dx at time = 0
surv_data <- surv_data %>% filter(time_to_event > 0)
long_data <- long_data %>% filter(RID %in% unique(surv_data$RID))

# extracting number of events with time_to_event != 0
table((surv_data %>% filter(time_to_event > 0))$event)

```

```

0    1
527 177

```

1.12.1 Checking the final diagnosis for the selected participants

```

final_diagnosis <- surv_data %>%
  left_join(
    diagnosis_data_clean %>%
      arrange(RID, EXAMDATE_dx) %>%
      group_by(RID) %>%
      slice_tail(n = 1) %>%
      ungroup() %>%
      transmute(
        RID,
        final_dx = DIAGNOSIS_dx
      ),
    by = "RID"
  )

table(final_diagnosis$final_dx, useNA = "ifany")

```

CN Dementia	MCI
240 168	296

1.12.2 Constructing baseline values of each biomarker, to include in the Cox PH models as covariates

```

# Creating a data frame with the first biomarker measurements
baseline <- long_data %>%
  group_by(RID) %>%
  arrange(RID, EXAMDATE_nfl) %>%
  mutate(baseline_value_nfl = first(PLASMA_NFL)) %>%
  arrange(RID, EXAMDATE_ptau) %>%
  mutate(baseline_value_ptau = first(PLASMAPTAU181)) %>%
  dplyr::select(RID, baseline_value_ptau, baseline_value_nfl) %>%
  ungroup() %>%
  distinct()

```

```
# merging the survival data with demographics and the baseline biomarker values
surv_data <- surv_data %>%
  merge(demographics_small, by = "RID") %>%
  merge(apoe %>% dplyr::select(RID, APOE4_status, genotype), by = "RID") %>%
  merge(baseline, by = "RID")
```

1.13 Table showing cohort characteristics and measurement distributions

```
# Filtering diagnosis data to cohort
dx_full_cohort <- diagnosis_data %>%
  filter(RID %in% surv_data$RID)

# Post-baseline diagnoses aligned to biomarker baseline
dx_post <- dx_full_cohort %>%
  left_join(
    long_data %>%
      dplyr::select(RID, EXAMDATE_baseline) %>%
      distinct(),
    by = "RID"
  ) %>%
  dplyr::filter(EXAMDATE_dx >= EXAMDATE_baseline)

table(dx_post$DIAGNOSIS_dx)
```

	CN Dementia	MCI
	1723	459
		2489

```
# Diagnosis distributions
dx_full_counts <- table(dx_full_cohort$DIAGNOSIS_dx)
dx_post_counts <- table(dx_post$DIAGNOSIS_dx)

# Biomarker measurement frequency per participant
ptau_table <- long_data %>%
  filter(!is.na(PLASMAPTAU181)) %>%
  count(RID) %>%
  count(n, name = "n_RID")

nfl_table <- long_data %>%
```

```

filter(!is.na(PLASMA_NFL)) %>%
count(RID) %>%
count(n, name = "n_RID")

# Diagnosis visits per participant (min/median/max)
dx_visits <- dx_full_cohort %>%
  filter(!is.na(DIAGNOSIS_dx)) %>%
  count(RID) %>%
  summarise(
    min = min(n),
    median = median(n),
    max = max(n)
  )

# Examination date ranges
ptau_range <- range(long_data$EXAMDATE_ptau, na.rm = TRUE)
nfl_range <- range(long_data$EXAMDATE_nfl, na.rm = TRUE)
dx_range <- range(dx_full_cohort$EXAMDATE_dx, na.rm = TRUE)

# Cohort summary table
cohort_summary_table <- tibble(
  Category = c(
    "Age at baseline, years (median [range])",
    "Sex distribution",
    "APOE 4 genotype distribution",
    "Plasma p-tau181 levels (median [range])",
    "Plasma NfL levels (median [range])",
    "p-tau181 examination dates (range)",
    "NfL examination dates (range)",
    "Diagnosis examination dates (range)",
    "All diagnoses distribution",
    "Post-baseline diagnoses distribution",
    "Participants by number of p-tau181 measurements",
    "Participants by number of NfL measurements",
    "Diagnosis visits per participant (median [range])"
  ),
  Value = c(
    paste0(median(surv_data$age_at_baseline, na.rm = TRUE),
      " (", min(surv_data$age_at_baseline, na.rm = TRUE),
      ", ", max(surv_data$age_at_baseline, na.rm = TRUE), ")"),
    paste(names(table(surv_data$PTGENDER)),

```



```

    table(surv_data$PTGENDER), collapse = "; "),

  paste(names(table(surv_data$genotype)),
    table(surv_data$genotype), collapse = "; "),

  paste0(median(long_data$PLASMAPTAU181, na.rm = TRUE),
    " (", min(long_data$PLASMAPTAU181, na.rm = TRUE),
    ", ", max(long_data$PLASMAPTAU181, na.rm = TRUE), ")"),

  paste0(median(long_data$PLASMA_NFL, na.rm = TRUE),
    " (", min(long_data$PLASMA_NFL, na.rm = TRUE),
    ", ", max(long_data$PLASMA_NFL, na.rm = TRUE), ")"),

  paste(ptau_range, collapse = " - "),
  paste(nfl_range, collapse = " - "),
  paste(dx_range, collapse = " - "),

  paste(names(dx_full_counts), dx_full_counts, collapse = "; "),
  paste(names(dx_post_counts), dx_post_counts, collapse = "; "),

  paste(ptau_table$n_RID, collapse = ", "),
  paste(nfl_table$n_RID, collapse = ", "),

  paste0(dx_visits$min, " (", dx_visits$median, ", ", dx_visits$max, ")")
)
)

cohort_summary_table

```

A tibble: 13 x 2

Category <chr>	Value <chr>
1 Age at baseline, years (median [range])	74 (55, 94)
2 Sex distribution	Female 315; Male 389
3 APOE 4 genotype distribution	2/2 1; 2/3 69; 2/4 13; 3/3~
4 Plasma p-tau181 levels (median [range])	15.479 (0.362, 238.324)
5 Plasma NfL levels (median [range])	36.5 (6.9, 408)
6 p-tau181 examination dates (range)	2007-02-01 - 2016-06-06
7 NfL examination dates (range)	2007-02-01 - 2016-06-06
8 Diagnosis examination dates (range)	2005-09-29 - 2025-11-07
9 All diagnoses distribution	CN 2469; Dementia 463; MCI~
10 Post-baseline diagnoses distribution	CN 1723; Dementia 459; MCI~

11 Participants by number of p-tau181 measurements 223, 303, 178
 12 Participants by number of NfL measurements 223, 303, 178
 13 Diagnosis visits per participant (median [range]) 3 (8, 20)

1.14 Checking outliers

```
# We identify "extreme" values

# Values greater than 3 sd above ptau mean
long_data[scale(long_data$PLASMAPTAU181) > 3, ]
```

	RID	VISCODE2	EXAMDATE_nfl	PLASMA_NFL	first_nfl	last_nfl	mean_nfl
89	125	m48	2010-03-02	46.2	46.2	46.2	46.2
114	159	m60	2011-04-28	36.2	36.2	36.2	36.2
115	159	m72	2012-03-05	36.5	36.5	36.5	36.5
117	159	m96	2014-03-12	38.9	38.9	38.9	38.9
118	159	m108	2015-03-05	48.2	48.2	48.2	48.2
316	605	m84	2013-07-16	53.5	53.5	53.5	53.5
318	605	m108	2015-06-30	65.3	65.3	65.3	65.3
372	709	m72	2012-08-09	59.3	59.3	59.3	59.3
473	951	m48	2010-12-17	128.1	128.1	128.1	128.1
495	989	m72	2012-06-29	34.8	34.8	34.8	34.8
798	2146	m12	2011-11-30	15.2	15.2	15.2	15.2
867	2200	m48	2015-02-05	19.9	19.9	19.9	19.9
948	2264	b1	2011-03-30	36.3	36.3	36.3	36.3
1010	2367	m48	2015-05-19	141.8	141.8	141.8	141.8
1198	4060	b1	2011-06-22	83.1	83.1	83.1	83.1
1199	4060	m12	2012-06-19	77.1	77.1	77.1	77.1
1200	4060	m24	2013-06-17	69.5	69.5	69.5	69.5
1201	4060	m48	2015-06-04	131.5	131.5	131.5	131.5
1359	4167	m36	2014-08-27	50.5	50.5	50.5	50.5
1360	4167	m48	2015-09-03	63.2	63.2	63.2	63.2
1375	4171	b1	2011-09-02	37.1	37.1	37.1	37.1
1377	4171	m24	2013-10-29	40.6	40.6	40.6	40.6
1512	4237	b1	2011-10-13	45.8	45.8	45.8	45.8
1761	4383	m24	2013-12-23	31.1	31.1	31.1	31.1
1762	4383	m36	2015-02-17	33.8	33.8	33.8	33.8
2016	4505	b1	2012-02-15	22.1	22.1	22.1	22.1
2018	4505	m24	2014-02-21	26.9	26.9	26.9	26.9
2232	4614	m12	2013-03-12	43.1	43.1	43.1	43.1

2289	4643	b1	2012-04-27	26.2	26.2	26.2	26.2
2356	4711	m24	2014-06-13	42.5	42.5	42.5	42.5
2534	4857	m12	2013-10-01	160.0	160.0	160.0	160.0
2573	4877	m24	2014-10-07	145.1	145.1	145.1	145.1
2596	4891	m36	2015-08-10	37.0	37.0	37.0	37.0
2754	5131	b1	2013-05-06	16.2	16.2	16.2	16.2
	EXAMDATE_ptau	PLASMAPTAU181	first_ptau	last_ptau	mean_ptau	EXAMDATE	
89	2010-03-02	76.067	76.067	76.067	76.067	2010-03-02	
114	2011-04-28	124.089	124.089	124.089	124.089	2011-04-28	
115	2012-03-05	136.848	136.848	136.848	136.848	2012-03-05	
117	2014-03-12	168.531	168.531	168.531	168.531	2014-03-12	
118	2015-03-05	238.324	238.324	238.324	238.324	2015-03-05	
316	2013-07-16	139.362	139.362	139.362	139.362	2013-07-16	
318	2015-06-30	109.514	109.514	109.514	109.514	2015-06-30	
372	2012-08-09	86.701	86.701	86.701	86.701	2012-08-09	
473	2010-12-17	83.658	83.658	83.658	83.658	2010-12-17	
495	2012-06-29	122.329	122.329	122.329	122.329	2012-06-29	
798	2011-11-30	69.362	69.362	69.362	69.362	2011-11-30	
867	2015-02-05	68.341	68.341	68.341	68.341	2015-02-05	
948	2011-03-30	67.556	67.556	67.556	67.556	2011-03-30	
1010	2015-05-19	57.540	57.540	57.540	57.540	2015-05-19	
1198	2011-06-22	58.593	58.593	58.593	58.593	2011-06-22	
1199	2012-06-19	64.773	64.773	64.773	64.773	2012-06-19	
1200	2013-06-17	57.576	57.576	57.576	57.576	2013-06-17	
1201	2015-06-04	83.520	83.520	83.520	83.520	2015-06-04	
1359	2014-08-27	64.958	64.958	64.958	64.958	2014-08-27	
1360	2015-09-03	61.861	61.861	61.861	61.861	2015-09-03	
1375	2011-08-29	121.405	121.405	121.405	121.405	2011-08-31	
1377	2013-10-29	95.447	95.447	95.447	95.447	2013-10-29	
1512	2011-10-13	60.020	60.020	60.020	60.020	2011-10-13	
1761	2013-12-23	83.682	83.682	83.682	83.682	2013-12-23	
1762	2015-02-17	62.412	62.412	62.412	62.412	2015-02-17	
2016	2012-02-15	59.610	59.610	59.610	59.610	2012-02-15	
2018	2014-02-21	59.248	59.248	59.248	59.248	2014-02-21	
2232	2013-03-12	71.836	71.836	71.836	71.836	2013-03-12	
2289	2012-04-27	72.258	72.258	72.258	72.258	2012-04-27	
2356	2014-06-13	72.440	72.440	72.440	72.440	2014-06-13	
2534	2013-10-01	86.961	86.961	86.961	86.961	2013-10-01	
2573	2014-10-07	61.970	61.970	61.970	61.970	2014-10-07	
2596	2015-08-10	60.836	60.836	60.836	60.836	2015-08-10	
2754	2013-05-06	148.037	148.037	148.037	148.037	2013-05-06	
	EXAMDATE_baseline	time_years	PTDOBY	PTGENDER	APOE4_status	genotype	
89	2010-03-02	0.0000000	1932	Male	0 alleles	3/3	

114	2011-04-28	0.0000000	1928	Female	0 alleles	3/3
115	2011-04-28	0.8542094	1928	Female	0 alleles	3/3
117	2011-04-28	2.8720055	1928	Female	0 alleles	3/3
118	2011-04-28	3.8521561	1928	Female	0 alleles	3/3
316	2007-06-27	6.0533881	1930	Female	1 allele	3/4
318	2007-06-27	8.0082136	1930	Female	1 allele	3/4
372	2012-08-09	0.0000000	1922	Male	1 allele	3/4
473	2010-12-17	0.0000000	1921	Male	0 alleles	3/3
495	2011-01-05	1.4811773	1942	Female	1 allele	2/4
798	2010-11-17	1.0349076	1944	Male	0 alleles	3/3
867	2010-12-15	4.1423682	1934	Female	0 alleles	3/3
948	2011-03-30	0.0000000	1941	Male	1 allele	3/4
1010	2011-05-25	3.9835729	1936	Male	1 allele	3/4
1198	2011-06-22	0.0000000	1926	Male	0 alleles	2/3
1199	2011-06-22	0.9938398	1926	Male	0 alleles	2/3
1200	2011-06-22	1.9876797	1926	Male	0 alleles	2/3
1201	2011-06-22	3.9507187	1926	Male	0 alleles	2/3
1359	2011-08-25	3.0061602	1942	Female	1 allele	3/4
1360	2011-08-25	4.0246407	1942	Female	1 allele	3/4
1375	2011-08-31	0.0000000	1942	Male	0 alleles	3/3
1377	2011-08-31	2.1629021	1942	Male	0 alleles	3/3
1512	2011-10-13	0.0000000	1930	Female	0 alleles	3/3
1761	2012-01-12	1.9466119	1939	Male	0 alleles	2/3
1762	2012-01-12	3.0992471	1939	Male	0 alleles	2/3
2016	2012-02-15	0.0000000	1931	Female	0 alleles	3/3
2018	2012-02-15	2.0177960	1931	Female	0 alleles	3/3
2232	2012-04-26	0.8761123	1944	Female	1 allele	3/4
2289	2012-04-27	0.0000000	1947	Female	0 alleles	3/3
2356	2012-06-01	2.0314853	1937	Female	0 alleles	3/3
2534	2012-08-13	1.1334702	1944	Male	1 allele	3/4
2573	2012-09-19	2.0479124	1928	Male	1 allele	3/4
2596	2012-08-23	2.9623546	1952	Male	2 alleles	4/4
2754	2013-05-06	0.0000000	1942	Male	0 alleles	3/3

age_at_visit

89	78
114	83
115	84
117	86
118	87
316	83
318	85
372	90
473	89

495	70
798	67
867	81
948	70
1010	79
1198	85
1199	86
1200	87
1201	89
1359	72
1360	73
1375	69
1377	71
1512	81
1761	74
1762	76
2016	81
2018	83
2232	69
2289	65
2356	77
2534	69
2573	86
2596	63
2754	71

```
# Values smaller than 3 sd below ptau mean
long_data[scale(long_data$PLASMAPTAU181) < -3, ]
```

```
[1] RID          VISCODE2      EXAMDATE_nfl  PLASMA_NFL
[5] first_nfl     last_nfl     mean_nfl      EXAMDATE_ptau
[9] PLASMAPTAU181 first_ptau   last_ptau     mean_ptau
[13] EXAMDATE      EXAMDATE_baseline time_years    PTDOBY
[17] PTGENDER      APOE4_status genotype      age_at_visit
<0 rows> (or 0-length row.names)
```

```
# Values greater than 3 sd above nfl mean
long_data[scale(long_data$PLASMA_NFL) > 3, ]
```

	RID	VISCODE2	EXAMDATE_nfl	PLASMA_NFL	first_nfl	last_nfl	mean_nfl
5	8	m60	2010-09-27	196.3	196.3	196.3	196.3

38	61	m96	2013-12-16	122.4	122.4	122.4	122.4
128	173	m96	2014-03-12	122.5	122.5	122.5	122.5
187	301	m96	2014-04-14	227.2	227.2	227.2	227.2
191	307	m72	2012-06-21	245.7	245.7	245.7	245.7
192	307	m84	2013-07-02	258.3	258.3	258.3	258.3
193	307	m96	2014-06-24	305.6	305.6	305.6	305.6
304	558	m72	2012-08-13	143.6	143.6	143.6	143.6
327	626	m48	2010-08-17	128.9	128.9	128.9	128.9
328	626	m60	2011-08-25	121.2	121.2	121.2	121.2
329	626	m72	2012-08-01	129.7	129.7	129.7	129.7
330	626	m84	2013-08-06	134.8	134.8	134.8	134.8
473	951	m48	2010-12-17	128.1	128.1	128.1	128.1
488	984	m96	2014-12-09	154.0	154.0	154.0	154.0
491	985	m72	2012-10-19	135.4	135.4	135.4	135.4
492	985	m84	2013-10-28	211.7	211.7	211.7	211.7
600	1268	m60	2012-03-13	142.0	142.0	142.0	142.0
886	2213	m24	2013-02-06	408.0	408.0	408.0	408.0
898	2220	b1	2011-01-26	151.8	151.8	151.8	151.8
1009	2367	m36	2014-05-21	203.6	203.6	203.6	203.6
1010	2367	m48	2015-05-19	141.8	141.8	141.8	141.8
1041	2381	m36	2014-06-02	124.5	124.5	124.5	124.5
1051	2390	m48	2015-07-20	122.2	122.2	122.2	122.2
1201	4060	m48	2015-06-04	131.5	131.5	131.5	131.5
1345	4158	m48	2015-09-03	175.6	175.6	175.6	175.6
1514	4237	m24	2013-10-22	177.2	177.2	177.2	177.2
1619	4299	m24	2013-11-04	230.5	230.5	230.5	230.5
1687	4343	m24	2013-11-22	142.4	142.4	142.4	142.4
1792	4392	m12	2013-01-07	152.8	152.8	152.8	152.8
1833	4410	b1	2012-01-13	282.2	282.2	282.2	282.2
1858	4426	m36	2015-03-09	122.6	122.6	122.6	122.6
2164	4579	m12	2013-06-04	199.2	199.2	199.2	199.2
2228	4613	m12	2013-04-10	196.5	196.5	196.5	196.5
2350	4689	m36	2015-05-20	138.5	138.5	138.5	138.5
2524	4844	m36	2015-11-19	128.2	128.2	128.2	128.2
2534	4857	m12	2013-10-01	160.0	160.0	160.0	160.0
2536	4857	m36	2015-08-06	133.5	133.5	133.5	133.5
2550	4869	m24	2014-09-25	154.9	154.9	154.9	154.9
2571	4877	b1	2012-09-19	124.4	124.4	124.4	124.4
2572	4877	m12	2013-09-09	140.1	140.1	140.1	140.1
2573	4877	m24	2014-10-07	145.1	145.1	145.1	145.1
2697	5031	m12	2014-04-17	151.1	151.1	151.1	151.1
EXAMDATE_ptau PLASMAPTAU181 first_ptau last_ptau mean_ptau EXAMDATE							
5	2010-09-27	18.305	18.305	18.305	18.305	18.305	2010-09-27

38	2013-12-16	31.044	31.044	31.044	31.044	2013-12-16
128	2014-03-05	28.325	28.325	28.325	28.325	2014-03-09
187	2014-04-14	13.750	13.750	13.750	13.750	2014-04-14
191	2012-06-21	37.148	37.148	37.148	37.148	2012-06-21
192	2013-07-02	48.334	48.334	48.334	48.334	2013-07-02
193	2014-06-24	48.784	48.784	48.784	48.784	2014-06-24
304	2012-08-13	30.057	30.057	30.057	30.057	2012-08-13
327	2010-08-17	17.559	17.559	17.559	17.559	2010-08-17
328	2011-08-25	13.133	13.133	13.133	13.133	2011-08-25
329	2012-08-01	21.652	21.652	21.652	21.652	2012-08-01
330	2013-08-06	22.261	22.261	22.261	22.261	2013-08-06
473	2010-12-17	83.658	83.658	83.658	83.658	2010-12-17
488	2014-12-09	35.915	35.915	35.915	35.915	2014-12-09
491	2012-10-19	25.049	25.049	25.049	25.049	2012-10-19
492	2013-10-28	33.615	33.615	33.615	33.615	2013-10-28
600	2012-03-13	27.982	27.982	27.982	27.982	2012-03-13
886	2012-12-19	10.075	10.075	10.075	10.075	2013-01-12
898	2011-01-26	8.012	8.012	8.012	8.012	2011-01-26
1009	2014-05-21	42.190	42.190	42.190	42.190	2014-05-21
1010	2015-05-19	57.540	57.540	57.540	57.540	2015-05-19
1041	2014-06-02	22.554	22.554	22.554	22.554	2014-06-02
1051	2015-07-20	8.513	8.513	8.513	8.513	2015-07-20
1201	2015-06-04	83.520	83.520	83.520	83.520	2015-06-04
1345	2015-09-03	51.773	51.773	51.773	51.773	2015-09-03
1514	2013-10-22	50.092	50.092	50.092	50.092	2013-10-22
1619	2013-11-04	8.081	8.081	8.081	8.081	2013-11-04
1687	2013-11-22	22.521	22.521	22.521	22.521	2013-11-22
1792	2013-01-07	21.521	21.521	21.521	21.521	2013-01-07
1833	2012-01-13	49.726	49.726	49.726	49.726	2012-01-13
1858	2015-03-09	14.152	14.152	14.152	14.152	2015-03-09
2164	2013-06-04	14.840	14.840	14.840	14.840	2013-06-04
2228	2013-04-10	10.890	10.890	10.890	10.890	2013-04-10
2350	2015-05-20	14.285	14.285	14.285	14.285	2015-05-20
2524	2015-11-19	25.219	25.219	25.219	25.219	2015-11-19
2534	2013-10-01	86.961	86.961	86.961	86.961	2013-10-01
2536	2015-08-06	49.016	49.016	49.016	49.016	2015-08-06
2550	2014-09-25	12.177	12.177	12.177	12.177	2014-09-25
2571	2012-09-19	52.246	52.246	52.246	52.246	2012-09-19
2572	2013-09-09	46.722	46.722	46.722	46.722	2013-09-09
2573	2014-10-07	61.970	61.970	61.970	61.970	2014-10-07
2697	2014-04-17	17.438	17.438	17.438	17.438	2014-04-17

	EXAMDATE_baseline	time_years	PTDOBY	PTGENDER	APOE4_status	genotype
5	2010-09-27	0.0000000	1921	Female	0 alleles	2/3

38	2011-02-08	2.8528405	1928	Female	0 alleles	3/3
128	2012-03-15	1.9822040	1932	Male	0 alleles	3/3
187	2011-04-27	2.9650924	1932	Female	0 alleles	3/3
191	2012-06-21	0.0000000	1927	Male	0 alleles	3/3
192	2012-06-21	1.0294319	1927	Male	0 alleles	3/3
193	2012-06-21	2.0068446	1927	Male	0 alleles	3/3
304	2010-07-20	2.0670773	1926	Male	0 alleles	3/3
327	2010-08-17	0.0000000	1923	Male	0 alleles	3/3
328	2010-08-17	1.0212183	1923	Male	0 alleles	3/3
329	2010-08-17	1.9575633	1923	Male	0 alleles	3/3
330	2010-08-17	2.9705681	1923	Male	0 alleles	3/3
473	2010-12-17	0.0000000	1921	Male	0 alleles	3/3
488	2010-12-15	3.9835729	1930	Male	1 allele	3/4
491	2010-11-30	1.8863792	1925	Male	1 allele	2/4
492	2010-11-30	2.9103354	1925	Male	1 allele	2/4
600	2011-04-25	0.8843258	1924	Male	1 allele	3/4
886	2010-12-14	2.0807666	1928	Female	0 alleles	3/3
898	2011-01-26	0.0000000	1928	Male	0 alleles	3/3
1009	2011-05-25	2.9897331	1936	Male	1 allele	3/4
1010	2011-05-25	3.9835729	1936	Male	1 allele	3/4
1041	2011-05-31	3.0061602	1942	Male	2 alleles	4/4
1051	2011-07-08	4.0328542	1923	Female	0 alleles	3/3
1201	2011-06-22	3.9507187	1926	Male	0 alleles	2/3
1345	2011-08-26	4.0219028	1927	Male	0 alleles	2/3
1514	2011-10-13	2.0260096	1930	Female	0 alleles	3/3
1619	2011-11-16	1.9685147	1934	Female	0 alleles	2/3
1687	2011-11-18	2.0123203	1932	Male	0 alleles	3/3
1792	2012-01-06	1.0047912	1929	Male	0 alleles	3/3
1833	2012-01-13	0.0000000	1942	Female	0 alleles	3/3
1858	2012-01-30	3.1047228	1934	Male	0 alleles	2/3
2164	2012-06-12	0.9774127	1927	Male	0 alleles	3/3
2228	2012-04-18	0.9774127	1944	Male	0 alleles	2/3
2350	2012-05-14	3.0143737	1942	Male	2 alleles	4/4
2524	2012-08-06	3.2854209	1927	Male	1 allele	2/4
2534	2012-08-13	1.1334702	1944	Male	1 allele	3/4
2536	2012-08-13	2.9787817	1944	Male	1 allele	3/4
2550	2012-09-10	2.0396988	1935	Male	0 alleles	2/3
2571	2012-09-19	0.0000000	1928	Male	1 allele	3/4
2572	2012-09-19	0.9719370	1928	Male	1 allele	3/4
2573	2012-09-19	2.0479124	1928	Male	1 allele	3/4
2697	2013-06-14	0.8405202	1932	Female	0 alleles	3/3

age_at_visit

5

89

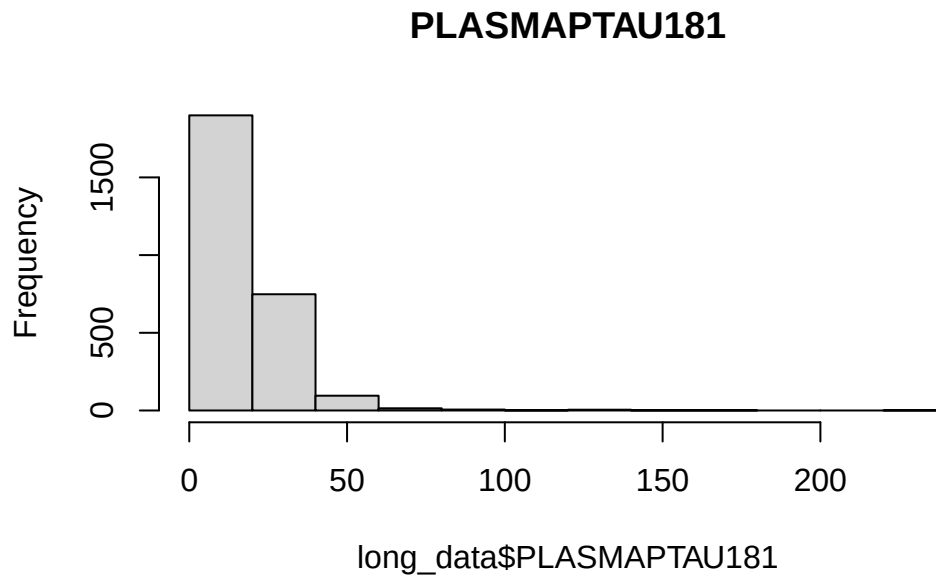
38	85
128	82
187	82
191	85
192	86
193	87
304	86
327	87
328	88
329	89
330	90
473	89
488	84
491	87
492	88
600	88
886	84
898	83
1009	78
1010	79
1041	72
1051	92
1201	89
1345	88
1514	83
1619	79
1687	81
1792	84
1833	70
1858	81
2164	86
2228	69
2350	73
2524	88
2534	69
2536	71
2550	79
2571	84
2572	85
2573	86
2697	82

```
# Values smaller than 3 sd below nfl mean
long_data[scale(long_data$PLASMA_NFL) < -3, ]
```

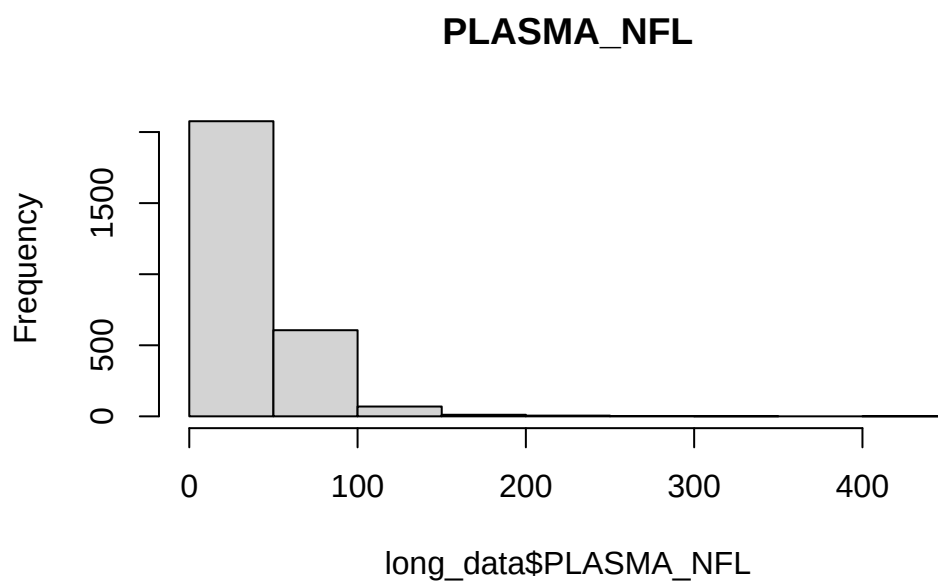
```
[1] RID          VISCODE2      EXAMDATE_nfl  PLASMA_NFL
[5] first_nfl    last_nfl     mean_nfl      EXAMDATE_ptau
[9] PLASMAPTAU181 first_ptau   last_ptau     mean_ptau
[13] EXAMDATE     EXAMDATE_baseline time_years    PTDOBYYY
[17] PTGENDER     APOE4_status genotype      age_at_visit
<0 rows> (or 0-length row.names)
```

1.15 Transformation of the biomarkers

```
# We check the distribution of each biomarker
hist(long_data$PLASMAPTAU181, main="PLASMAPTAU181")
```

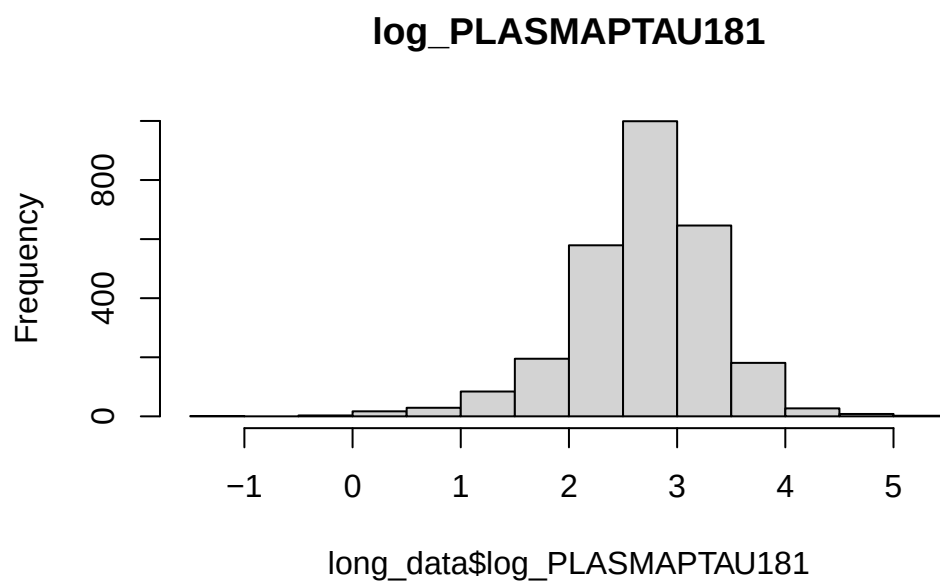


```
hist(long_data$PLASMA_NFL, main="PLASMA_NFL")
```

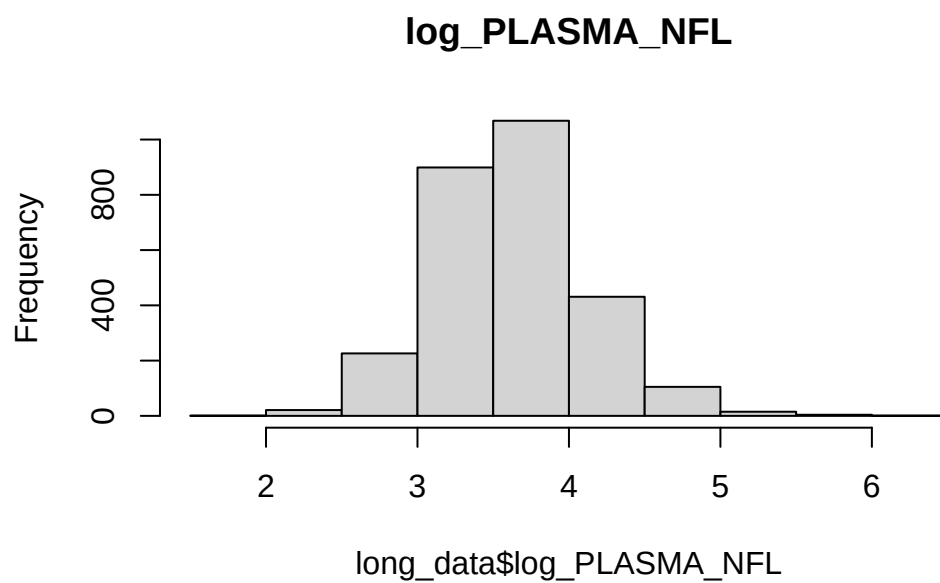


```
# We log transform the biomarkers
long_data$log_PLASMAPTAU181 <- log(long_data$PLASMAPTAU181)
long_data$log_PLASMA_NFL <- log(long_data$PLASMA_NFL)
```

```
# We re-check the distribution of each biomarker
hist(long_data$log_PLASMAPTAU181, main="log_PLASMAPTAU181")
```



```
hist(long_data$log_PLASMA_NFL, main="log_PLASMA_NFL")
```



```
# Transforming the baseline values in the survival data
surv_data$baseline_value_ptau <- log(surv_data$baseline_value_ptau)
surv_data$baseline_value_nfl <- log(surv_data$baseline_value_nfl)
```

1.16 Rechecking longitudinal and survival data lengths

```
# We check the number of individuals in each dataset
length(unique(long_data$RID)) # longitudinal data
```

```
[1] 704
```

```
length(unique(surv_data$RID)) # survival data
```

```
[1] 704
```

1.17 Splitting data into in-sample and out-sample

```
# Total number of individuals
RID <- unique(surv_data$RID)

# Number of individuals to test the model on
n_out_sample <- length(RID) * (1/4)

# Out-sample individuals
RID_out_sample <- sample(RID, n_out_sample)

# Longitudinal data with in-sample individuals
long_data_in <- long_data %>% filter(!RID %in% RID_out_sample)

# Longitudinal data with out-sample individuals
long_data_out <- long_data %>% filter(RID %in% RID_out_sample)

# Survival data with in-sample individuals
surv_data_in <- surv_data %>% filter(!RID %in% RID_out_sample)

# Survival data with out-sample individuals
surv_data_out <- surv_data %>% filter(RID %in% RID_out_sample)
```

1.18 Table showing number of participants, event distribution, and event rates in in-sample and out-of-sample data

```
event_table <- data.frame(  
  Dataset = c("In-sample", "Out-of-sample"),  
  
  # Number of unique participants in each dataset  
  Participants = c(  
    length(unique(surv_data_in$RID)),  
    length(unique(surv_data_out$RID))  
  ),  
  
  # Number of events (event == 1)  
  Events = c(  
    sum(surv_data_in$event == 1),  
    sum(surv_data_out$event == 1)  
  ),  
  
  # Number of censored observations (event == 0)  
  Censored = c(  
    sum(surv_data_in$event == 0),  
    sum(surv_data_out$event == 0)  
  ),  
  
  # Event rate = proportion of participants experiencing the event  
  Event_rate = c(  
    sum(surv_data_in$event == 1) / length(unique(surv_data_in$RID)),  
    sum(surv_data_out$event == 1) / length(unique(surv_data_out$RID))  
  )  
)  
event_table
```

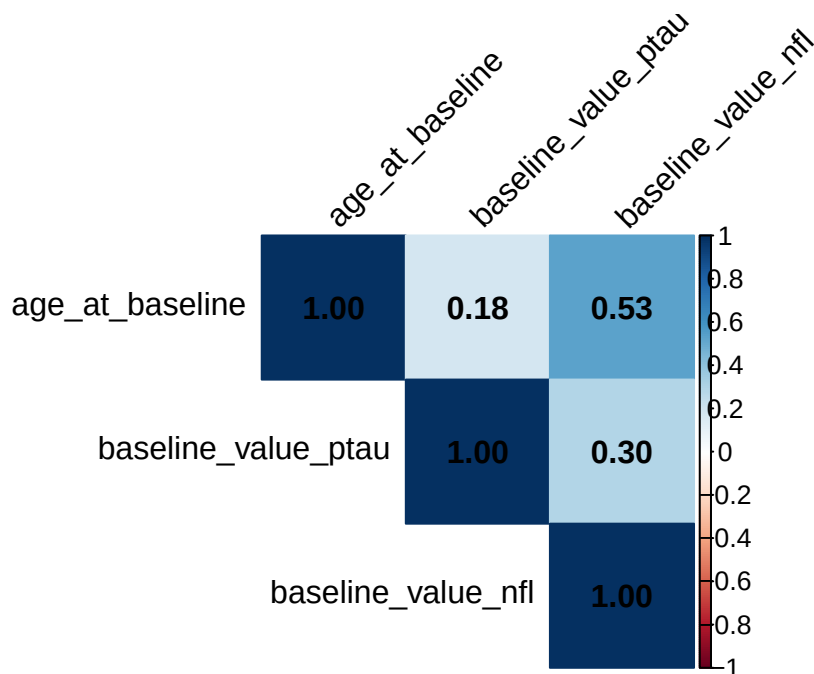
	Dataset	Participants	Events	Censored	Event_rate
1	In-sample	528	128	400	0.2424242
2	Out-of-sample	176	49	127	0.2784091

1.19 Checking correlation between the log-transformed biomarkers as well as the numerical variables going to be used in the models

```
# Saving the numerical variables used in the Cox PH models
cox_vars <- surv_data_in[, c("age_at_baseline",
                             "baseline_value_ptau",
                             "baseline_value_nfl")]

# Computing correlation matrix for Cox model variables
cox_cor <- cor(cox_vars, use = "complete.obs")

# Visualising correlations
corrplot(cox_cor,
          method = "color",
          type = "upper",
          addCoef.col = "black",
          tl.col = "black",
          tl.srt = 45)
```



```
# Saving the numerical variables used in the LME models
lme_vars <- long_data_in[, c("time_years",
                             "log_PLASMAPTAU181",
```

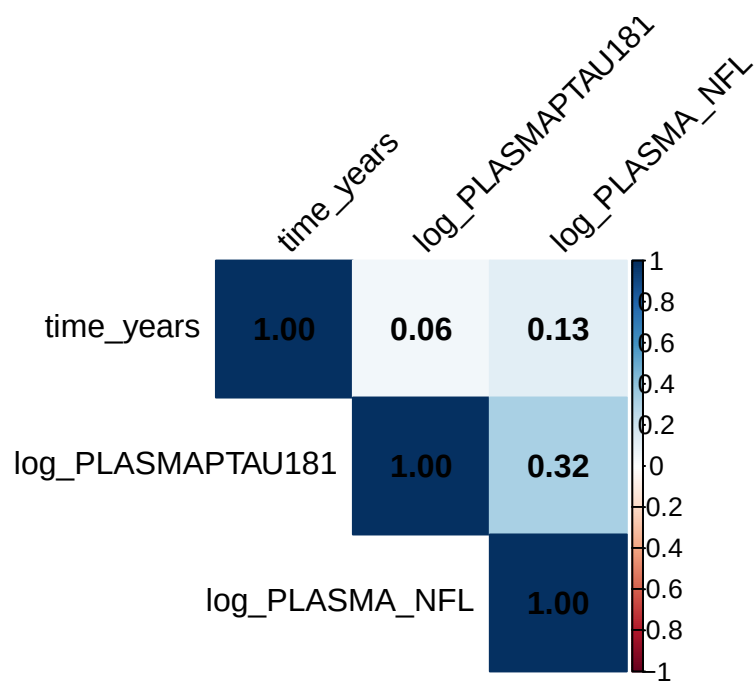
```

    "log_PLASMA_NFL")]

# Computing correlation matrix for LME model variables
lme_cor <- cor(lme_vars, use = "complete.obs")

# Visualising correlations
corrplot(lme_cor,
  method = "color",
  type = "upper",
  addCoef.col = "black",
  tl.col = "black",
  tl.srt = 45)

```



2 Longitudinal models

2.1 Comparing link functions for longitudinal models

2.1.1 PTAU


```
# checking the mean and variance for ptau181
mean(long_data$PLASMAPTAU181)
```

```
[1] 18.00103
```

```
var(long_data$PLASMAPTAU181)
```

```
[1] 172.8281
```

2.1.1.1 Fitting the models

```
# GLMM
fit_gamma_ptau <- glmmTMB(PLASMAPTAU181 ~ time_years + (1|RID),
  data = long_data_in,
  family = Gamma(link = "log"))

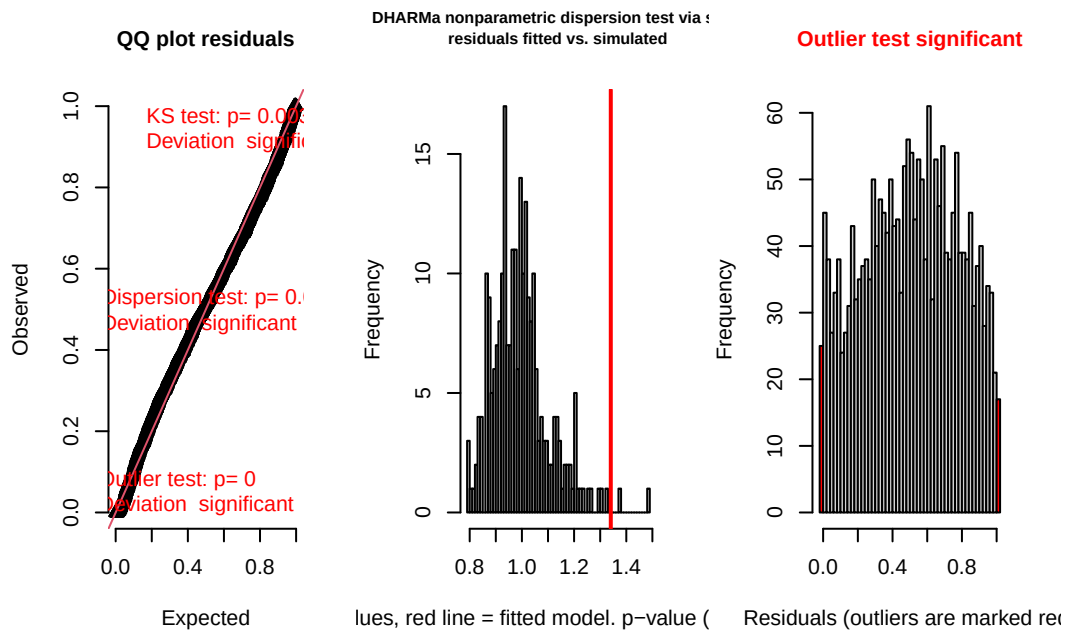
# LME approach with log transformed data
fit_lme_log_ptau <- lmer(log(PLASMAPTAU181) ~ time_years + (1|RID),
  data = long_data_in)

# LME approach with log transformed data
fit_lme_cubed_ptau <- lmer(sqrt(sqrt(PLASMAPTAU181)) ~ time_years + (1|RID),
  data = long_data_in)

# Lognormal - similar to log-transforming then fitting LME
fit_lognormal_ptau <- glmmTMB(PLASMAPTAU181 ~ time_years + (1|RID),
  data = long_data_in,
  family = lognormal(link = "log"))

# Student-t
#fit_student_ptau <- glmmTMB(PLASMAPTAU181 ~ time_years + (1|RID),
  #data = long_data_in,
  #family = students.t(df = 3, link = "identity"))

# residual tests for each model
gamma_residuals_ptau <- simulateResiduals(fit_gamma_ptau)
testResiduals(gamma_residuals_ptau)
```



`$uniformity`

Asymptotic one-sample Kolmogorov-Smirnov test

```
data: simulationOutput$scaledResiduals
D = 0.039402, p-value = 0.003193
alternative hypothesis: two-sided
```

`$dispersion`

DHARMA nonparametric dispersion test via sd of residuals fitted vs. simulated

```
data: simulationOutput
dispersion = 1.3467, p-value = 0.016
alternative hypothesis: two.sided
```

`$outliers`

DHARMA outlier test based on exact binomial test with approximate expectations

```

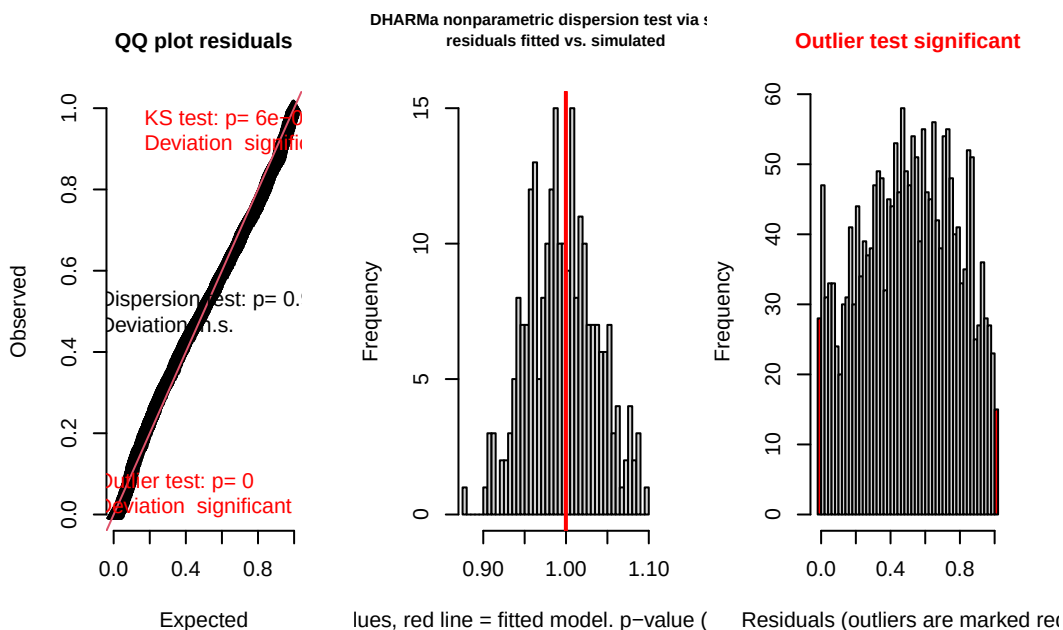
data: simulationOutput
outliers at both margin(s) = 42, observations = 2074, p-value =
9.497e-08
alternative hypothesis: true probability of success is not equal to 0.007968127
95 percent confidence interval:
 0.01463296 0.02727537
sample estimates:
frequency of outliers (expected: 0.00796812749003984 )
                                0.02025072

```

```

lme_residuals_log_ptau <- simulateResiduals(fit_lme_log_ptau)
testResiduals(lme_residuals_log_ptau)

```



`$uniformity`

Asymptotic one-sample Kolmogorov-Smirnov test

```

data: simulationOutput$scaledResiduals
D = 0.044224, p-value = 0.0005996
alternative hypothesis: two-sided

```

```
$dispersion
```

```
DHARMA nonparametric dispersion test via sd of residuals fitted vs.  
simulated
```

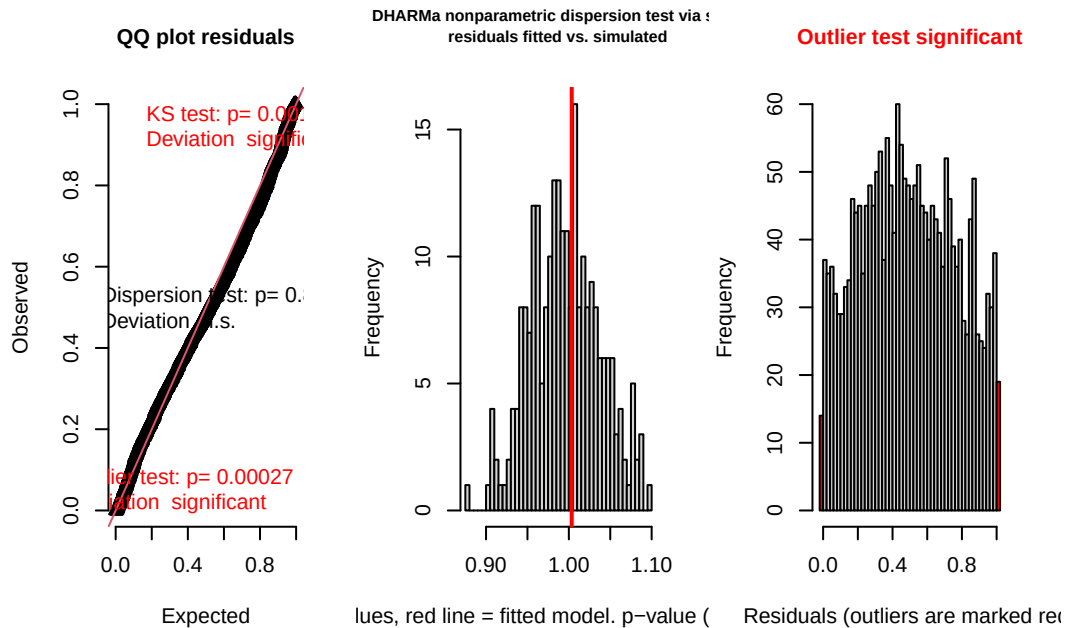
```
data: simulationOutput  
dispersion = 1.0042, p-value = 0.912  
alternative hypothesis: two.sided
```

```
$outliers
```

```
DHARMA outlier test based on exact binomial test with approximate  
expectations
```

```
data: simulationOutput  
outliers at both margin(s) = 43, observations = 2074, p-value =  
3.557e-08  
alternative hypothesis: true probability of success is not equal to 0.007968127  
95 percent confidence interval:  
0.01504413 0.02782634  
sample estimates:  
frequency of outliers (expected: 0.00796812749003984 )  
0.02073288
```

```
lme_residuals_cubed_ptau <- simulateResiduals(fit_lme_cubed_ptau)  
testResiduals(lme_residuals_cubed_ptau)
```



`$uniformity`

Asymptotic one-sample Kolmogorov-Smirnov test

```
data: simulationOutput$scaledResiduals
D = 0.041724, p-value = 0.001462
alternative hypothesis: two-sided
```

`$dispersion`

DHARMA nonparametric dispersion test via sd of residuals fitted vs. simulated

```
data: simulationOutput
dispersion = 1.0083, p-value = 0.864
alternative hypothesis: two.sided
```

`$outliers`

DHARMA outlier test based on exact binomial test with approximate expectations

```

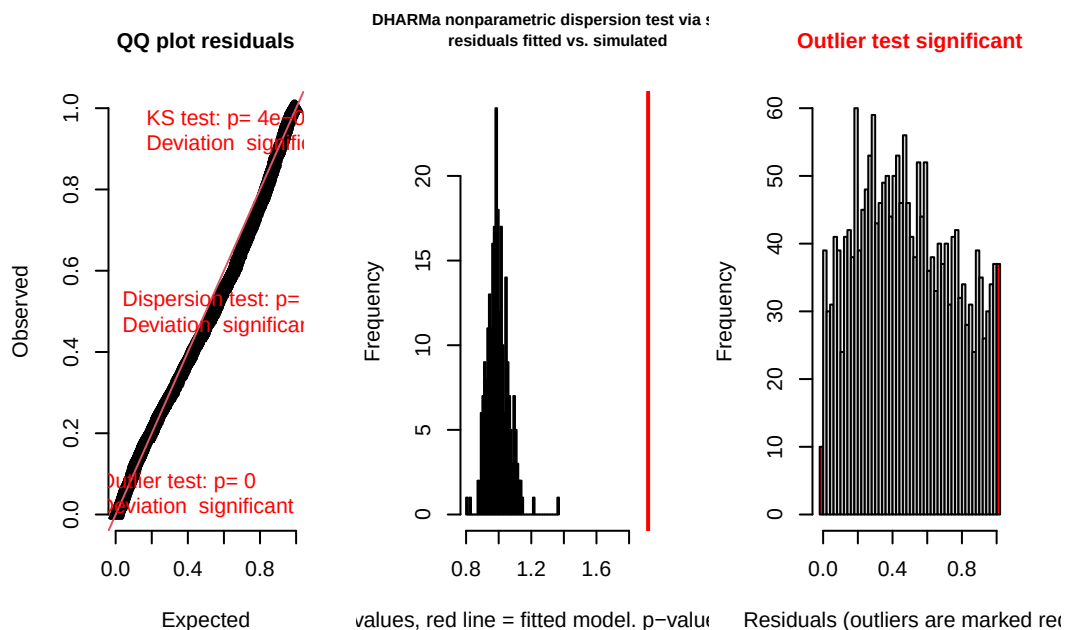
data: simulationOutput
outliers at both margin(s) = 33, observations = 2074, p-value =
0.0002738
alternative hypothesis: true probability of success is not equal to 0.007968127
95 percent confidence interval:
 0.01097725 0.02227335
sample estimates:
frequency of outliers (expected: 0.00796812749003984 )
                                0.01591128

```

```

lognormal_residuals_ptau <- simulateResiduals(fit_lognormal_ptau)
testResiduals(lognormal_residuals_ptau)

```



\$uniformity

Asymptotic one-sample Kolmogorov-Smirnov test

```

data: simulationOutput$scaledResiduals
D = 0.05097, p-value = 4.178e-05
alternative hypothesis: two-sided

```

```
$dispersion
```

```
DHARMa nonparametric dispersion test via sd of residuals fitted vs.  
simulated
```

```
data: simulationOutput  
dispersion = 1.9249, p-value < 2.2e-16  
alternative hypothesis: two.sided
```

```
$outliers
```

```
DHARMa outlier test based on exact binomial test with approximate  
expectations
```

```
data: simulationOutput  
outliers at both margin(s) = 47, observations = 2074, p-value =  
5.599e-10  
alternative hypothesis: true probability of success is not equal to 0.007968127  
95 percent confidence interval:  
0.01669722 0.03002208  
sample estimates:  
frequency of outliers (expected: 0.00796812749003984 )  
0.02266152
```

```
#student_residuals_ptau <- simulateResiduals(fit_student_ptau)  
#testResiduals(student_residuals_ptau)
```

2.1.1.2 Shapiro-Wilk test

```
# Shapiro-Wilk test of normality of the residuals  
shapiro.test(gamma_residuals_ptau$fittedResiduals)
```

```
Shapiro-Wilk normality test
```

```
data: gamma_residuals_ptau$fittedResiduals  
W = 0.65447, p-value < 2.2e-16
```

```
shapiro.test(lme_residuals_log_ptau$fittedResiduals)
```

Shapiro-Wilk normality test

```
data: lme_residuals_log_ptau$fittedResiduals  
W = 0.93881, p-value < 2.2e-16
```

```
shapiro.test(lme_residuals_cubed_ptau$fittedResiduals)
```

Shapiro-Wilk normality test

```
data: lme_residuals_cubed_ptau$fittedResiduals  
W = 0.94131, p-value < 2.2e-16
```

```
shapiro.test(lognormal_residuals_ptau$fittedResiduals)
```

Shapiro-Wilk normality test

```
data: lognormal_residuals_ptau$fittedResiduals  
W = 0.39621, p-value < 2.2e-16
```

```
#shapiro.test(student_residuals_ptau$fittedResiduals)
```

2.1.2 NFL

```
# checking the mean and variance for nfl  
mean(long_data$PLASMA_NFL)
```

```
[1] 42.51267
```

```
var(long_data$PLASMA_NFL)
```

```
[1] 676.8582
```


2.1.2.1 Fitting the models

```
# GLMM
fit_gamma_nfl <- glmmTMB(PLASMA_NFL ~ time_years + (1|RID),
                        data = long_data_in,
                        family = Gamma(link = "log"))

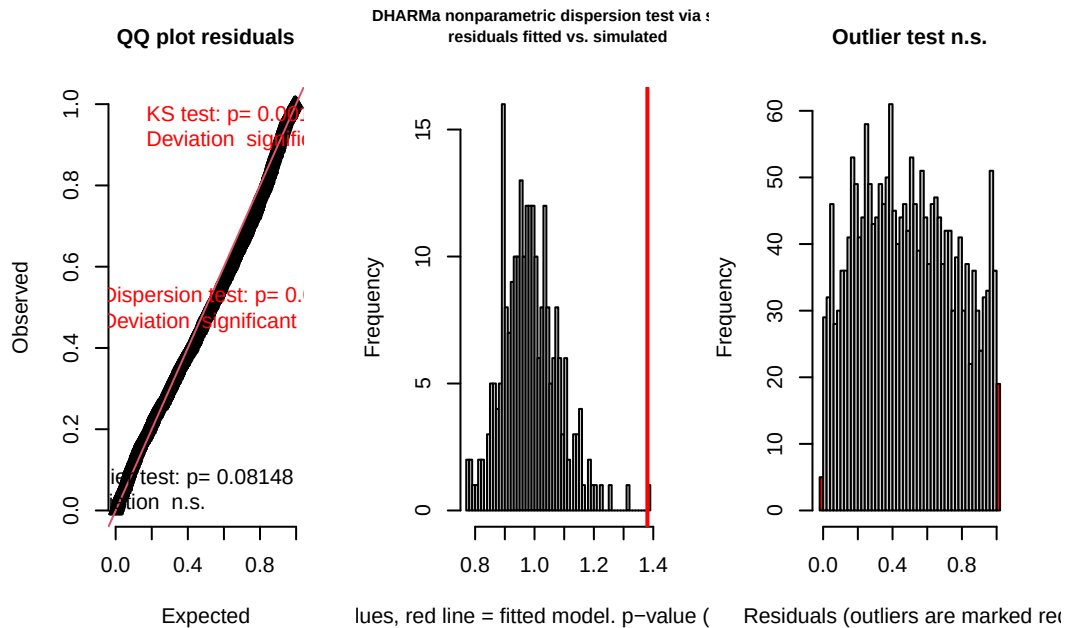
# LME approach with log transformed data
fit_lme_log_nfl <- lmer(log(PLASMA_NFL) ~ time_years + (1|RID),
                       data = long_data_in)

# LME approach with log transformed data
fit_lme_cubed_nfl <- lmer(sqrt(sqrt(PLASMA_NFL)) ~ time_years + (1|RID),
                         data = long_data_in)

# Lognormal - similar to log-transforming then fitting LME
fit_lognormal_nfl <- glmmTMB(PLASMA_NFL ~ time_years + (1|RID),
                             data = long_data_in,
                             family = lognormal(link = "log"))

# Student-t
#fit_student_nfl <- glmmTMB(PLASMA_NFL ~ time_years + (1|RID),
#                           #data = long_data_in,
#                           #family = students.t(df = 3, link = "identity"))

# residual tests for each model
gamma_residuals_nfl <- simulateResiduals(fit_gamma_nfl)
testResiduals(gamma_residuals_nfl)
```



`$uniformity`

Asymptotic one-sample Kolmogorov-Smirnov test

```
data: simulationOutput$scaledResiduals
D = 0.04201, p-value = 0.001324
alternative hypothesis: two-sided
```

`$dispersion`

DHARMA nonparametric dispersion test via sd of residuals fitted vs. simulated

```
data: simulationOutput
dispersion = 1.4011, p-value = 0.008
alternative hypothesis: two.sided
```

`$outliers`

DHARMA outlier test based on exact binomial test with approximate expectations

```

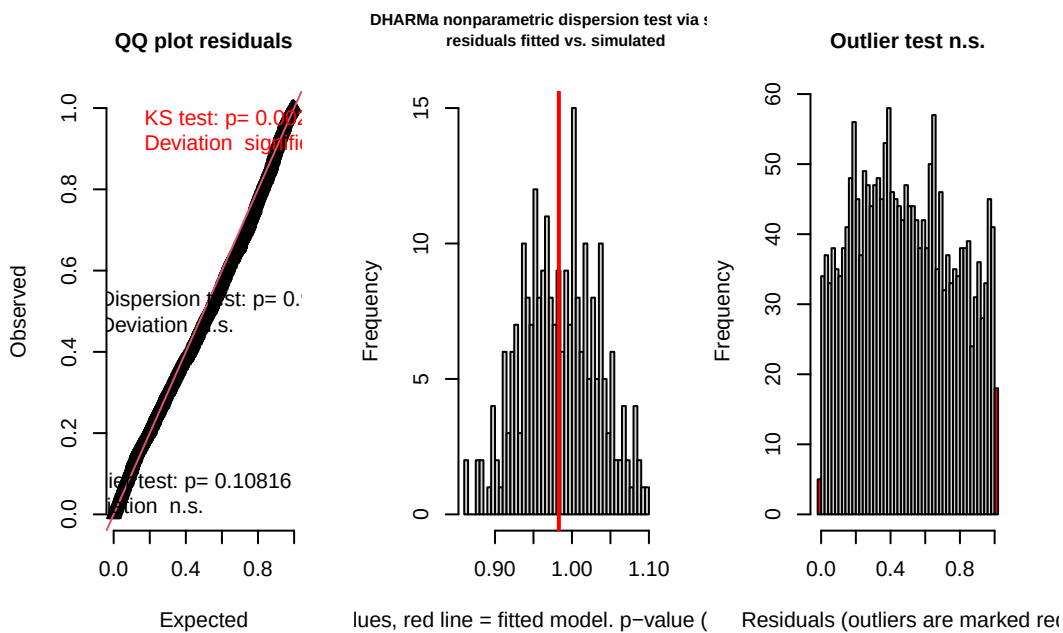
data: simulationOutput
outliers at both margin(s) = 24, observations = 2074, p-value = 0.08148
alternative hypothesis: true probability of success is not equal to 0.007968127
95 percent confidence interval:
 0.007427981 0.017169334
sample estimates:
frequency of outliers (expected: 0.00796812749003984 )
                                0.01157184

```

```

lme_residuals_log_nfl <- simulateResiduals(fit_lme_log_nfl)
testResiduals(lme_residuals_log_nfl)

```



`$uniformity`

Asymptotic one-sample Kolmogorov-Smirnov test

```

data: simulationOutput$scaledResiduals
D = 0.039614, p-value = 0.002979
alternative hypothesis: two-sided

```

```
$dispersion
```

```
DHARMA nonparametric dispersion test via sd of residuals fitted vs.  
simulated
```

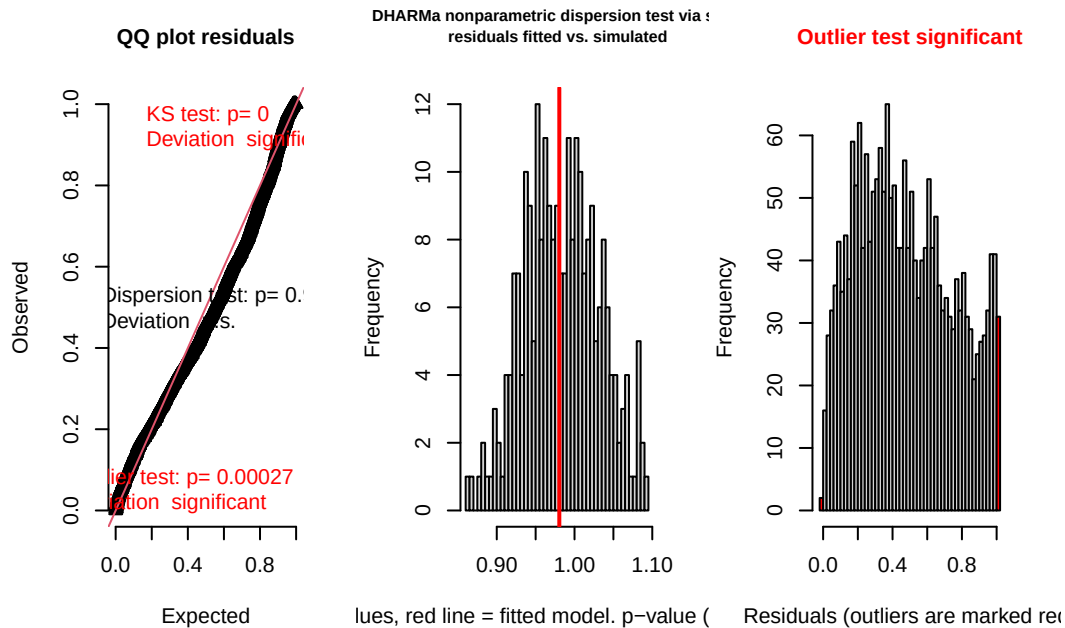
```
data: simulationOutput  
dispersion = 0.99872, p-value = 0.992  
alternative hypothesis: two.sided
```

```
$outliers
```

```
DHARMA outlier test based on exact binomial test with approximate  
expectations
```

```
data: simulationOutput  
outliers at both margin(s) = 23, observations = 2074, p-value = 0.1082  
alternative hypothesis: true probability of success is not equal to 0.007968127  
95 percent confidence interval:  
0.007042534 0.016593754  
sample estimates:  
frequency of outliers (expected: 0.00796812749003984 )  
0.01108968
```

```
lme_residuals_cubed_nfl <- simulateResiduals(fit_lme_cubed_nfl)  
testResiduals(lme_residuals_cubed_nfl)
```



`$uniformity`

Asymptotic one-sample Kolmogorov-Smirnov test

```
data: simulationOutput$scaledResiduals
D = 0.063919, p-value = 8.729e-08
alternative hypothesis: two-sided
```

`$dispersion`

DHARMA nonparametric dispersion test via sd of residuals fitted vs. simulated

```
data: simulationOutput
dispersion = 0.99617, p-value = 0.968
alternative hypothesis: two.sided
```

`$outliers`

DHARMA outlier test based on exact binomial test with approximate expectations

```

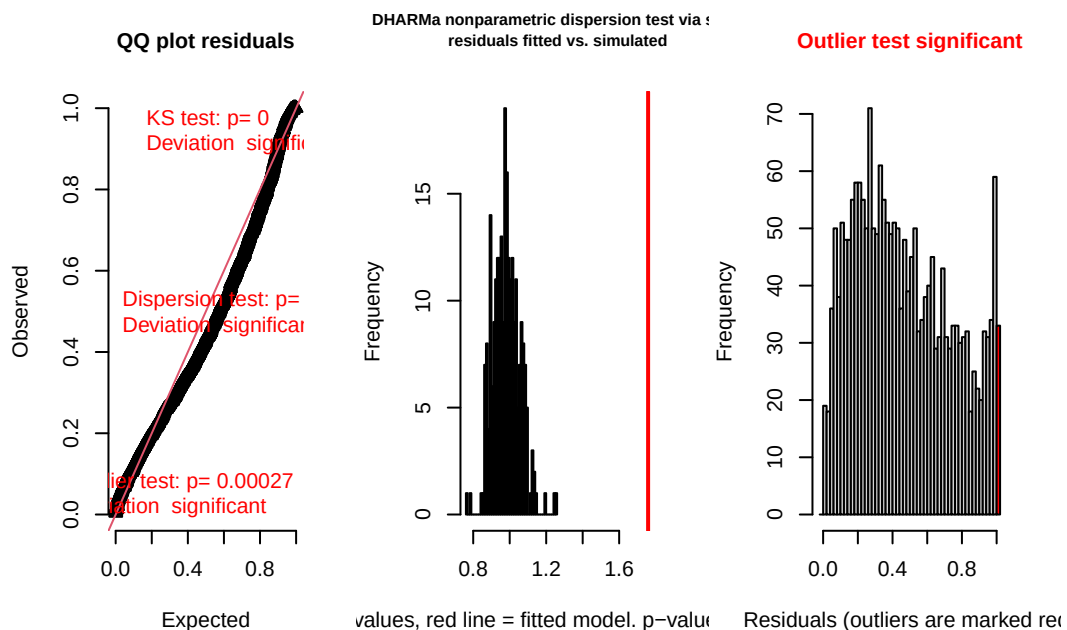
data: simulationOutput
outliers at both margin(s) = 33, observations = 2074, p-value =
0.0002738
alternative hypothesis: true probability of success is not equal to 0.007968127
95 percent confidence interval:
 0.01097725 0.02227335
sample estimates:
frequency of outliers (expected: 0.00796812749003984 )
                                0.01591128

```

```

lognormal_residuals_nfl <- simulateResiduals(fit_lognormal_nfl)
testResiduals(lognormal_residuals_nfl)

```



\$uniformity

Asymptotic one-sample Kolmogorov-Smirnov test

```

data: simulationOutput$scaledResiduals
D = 0.082326, p-value = 1.235e-12
alternative hypothesis: two-sided

```

```
$dispersion
```

```
DHARMa nonparametric dispersion test via sd of residuals fitted vs.  
simulated
```

```
data: simulationOutput  
dispersion = 1.792, p-value < 2.2e-16  
alternative hypothesis: two.sided
```

```
$outliers
```

```
DHARMa outlier test based on exact binomial test with approximate  
expectations
```

```
data: simulationOutput  
outliers at both margin(s) = 33, observations = 2074, p-value =  
0.0002738  
alternative hypothesis: true probability of success is not equal to 0.007968127  
95 percent confidence interval:  
0.01097725 0.02227335  
sample estimates:  
frequency of outliers (expected: 0.00796812749003984 )  
0.01591128
```

```
#student_residuals_nfl <- simulateResiduals(fit_student_nfl)  
#testResiduals(student_residuals_nfl)
```

2.1.2.2 Shapiro-Wilk test

```
# Shapiro-Wilk test of normality of the residuals  
shapiro.test(gamma_residuals_nfl$fittedResiduals)
```

```
Shapiro-Wilk normality test
```

```
data: gamma_residuals_nfl$fittedResiduals  
W = 0.58139, p-value < 2.2e-16
```

```
shapiro.test(lme_residuals_log_nfl$fittedResiduals)
```

Shapiro-Wilk normality test

```
data: lme_residuals_log_nfl$fittedResiduals  
W = 0.90784, p-value < 2.2e-16
```

```
shapiro.test(lme_residuals_cubed_nfl$fittedResiduals)
```

Shapiro-Wilk normality test

```
data: lme_residuals_cubed_nfl$fittedResiduals  
W = 0.85378, p-value < 2.2e-16
```

```
shapiro.test(lognormal_residuals_nfl$fittedResiduals)
```

Shapiro-Wilk normality test

```
data: lognormal_residuals_nfl$fittedResiduals  
W = 0.39437, p-value < 2.2e-16
```

```
#shapiro.test(student_residuals_nfl$fittedResiduals)
```

3 Fitted LME models

3.1 PTAU

3.1.0.1 With all genotype levels

- None of the coefficients are significant for any of the levels for genotype

```
# Longitudinal model for log p-tau181 with normal distribution  
long_model_ptau_full <- lme(  
  fixed = log_PLASMAPTAU181 ~ time_years + PTGENDER + genotype,  
  random = ~ time_years | RID,  
  data = long_data_in  
)
```



```
summary(long_model_ptau_full)
```

Linear mixed-effects model fit by REML

Data: long_data_in

	AIC	BIC	logLik
	2966.918	3028.89	-1472.459

Random effects:

Formula: ~time_years | RID

Structure: General positive-definite, Log-Cholesky parametrization

	StdDev	Corr
(Intercept)	0.44692120	(Intr)
time_years	0.03674755	-0.046
Residual	0.38473386	

Fixed effects: log_PLASMAPTAU181 ~ time_years + PTGENDER + genotype

	Value	Std.Error	DF	t-value	p-value
(Intercept)	2.5052814	0.07433678	1545	33.70177	0.0000
time_years	0.0280568	0.00681553	1545	4.11660	0.0000
PTGENDERMale	0.1440026	0.04283348	522	3.36192	0.0008
genotype2/4	0.3741022	0.16286738	522	2.29697	0.0220
genotype3/3	0.0104219	0.07529241	522	0.13842	0.8900
genotype3/4	0.1818469	0.07978823	522	2.27912	0.0231
genotype4/4	0.3672691	0.10989807	522	3.34191	0.0009

Correlation:

	(Intr)	tm_yrs	PTGEND	gnt2/4	gnt3/3	gnt3/4
time_years	-0.134					
PTGENDERMale	-0.336	0.006				
genotype2/4	-0.407	-0.006	0.033			
genotype3/3	-0.868	-0.010	0.031	0.393		
genotype3/4	-0.813	-0.005	0.008	0.370	0.801	
genotype4/4	-0.593	0.002	0.013	0.269	0.581	0.548

Standardized Within-Group Residuals:

	Min	Q1	Med	Q3	Max
	-5.35708358	-0.39126092	0.04771697	0.44370728	4.49604683

Number of Observations: 2074

Number of Groups: 528

3.1.0.2 APOE4 status

- All three levels of APOE4_status are significant

```
# Longitudinal model for log p-tau181 with APOE4 status
long_model_ptau <- lme(
  fixed = log_PLASMAPTAU181 ~ time_years + PTGENDER + APOE4_status,
  random = ~ time_years | RID,
  data = long_data_in
)
```

```
summary(long_model_ptau)
```

Linear mixed-effects model fit by REML

Data: long_data_in

	AIC	BIC	logLik
	2959.265	3009.979	-1470.633

Random effects:

Formula: ~time_years | RID

Structure: General positive-definite, Log-Cholesky parametrization

	StdDev	Corr
(Intercept)	0.44713367	(Intr)
time_years	0.03663542	-0.053
Residual	0.38475714	

Fixed effects: log_PLASMAPTAU181 ~ time_years + PTGENDER + APOE4_status

	Value	Std.Error	DF	t-value	p-value
(Intercept)	2.5150563	0.03691854	1545	68.12449	0e+00
time_years	0.0280923	0.00681321	1545	4.12320	0e+00
PTGENDERMale	0.1421149	0.04277646	524	3.32227	1e-03
APOE4_status1 allele	0.1859964	0.04668609	524	3.98398	1e-04
APOE4_status2 alleles	0.3585102	0.08938962	524	4.01065	1e-04

Correlation:

	(Intr)	tm_yrs	PTGEN	APOE4_1a
time_years	-0.287			
PTGENDERMale	-0.622	0.007		
APOE4_status1 allele	-0.409	0.003	-0.021	
APOE4_status2 alleles	-0.219	0.009	-0.006	0.174

Standardized Within-Group Residuals:

	Min	Q1	Med	Q3	Max
	-5.36479900	-0.38395864	0.04537558	0.44489192	4.49569664

Number of Observations: 2074

Number of Groups: 528

3.1.0.3 Determining random effects

- Fitting the model with ML estimation in order to compare models using likelihood ratio test
- The results give support to fitting the simpler model with only a random intercept. However, *jm()* does not currently support model fitting for univariate longitudinal data with only a random intercept.

```
m1 <- lmer(log_PLASMAPTAU181 ~ time_years + PTGENDER + APOE4_status + (1 | RID),
           data = long_data_in, REML = FALSE)

m2 <- lmer(log_PLASMAPTAU181 ~ time_years + PTGENDER + APOE4_status + (time_years | RID),
           data = long_data_in, REML = FALSE)
```

```
anova(m1, m2, type = "LRT")
```

Data: long_data_in

Models:

```
m1: log_PLASMAPTAU181 ~ time_years + PTGENDER + APOE4_status + (1 | RID)
m2: log_PLASMAPTAU181 ~ time_years + PTGENDER + APOE4_status + (time_years | RID)
```

	npars	AIC	BIC	logLik	-2*log(L)	Chisq	Df	Pr(>Chisq)
m1	7	2931.0	2970.4	-1458.5	2917.0			
m2	9	2933.5	2984.2	-1457.7	2915.5	1.4879	2	0.4752

3.2 NFL

3.2.0.1 With all genotype levels

```
# Longitudinal model for log NfL with normal distribution, all genotype levels
long_model_nfl_full <- lme(
  log_PLASMA_NFL ~ time_years + PTGENDER + genotype,
  random = ~ time_years | RID,
  data = long_data_in
)
```

```
summary(long_model_nfl_full)
```

Linear mixed-effects model fit by REML

Data: long_data_in

	AIC	BIC	logLik
	1078.22	1140.193	-528.1101

Random effects:

Formula: ~time_years | RID

Structure: General positive-definite, Log-Cholesky parametrization

	StdDev	Corr
(Intercept)	0.42104921	(Intr)
time_years	0.02557563	0.396
Residual	0.21404907	

Fixed effects: log_PLASMA_NFL ~ time_years + PTGENDER + genotype

	Value	Std.Error	DF	t-value	p-value
(Intercept)	3.513205	0.06739490	1545	52.12864	0.0000
time_years	0.044734	0.00389577	1545	11.48259	0.0000
PTGENDERMale	0.042819	0.03920154	522	1.09227	0.2752
genotype2/4	-0.008140	0.14926341	522	-0.05453	0.9565
genotype3/3	0.007659	0.06880584	522	0.11132	0.9114
genotype3/4	0.023745	0.07291852	522	0.32564	0.7448
genotype4/4	0.091171	0.10042788	522	0.90782	0.3644

Correlation:

	(Intr)	tm_yrs	PTGEND	gnt2/4	gnt3/3	gnt3/4
time_years	-0.051					
PTGENDERMale	-0.338	0.000				
genotype2/4	-0.410	0.000	0.033			
genotype3/3	-0.875	-0.001	0.029	0.392		
genotype3/4	-0.819	-0.001	0.008	0.369	0.800	
genotype4/4	-0.597	0.000	0.014	0.268	0.581	0.548

Standardized Within-Group Residuals:

	Min	Q1	Med	Q3	Max
	-4.39930890	-0.47163516	-0.04482101	0.40535786	7.00732964

Number of Observations: 2074

Number of Groups: 528

3.2.0.2 APOE4 status

- None of the levels are significant for log NfL and neither is gender

```
# Longitudinal model for log NfL with normal distribution, APOE4_status
long_model_nfl <- lme(
  fixed = log_PLASMA_NFL ~ time_years + PTGENDER + APOE4_status,
  random = ~ time_years | RID,
  data = long_data_in
)
```

```
summary(long_model_nfl)
```

Linear mixed-effects model fit by REML

Data: long_data_in

	AIC	BIC	logLik
	1068.67	1119.383	-525.3349

Random effects:

Formula: ~time_years | RID

Structure: General positive-definite, Log-Cholesky parametrization

	StdDev	Corr
(Intercept)	0.42021509	(Intr)
time_years	0.02556776	0.395
Residual	0.21405070	

Fixed effects: log_PLASMA_NFL ~ time_years + PTGENDER + APOE4_status

	Value	Std.Error	DF	t-value	p-value
(Intercept)	3.519630	0.03254041	1545	108.16185	0.0000
time_years	0.044734	0.00389566	1545	11.48314	0.0000
PTGENDERMale	0.042958	0.03909331	524	1.09886	0.2723
APOE4_status1 allele	0.015096	0.04266195	524	0.35385	0.7236
APOE4_status2 alleles	0.084709	0.08158250	524	1.03832	0.2996

Correlation:

	(Intr)	tm_yrs	PTGEND	APOE4_1a
time_years	-0.108			
PTGENDERMale	-0.645	0.000		
APOE4_status1 allele	-0.424	0.000	-0.019	
APOE4_status2 alleles	-0.226	0.000	-0.004	0.174

Standardized Within-Group Residuals:

	Min	Q1	Med	Q3	Max
	-4.39896743	-0.47153732	-0.04516553	0.40568566	7.00828129

Number of Observations: 2074
Number of Groups: 528

3.2.0.3 Removing APOE4 status

```
# Longitudinal model for log NfL without APOE4_status
long_model_nfl_reduced1 <- lme(
  log_PLASMA_NFL ~ time_years + PTGENDER,
  random = ~ time_years | RID,
  data = long_data_in
)
```

```
summary(long_model_nfl_reduced1)
```

Linear mixed-effects model fit by REML

Data: long_data_in

	AIC	BIC	logLik
	1058.101	1097.551	-522.0505

Random effects:

Formula: ~time_years | RID

Structure: General positive-definite, Log-Cholesky parametrization

	StdDev	Corr
(Intercept)	0.41972831	(Intr)
time_years	0.02558726	0.399
Residual	0.21404427	

Fixed effects: log_PLASMA_NFL ~ time_years + PTGENDER

	Value	Std.Error	DF	t-value	p-value
(Intercept)	3.529479	0.02900723	1545	121.67584	0.0000
time_years	0.044735	0.00389588	1545	11.48251	0.0000
PTGENDERMale	0.043282	0.03905378	526	1.10826	0.2683

Correlation:

	(Intr)	tm_yrs
time_years	-0.120	
PTGENDERMale	-0.732	0.000

Standardized Within-Group Residuals:

	Min	Q1	Med	Q3	Max
	-4.39873980	-0.47089470	-0.04654747	0.40374733	7.00634940

Number of Observations: 2074
Number of Groups: 528

3.2.0.4 Removing PTGENDER

```
# Longitudinal model for log NFL without covariates
long_model_nfl_reduced2 <- lme(
  log_PLASMA_NFL ~ time_years,
  random = ~ time_years | RID,
  data = long_data_in
)
```

```
summary(long_model_nfl_reduced2)
```

Linear mixed-effects model fit by REML

Data: long_data_in

	AIC	BIC	logLik
	1052.681	1086.498	-520.3404

Random effects:

Formula: ~time_years | RID

Structure: General positive-definite, Log-Cholesky parametrization

	StdDev	Corr
(Intercept)	0.41991560	(Intr)
time_years	0.02556291	0.396
Residual	0.21405026	

Fixed effects: log_PLASMA_NFL ~ time_years

	Value	Std.Error	DF	t-value	p-value
(Intercept)	3.553015	0.019767432	1545	179.74083	0
time_years	0.044733	0.003895573	1545	11.48297	0

Correlation:

	(Intr)
time_years	-0.177

Standardized Within-Group Residuals:

	Min	Q1	Med	Q3	Max
	-4.39641091	-0.47112608	-0.04294366	0.40493297	7.00156348

Number of Observations: 2074
Number of Groups: 528

3.2.0.5 Determining the random effects

- The results of the likelihood-ratio test support including a random slope as well as a random intercept for the NfL model

```
m1 <- lmer(log_PLASMA_NFL ~ time_years + (1 | RID),
           data = long_data_in, REML = FALSE)

m2 <- lmer(log_PLASMA_NFL ~ time_years + (time_years | RID),
           data = long_data_in, REML = FALSE)
```

```
anova(m1, m2, type = "LRT")
```

Data: long_data_in

Models:

m1: log_PLASMA_NFL ~ time_years + (1 | RID)

m2: log_PLASMA_NFL ~ time_years + (time_years | RID)

	npars	AIC	BIC	logLik	-2*log(L)	Chisq	Df	Pr(>Chisq)
m1	4	1043.9	1066.4	-517.94	1035.9			
m2	6	1037.4	1071.2	-512.69	1025.4	10.506	2	0.005231 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Table of results from the likelihood ratio tests between models with only a random intercept

```
long_re_lrt_table <- data.frame(
  "Model" = c("ptau-181", "NfL"),
  "Test statistic" = c(1.488, 10.506),
  "P-value" = c(0.475, 0.005),
  check.names = FALSE
)
```

Mixed object for multivariate joint model

```
mixed <- list(long_model_ptau, long_model_nfl)
```

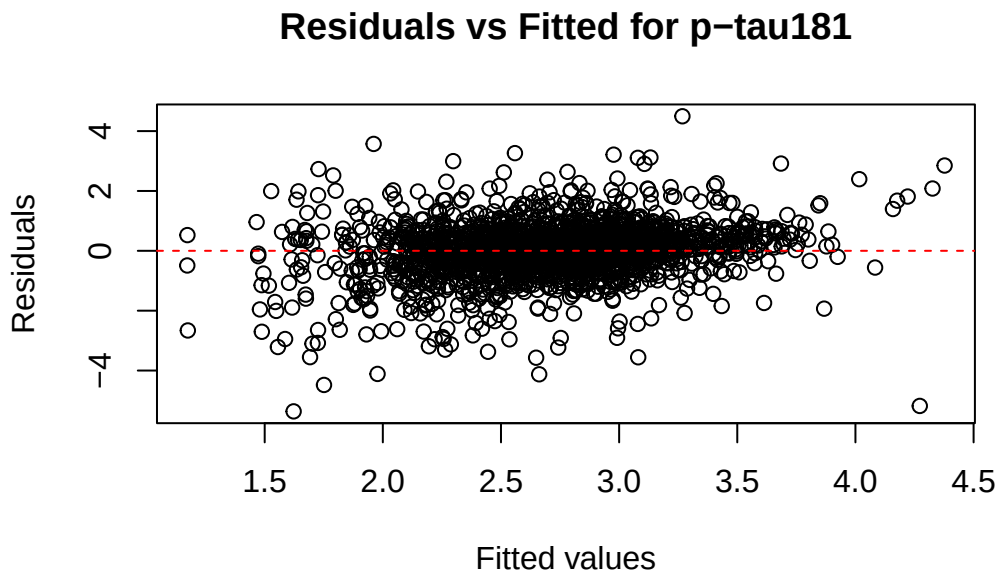
3.3 Residual check

3.3.0.1 Residuals vs fitted values


```
# residuals of the p-tau181 LME model
res_ptau <- residuals(long_model_ptau, type = "pearson")

# fitted values of the p-tau181 LME model
fitted_vals_ptau <- fitted(long_model_ptau)

# plot of residuals vs fitted values
plot(fitted_vals_ptau, res_ptau,
     xlab = "Fitted values",
     ylab = "Residuals",
     main = "Residuals vs Fitted for p-tau181")
abline(h = 0, col = "red", lty = 2)
```



```
# residuals of the NfL LME model
res_nfl <- residuals(long_model_nfl, type = "pearson")

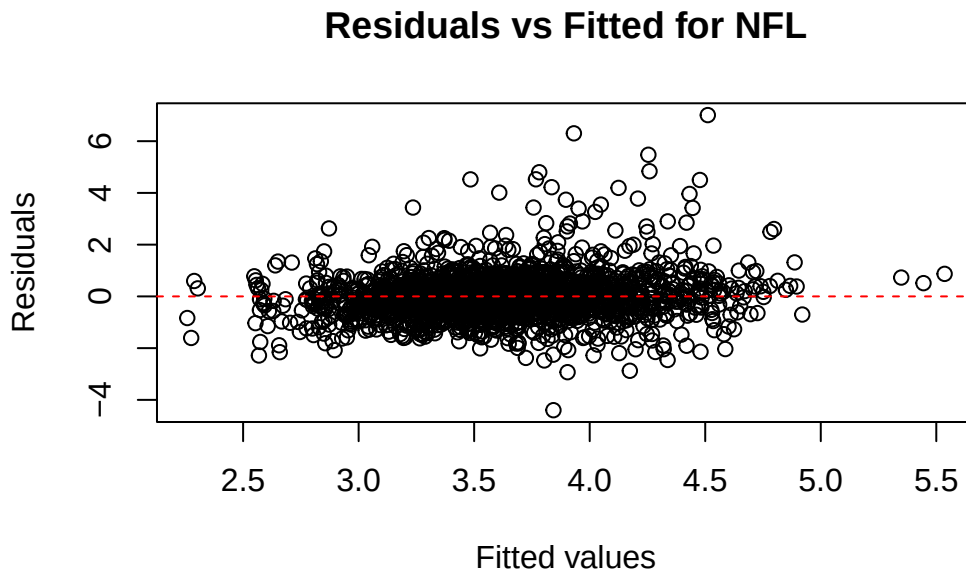
# fitted values of the NfL LME model
fitted_vals_nfl <- fitted(long_model_nfl)

# plot of residuals vs fitted values
plot(fitted_vals_nfl, res_nfl,
     xlab = "Fitted values",
```

```

ylab = "Residuals",
main = "Residuals vs Fitted for NFL")
abline(h = 0, col = "red", lty = 2)

```



3.3.0.2 Shapiro-Wilk test

```

# Shapiro-Wilk test of normality of the residuals
shapiro.test(res_ptau)

```

Shapiro-Wilk normality test

```

data:  res_ptau
W = 0.9387, p-value < 2.2e-16

```

```

# Shapiro-Wilk test of normality of the residuals
shapiro.test(res_nfl)

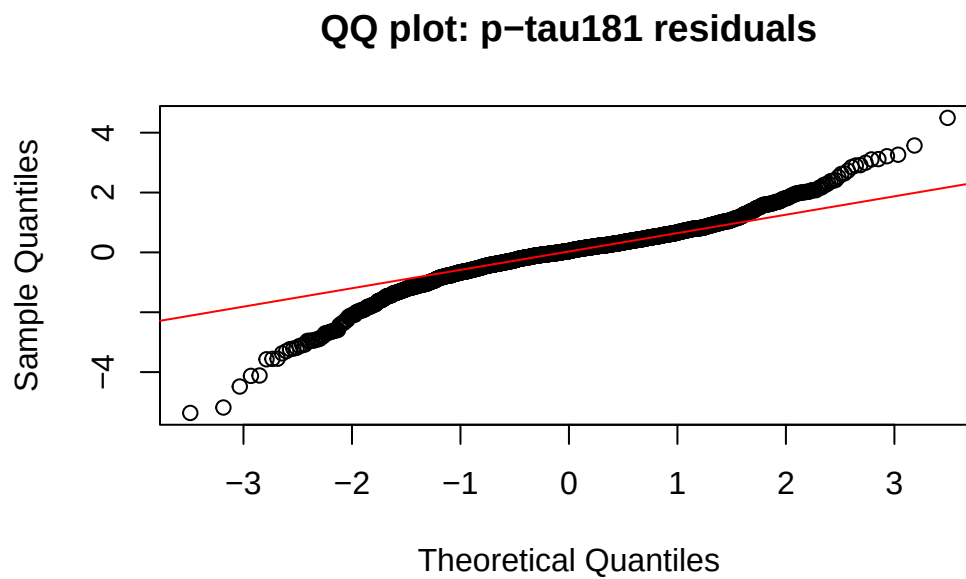
```

Shapiro-Wilk normality test

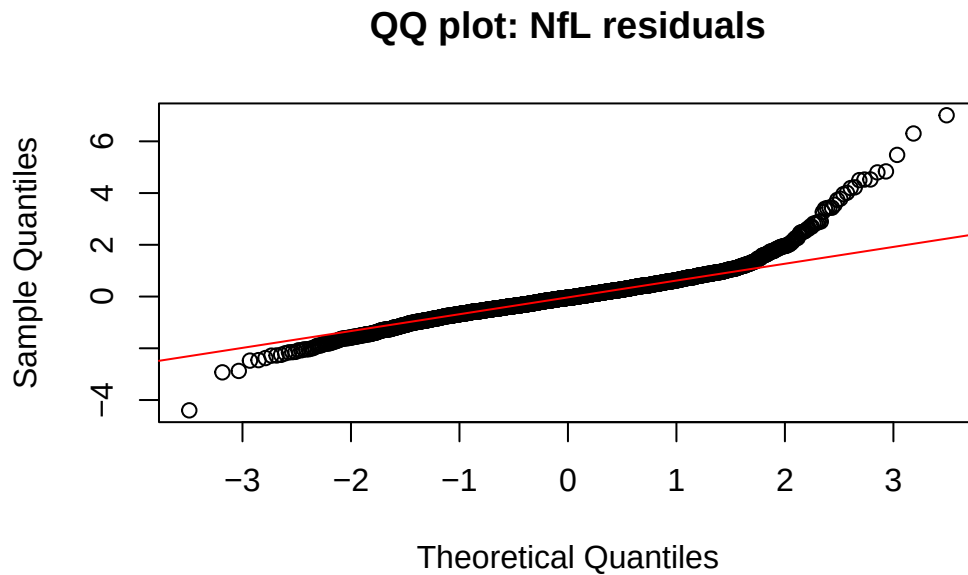
```
data: res_nfl  
W = 0.90607, p-value < 2.2e-16
```

3.3.0.3 QQ plots

```
# For P-tau  
qqnorm(res_ptau, main = "QQ plot: p-tau181 residuals")  
qqline(res_ptau, col = "red")
```



```
# For NFL  
qqnorm(res_nfl, main = "QQ plot: NfL residuals")  
qqline(res_nfl, col = "red")
```



4 Survival models

4.1 For the different cox PH models and the separate joint models

4.2 Fitted standard Cox PH model (for the joint models)

4.2.0.1 With all genotype levels

```
# Cox survival-models, with covariates
# All genotype levels
surv_model_full <- coxph(
  Surv(time_to_event, event) ~ age_at_baseline + PTGENDER + genotype,
  data = surv_data_in,
  x = TRUE
)

summary(surv_model_full)
```

Call:

```
survival::coxph(formula = Surv(time_to_event, event) ~ age_at_baseline +
  PTGENDER + genotype, data = surv_data_in, x = TRUE)
```

n= 528, number of events= 128

	coef	exp(coef)	se(coef)	z	Pr(> z)	
age_at_baseline	0.03972	1.04052	0.01272	3.123	0.001788	**
PTGENDERMale	-0.03419	0.96638	0.18131	-0.189	0.850412	
genotype2/3	-1.68047	0.18629	0.43224	-3.888	0.000101	***
genotype2/4	-0.24061	0.78615	0.51029	-0.472	0.637272	
genotype3/3	-1.61769	0.19836	0.28660	-5.644	1.66e-08	***
genotype3/4	-0.78381	0.45666	0.28329	-2.767	0.005661	**
genotype4/4	NA	NA	0.00000	NA	NA	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
age_at_baseline	1.0405	0.9611	1.01491	1.0668
PTGENDERMale	0.9664	1.0348	0.67736	1.3787
genotype2/3	0.1863	5.3681	0.07985	0.4346
genotype2/4	0.7861	1.2720	0.28917	2.1373
genotype3/3	0.1984	5.0415	0.11311	0.3479
genotype3/4	0.4567	2.1898	0.26209	0.7957
genotype4/4	NA	NA	NA	NA

Concordance= 0.667 (se = 0.028)

Likelihood ratio test= 43.29 on 6 df, p=1e-07

Wald test = 46.46 on 6 df, p=2e-08

Score (logrank) test = 50.83 on 6 df, p=3e-09

4.2.0.2 APOE4 status

```
# Cox survival-models, with covariates
# With APOE4_status
surv_model <- coxph(
  Surv(time_to_event, event) ~ age_at_baseline + PTGENDER + APOE4_status,
  data = surv_data_in,
  x = TRUE
)

summary(surv_model)
```

Call:

```
survival::coxph(formula = Surv(time_to_event, event) ~ age_at_baseline +
  PTGENDER + APOE4_status, data = surv_data_in, x = TRUE)
```

n= 528, number of events= 128

	coef	exp(coef)	se(coef)	z	Pr(> z)
age_at_baseline	0.03893	1.03969	0.01268	3.070	0.00214 **
PTGENDERMale	-0.03738	0.96331	0.18128	-0.206	0.83665
APOE4_status1 allele	0.88116	2.41370	0.19170	4.597	4.29e-06 ***
APOE4_status2 alleles	1.62409	5.07381	0.28155	5.768	8.01e-09 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
age_at_baseline	1.0397	0.9618	1.0142	1.066
PTGENDERMale	0.9633	1.0381	0.6753	1.374
APOE4_status1 allele	2.4137	0.4143	1.6577	3.514
APOE4_status2 alleles	5.0738	0.1971	2.9220	8.810

Concordance= 0.668 (se = 0.027)

Likelihood ratio test= 42.12 on 4 df, p=2e-08

Wald test = 45.02 on 4 df, p=4e-09

Score (logrank) test = 49.14 on 4 df, p=5e-10

4.3 Fitted Cox PH models using baseline values of each biomarker

4.3.1 PTAU

```
# Cox survival-models, with covariates including baseline values of each biomarker
# For p-tau181 with PTGENDER
surv_model_ptau_b1 <- coxph(
  Surv(time_to_event, event) ~
    age_at_baseline + PTGENDER + APOE4_status + baseline_value_ptau,
  data = surv_data_in,
  x = TRUE
)

summary(surv_model_ptau_b1)
```

Call:

```
survival::coxph(formula = Surv(time_to_event, event) ~ age_at_baseline +
  PTGENDER + APOE4_status + baseline_value_ptau, data = surv_data_in,
  x = TRUE)
```

n= 528, number of events= 128

	coef	exp(coef)	se(coef)	z	Pr(> z)
age_at_baseline	0.02839	1.02879	0.01288	2.203	0.0276 *
PTGENDERMale	-0.06761	0.93462	0.18082	-0.374	0.7085
APOE4_status1 allele	0.78033	2.18220	0.19307	4.042	5.31e-05 ***
APOE4_status2 alleles	1.40485	4.07493	0.28425	4.942	7.72e-07 ***
baseline_value_ptau	0.64115	1.89866	0.15349	4.177	2.95e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
age_at_baseline	1.0288	0.9720	1.0031	1.055
PTGENDERMale	0.9346	1.0700	0.6557	1.332
APOE4_status1 allele	2.1822	0.4583	1.4947	3.186
APOE4_status2 alleles	4.0749	0.2454	2.3343	7.113
baseline_value_ptau	1.8987	0.5267	1.4054	2.565

Concordance= 0.7 (se = 0.025)

Likelihood ratio test= 59.41 on 5 df, p=2e-11

Wald test = 60.71 on 5 df, p=9e-12

Score (logrank) test = 64.35 on 5 df, p=2e-12

4.3.2 NfL

```
# For baseline NfL with covariates
surv_model_nfl_bl <- coxph(
  Surv(time_to_event, event) ~
    age_at_baseline + PTGENDER + APOE4_status + baseline_value_nfl,
  data = surv_data_in,
  x = TRUE
)

summary(surv_model_nfl_bl)
```

Call:

```
survival::coxph(formula = Surv(time_to_event, event) ~ age_at_baseline +
  PTGENDER + APOE4_status + baseline_value_nfl, data = surv_data_in,
  x = TRUE)
```

n= 528, number of events= 128

	coef	exp(coef)	se(coef)	z	Pr(> z)
age_at_baseline	0.01457	1.01468	0.01462	0.997	0.318913
PTGENDERMale	-0.04055	0.96026	0.18160	-0.223	0.823323
APOE4_status1 allele	0.82393	2.27945	0.19286	4.272	1.94e-05 ***
APOE4_status2 alleles	1.56046	4.76102	0.28566	5.463	4.69e-08 ***
baseline_value_nfl	0.75765	2.13325	0.22583	3.355	0.000794 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
age_at_baseline	1.0147	0.9855	0.9860	1.044
PTGENDERMale	0.9603	1.0414	0.6727	1.371
APOE4_status1 allele	2.2794	0.4387	1.5619	3.327
APOE4_status2 alleles	4.7610	0.2100	2.7199	8.334
baseline_value_nfl	2.1333	0.4688	1.3703	3.321

Concordance= 0.676 (se = 0.026)

Likelihood ratio test= 53.03 on 5 df, p=3e-10

Wald test = 54.28 on 5 df, p=2e-10

Score (logrank) test = 59.53 on 5 df, p=2e-11

4.3.3 PTAU and NfL

```
# Including both p-tau181 and NfL
surv_model_both_bl <- coxph(
  Surv(time_to_event, event) ~
    age_at_baseline + PTGENDER + APOE4_status + baseline_value_ptau + baseline_value_nfl,
  data = surv_data_in,
  x = TRUE
)

summary(surv_model_both_bl)
```

Call:


```
survival::coxph(formula = Surv(time_to_event, event) ~ age_at_baseline +
  PTGENDER + APOE4_status + baseline_value_ptau + baseline_value_nfl,
  data = surv_data_in, x = TRUE)
```

n= 528, number of events= 128

	coef	exp(coef)	se(coef)	z	Pr(> z)
age_at_baseline	0.01033	1.01039	0.01452	0.711	0.476787
PTGENDERMale	-0.07900	0.92404	0.18142	-0.435	0.663233
APOE4_status1 allele	0.73920	2.09426	0.19460	3.799	0.000146 ***
APOE4_status2 alleles	1.38974	4.01380	0.28683	4.845	1.27e-06 ***
baseline_value_ptau	0.58000	1.78604	0.15925	3.642	0.000270 ***
baseline_value_nfl	0.60749	1.83582	0.23000	2.641	0.008259 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
age_at_baseline	1.010	0.9897	0.9820	1.040
PTGENDERMale	0.924	1.0822	0.6475	1.319
APOE4_status1 allele	2.094	0.4775	1.4302	3.067
APOE4_status2 alleles	4.014	0.2491	2.2877	7.042
baseline_value_ptau	1.786	0.5599	1.3072	2.440
baseline_value_nfl	1.836	0.5447	1.1697	2.881

Concordance= 0.702 (se = 0.025)

Likelihood ratio test= 66.2 on 6 df, p=2e-12

Wald test = 65.14 on 6 df, p=4e-12

Score (logrank) test = 70.45 on 6 df, p=3e-13

4.3.4 Test of the assumption of proportional hazards

```
# Extracting p-values from cox.zph models
extract_ph_p <- function(model, model_name) {
  test <- cox.zph(model)
  df <- as.data.frame(test$table)

  data.frame(
    variable = rownames(df),
    p_value = round(df$p, 3),
    model = model_name
  )
}
```

```

}

# Combining all Cox PH models into one dataset
ph_all <- rbind(
  extract_ph_p(surv_model, "Standard"),
  extract_ph_p(surv_model_ptau_bl, "p-tau181 baseline"),
  extract_ph_p(surv_model_nfl_bl, "NfL baseline"),
  extract_ph_p(surv_model_both_bl, "Both baseline")
)

# Forming a table where each model has its own column
table_ph <- ph_all %>%
  dplyr::select(variable, model, p_value) %>%
  tidyr::pivot_wider(names_from = model, values_from = p_value)

# Displaying final table
table_ph

```

```

# A tibble: 6 x 5
  variable      Standard `p-tau181 baseline` `NfL baseline` `Both baseline`
  <chr>          <dbl>          <dbl>          <dbl>          <dbl>
1 age_at_baseline 0.002          0.002          0.003          0.002
2 PTGENDER        0.803          0.835          0.827          0.888
3 APOE4_status    0.158          0.173          0.127          0.154
4 GLOBAL          0.013          0.028          0.031          0.057
5 baseline_value_pt~ NA              0.698          NA              0.632
6 baseline_value_nfl NA              NA              0.14            0.113

```

4.3.4.1 Removing age-at-baseline due to evidence of non-proportional hazards

```

# Cox survival-models with covariates but excluding age-at-baseline
# Standard cox model
surv_model_final <- coxph(
  Surv(time_to_event, event) ~ PTGENDER + APOE4_status,
  data = surv_data_in,
  x = TRUE
)

# Including p-tau181
surv_model_ptau_bl_final <- coxph(
  Surv(time_to_event, event) ~

```

```

    PTGENDER + APOE4_status + baseline_value_ptau,
  data = surv_data_in,
  x = TRUE
)

# Including NfL
surv_model_nfl_bl_final <- coxph(
  Surv(time_to_event, event) ~
    PTGENDER + APOE4_status + baseline_value_nfl,
  data = surv_data_in,
  x = TRUE
)

# Including both p-tau181 and NfL
surv_model_both_bl_final <- coxph(
  Surv(time_to_event, event) ~
    PTGENDER + APOE4_status + baseline_value_ptau + baseline_value_nfl,
  data = surv_data_in,
  x = TRUE
)

```

4.3.4.2 Final test of the assumption of proportional hazards

```

# Combining all final Cox PH models into one dataset
ph_all_final <- rbind(
  extract_ph_p(surv_model_final, "Standard"),
  extract_ph_p(surv_model_ptau_bl_final, "p-tau181 baseline"),
  extract_ph_p(surv_model_nfl_bl_final, "NfL baseline"),
  extract_ph_p(surv_model_both_bl_final, "Both baseline")
)

# Forming a table where each final model has its own column
table_ph_final <- ph_all_final %>%
  dplyr::select(variable, model, p_value) %>%
  tidyr::pivot_wider(names_from = model, values_from = p_value)

# Displaying final PH table
table_ph_final

```

```

# A tibble: 5 x 5
  variable      Standard `p-tau181 baseline` `NfL baseline` `Both baseline`

```

	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	PTGENDER	0.864	0.875	0.849	0.907
2	APOE4_status	0.178	0.186	0.134	0.158
3	GLOBAL	0.324	0.448	0.163	0.266
4	baseline_value_pt~	NA	0.692	NA	0.623
5	baseline_value_nfl	NA	NA	0.134	0.108

4.3.5 Removing PTGENDER to run likelihood-ratio tests

```
# Cox survival-models, with covariates including baseline values of each biomarker

# For p-tau181 without PTGENDER
surv_model_reduced <- coxph(
  Surv(time_to_event, event) ~ APOE4_status,
  data = surv_data_in,
  x = TRUE
)

anova_surv <- anova(surv_model_final, surv_model_reduced, test = "LRT")
chi_surv <- anova_surv[2,2]

# For p-tau181 without PTGENDER
surv_model_ptau_bl_reduced <- coxph(
  Surv(time_to_event, event) ~ APOE4_status + baseline_value_ptau,
  data = surv_data_in,
  x = TRUE
)

anova_ptau <- anova(surv_model_ptau_bl_final, surv_model_ptau_bl_reduced, test = "LRT")
chi_ptau <- anova_ptau[2,2]

# For NfL without PTGENDER
surv_model_nfl_bl_reduced <- coxph(
  Surv(time_to_event, event) ~ APOE4_status + baseline_value_nfl,
  data = surv_data_in,
  x = TRUE
)

anova_nfl <- anova(surv_model_nfl_bl_final, surv_model_nfl_bl_reduced, test = "LRT")
chi_nfl <- anova_nfl[2,2]
```

```
# Without PTGENDER
surv_model_both_bl_reduced <- coxph(
  Surv(time_to_event, event) ~ APOE4_status + baseline_value_ptau + baseline_value_nfl,
  data = surv_data_in,
  x = TRUE
)

anova_both <- anova(surv_model_both_bl_final, surv_model_both_bl_reduced, test = "LRT")
chi_both <- anova_both[2,2]
```

```
# Viewing the LRT results
anova_ptau
```

Analysis of Deviance Table

```
Cox model: response is Surv(time_to_event, event)
Model 1: ~ PTGENDER + APOE4_status + baseline_value_ptau
Model 2: ~ APOE4_status + baseline_value_ptau
loglik Chisq Df Pr(>|Chi|)
1 -709.74
2 -709.74 7e-04 1 0.9795
```

```
anova_nfl
```

Analysis of Deviance Table

```
Cox model: response is Surv(time_to_event, event)
Model 1: ~ PTGENDER + APOE4_status + baseline_value_nfl
Model 2: ~ APOE4_status + baseline_value_nfl
loglik Chisq Df Pr(>|Chi|)
1 -710.98
2 -710.98 0.0037 1 0.9514
```

```
anova_both
```

Analysis of Deviance Table

```
Cox model: response is Surv(time_to_event, event)
Model 1: ~ PTGENDER + APOE4_status + baseline_value_ptau + baseline_value_nfl
Model 2: ~ APOE4_status + baseline_value_ptau + baseline_value_nfl
loglik Chisq Df Pr(>|Chi|)
1 -704.15
2 -704.21 0.1184 1 0.7308
```

```
# Table of results from the likelihood ratio tests for cox models with and without PTGENDER
surv_lrt_table <- data.frame(
  "Model" = c("ptau-181 baseline", "NfL baseline", "ptau-181 and NfL baseline"),
  "Test statistic" = c(chi_ptau, chi_nfl, chi_both),
  "P-value" = c(0.98, 0.95, 0.73),
  check.names = FALSE
)
```

4.3.6 Cox model comparison table

```
# Building Cox model comparison table
# Extracting model output
extract_model_cox <- function(model) {
  s <- summary(model)

  data.frame(
    Variable = rownames(s$coefficients),
    HR = exp(s$coefficients[, "coef"]),
    CI_low = s$conf.int[, "lower .95"],
    CI_high = s$conf.int[, "upper .95"],
    P = s$coefficients[, "Pr(>|z|)"],
    row.names = NULL
  )
}

# Formatting HR and CI
format_hr_ci <- function(hr, low, high) {
  paste0(round(hr, 3), " (", round(low, 3), "-", round(high, 3), ")")
}

# Formatting P-values
format_p <- function(x) {
  ifelse(is.na(x), NA,
    ifelse(x < 0.001, "<0.001", round(x, 3)))
}

# Extracting models
ptau_cox <- extract_model_cox(surv_model_ptau_bl_final)
nfl_cox <- extract_model_cox(surv_model_nfl_bl_final)
both_cox <- extract_model_cox(surv_model_both_bl_final)
```

```

# Building variable set
all_vars_cox <- unique(c(ptau_cox$Variable, nfl_cox$Variable, both_cox$Variable))

# Value extraction
get_value <- function(df, var, col) {
  out <- rep(NA, length(var))

  for (i in seq_along(var)) {
    idx <- which(df$Variable == var[i])
    if (length(idx) == 1) {
      out[i] <- df[idx, col]
    }
  }
  out
}

# Building final table
table_cox <- data.frame(
  Variable = all_vars_cox,

  "PTau HR (CI)" = format_hr_ci(
    get_value(ptau_cox, all_vars_cox, "HR"),
    get_value(ptau_cox, all_vars_cox, "CI_low"),
    get_value(ptau_cox, all_vars_cox, "CI_high")
  ),
  "PTau P" = format_p(get_value(ptau_cox, all_vars_cox, "P")),

  "NfL HR (CI)" = format_hr_ci(
    get_value(nfl_cox, all_vars_cox, "HR"),
    get_value(nfl_cox, all_vars_cox, "CI_low"),
    get_value(nfl_cox, all_vars_cox, "CI_high")
  ),
  "NfL P" = format_p(get_value(nfl_cox, all_vars_cox, "P")),

  "Both HR (CI)" = format_hr_ci(
    get_value(both_cox, all_vars_cox, "HR"),
    get_value(both_cox, all_vars_cox, "CI_low"),
    get_value(both_cox, all_vars_cox, "CI_high")
  ),
  "Both P" = format_p(get_value(both_cox, all_vars_cox, "P")),

  row.names = NULL,

```

```

    check.names = FALSE
  )

# Showing table
table_cox

```

	Variable	PTau HR (CI)	PTau P	NfL HR (CI)	NfL P
1	PTGENDERMale	0.995 (0.702-1.412)	0.979	0.989 (0.696-1.405)	0.951
2	APOE4_status1 allele	2.039 (1.403-2.963)	<0.001	2.209 (1.521-3.207)	<0.001
3	APOE4_status2 alleles	3.617 (2.095-6.245)	<0.001	4.442 (2.579-7.652)	<0.001
4	baseline_value_ptau	2.043 (1.521-2.744)	<0.001	NA (NA-NA)	<NA>
5	baseline_value_nfl	NA (NA-NA)	<NA>	2.39 (1.635-3.494)	<0.001

	Both HR (CI)	Both P
1	0.94 (0.661-1.337)	0.731
2	2.038 (1.402-2.963)	<0.001
3	3.833 (2.216-6.628)	<0.001
4	1.808 (1.323-2.469)	<0.001
5	1.982 (1.334-2.946)	<0.001

4.3.7 Concordance Index Table

```

# C-index table
c_index_table <- data.frame(
  "Model" = c("Cox ptau-181", "Cox NfL", "Cox ptau-181 & NfL"),
  "C-Index" = c(0.679, 0.673, 0.681),
  check.names = FALSE)

```

5 Two separate joint models

5.1 Fitted joint model for p-tau181

```

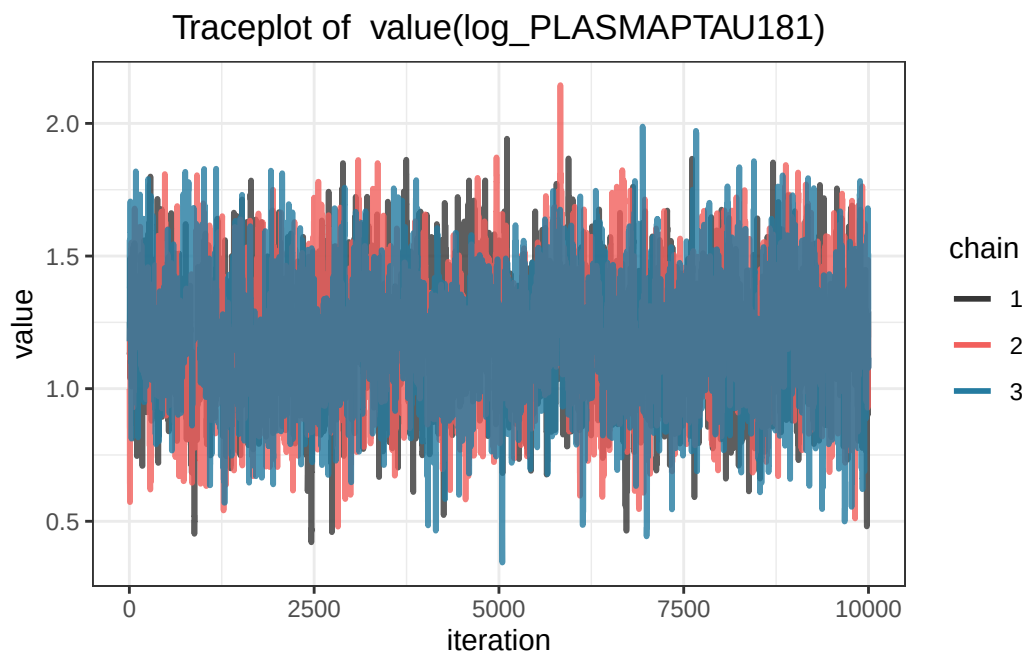
# Joint model for p-tau181
joint_model_ptau <- jm(
  surv_model_final,
  long_model_ptau,
  time_var = "time_years",
  n_chains = 3L,

```



```
n_iter = 11000L,
n_burnin = 1000L
)
```

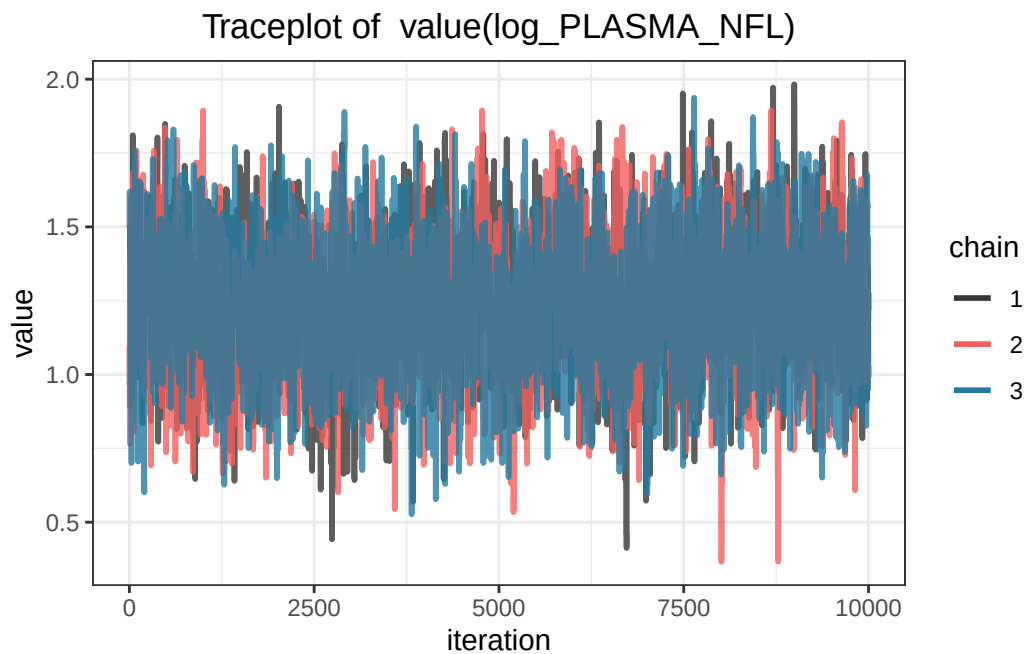
```
# Checking convergence
ggtraceplot(joint_model_ptau, "alphas")
```



5.2 Fitted joint model for NfL

```
# Joint model for NfL
joint_model_nfl <- jm(
  surv_model_final,
  long_model_nfl,
  time_var = "time_years",
  n_chains = 3L,
  n_iter = 11000L,
  n_burnin = 1000L
)
```

```
# Checking convergence
ggtraceplot(joint_model_nfl, "alphas")
```



6 Multivariate joint model

6.1 To check if the models have stable random slopes

```
VarCorr(long_model_ptau)
```

```
RID = pdLogChol(time_years)
      Variance StdDev  Corr
(Intercept) 0.199928519 0.44713367 (Intr)
time_years  0.001342154 0.03663542 -0.053
Residual    0.148038053 0.38475714
```

```
VarCorr(long_model_nfl)
```

```
RID = pdLogChol(time_years)
      Variance      StdDev      Corr
(Intercept) 0.1765807260 0.42021509 (Intr)
time_years  0.0006537102 0.02556776 0.395
Residual    0.0458177004 0.21405070
```

For both models, the slope variance is very low (lme: 0.0011, mixed_model: 0.036), which indicates that individual biomarker trajectories over time show minimal variation → We skip the random slope in fForms:

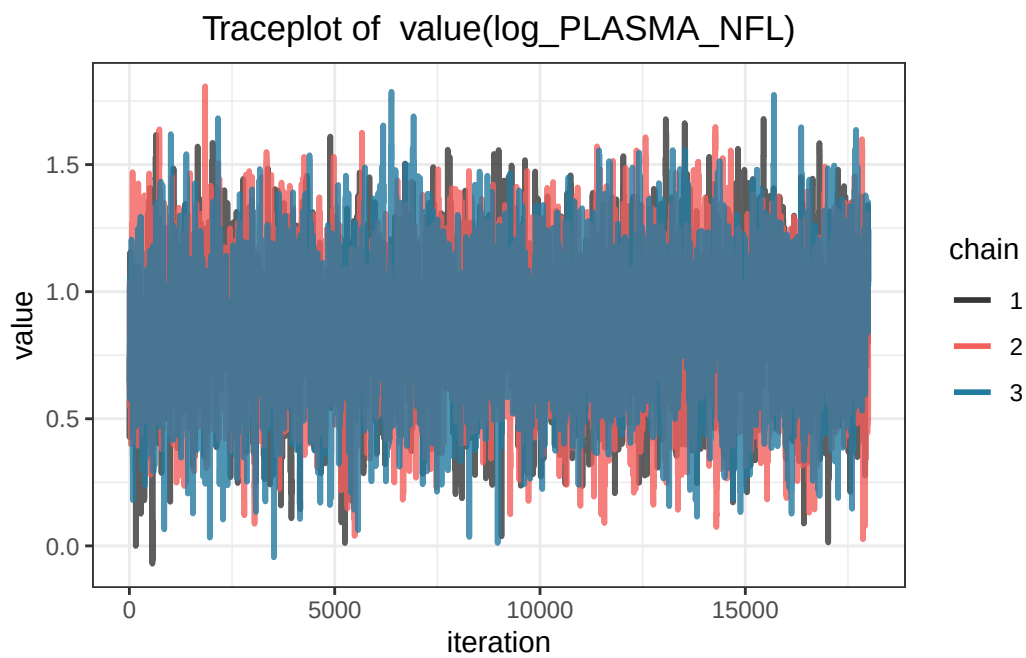
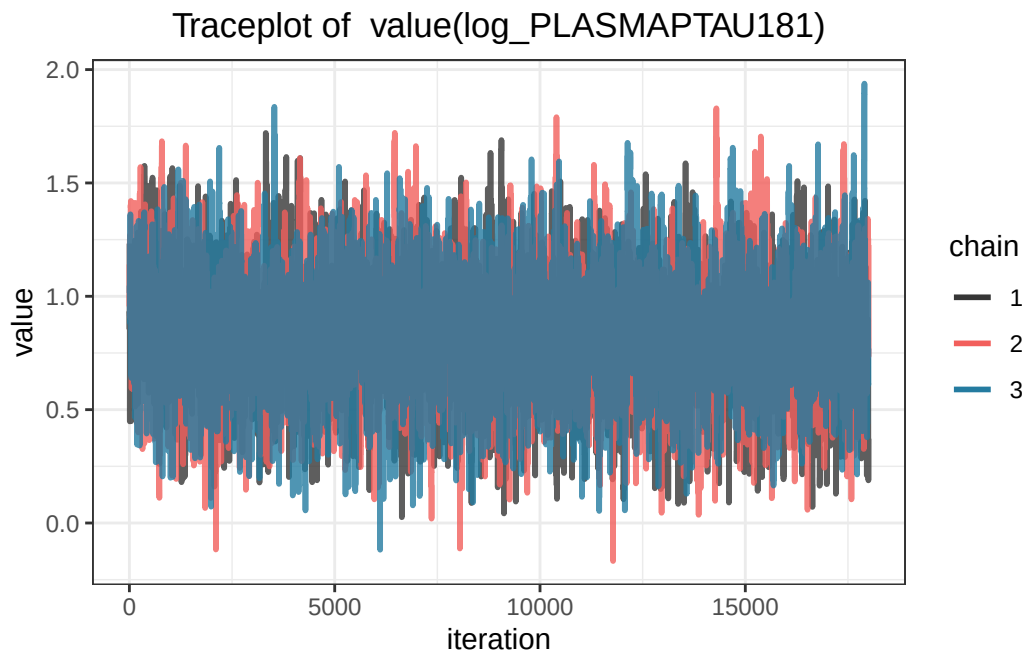
6.2 fForms

```
fForms <- list(
  "log_PLASMAPTAU181" = ~ value(log_PLASMAPTAU181),
  "log_PLASMA_NFL" = ~ value(log_PLASMA_NFL)
)
```

6.3 Fitted multivariate joint model

```
multi_joint_model <- jm(surv_model_final, mixed ,
  time_var = "time_years",
  functional_forms = fForms,
  n_chains = 3L,
  n_iter = 20000L,
  n_burnin = 2000L
)
```

```
# Checking convergence
ggtraceplot(multi_joint_model, "alphas")
```



6.3.1 Joint model comparison table

```
# Building joint model comparison table
# Extracting models (JMbates2 joint models use log-HR scale with HR = exp(Mean)), where Mean
extract_model_jm <- function(model) {

  s <- summary(model)$Survival

  data.frame(
    Variable = rownames(s),

    HR = exp(s[, "Mean"]),
    CI_low = exp(s[, "2.5%"]),
    CI_high = exp(s[, "97.5%"]),

    P = s[, "P"],
    row.names = NULL
  )
}

# Extracting models
ptau_jm <- extract_model_jm(joint_model_ptau)
nfl_jm <- extract_model_jm(joint_model_nfl)
both_jm <- extract_model_jm(multi_joint_model)

# Building variable set
all_vars_jm <- unique(c(ptau_jm$Variable, nfl_jm$Variable, both_jm$Variable))

# Building final table
table_jm <- data.frame(
  Variable = all_vars_jm,

  "PTau HR (CI)" = format_hr_ci(
    get_value(ptau_jm, all_vars_jm, "HR"),
    get_value(ptau_jm, all_vars_jm, "CI_low"),
    get_value(ptau_jm, all_vars_jm, "CI_high")
  ),
  "PTau P" = format_p(get_value(ptau_jm, all_vars_jm, "P")),

  "NfL HR (CI)" = format_hr_ci(
    get_value(nfl_jm, all_vars_jm, "HR"),
    get_value(nfl_jm, all_vars_jm, "CI_low"),
```

```

    get_value(nfl_jm, all_vars_jm, "CI_high")
  ),
  "NfL P" = format_p(get_value(nfl_jm, all_vars_jm, "P")),

  "Both HR (CI)" = format_hr_ci(
    get_value(both_jm, all_vars_jm, "HR"),
    get_value(both_jm, all_vars_jm, "CI_low"),
    get_value(both_jm, all_vars_jm, "CI_high")
  ),
  "Both P" = format_p(get_value(both_jm, all_vars_jm, "P")),

  row.names = NULL,
  check.names = FALSE
)

# Showing table
table_jm

```

	Variable	PTau HR (CI)	PTau P	NfL HR (CI)
1	PTGENDERMale	0.929 (0.575-1.515)	0.771 0.967	(0.605-1.586)
2	APOE4_status1 allele	1.901 (1.284-2.838)	0.002 2.313	(1.547-3.449)
3	APOE4_status2 alleles	3.174 (1.771-5.434)	<0.001 4.578	(2.555-7.789)
4	value(log_PLASMAPTAU181)	3.332 (2.172-5.116)	<0.001	NA (NA-NA)
5	value(log_PLASMA_NFL)	NA (NA-NA)	<NA> 3.454	(2.308-5.194)
	NfL P	Both HR (CI)	Both P	
1	0.881 0.89	(0.547-1.443)	0.65	
2	<0.001 2.023	(1.344-3.037)	<0.001	
3	<0.001 3.69	(2.018-6.508)	<0.001	
4	<NA> 2.296	(1.41-3.785)	<0.001	
5	<0.001 2.363	(1.457-3.812)	<0.001	

6.4 Rhat value comparison table

```

# Extracting Rhat for each model (survival component)
extract_rhat_surv <- function(model) {

  s <- summary(model)$Survival

  data.frame(
    Variable = rownames(s),

```

```

    Rhat = s[, "Rhat"],
    row.names = NULL
  )
}

# Applying to each model
ptau_surv <- extract_rhat_surv(joint_model_ptau)
nfl_surv <- extract_rhat_surv(joint_model_nfl)
multi_surv <- extract_rhat_surv(multi_joint_model)

# Combining with model labels to form a table
rhat_survival_table <- bind_rows(
  ptau_surv %>% mutate(Model = "p-tau181"),
  nfl_surv %>% mutate(Model = "NfL"),
  multi_surv %>% mutate(Model = "Multivariate")
) %>%
  pivot_wider(
    names_from = Model,
    values_from = Rhat
  ) %>%
  mutate(across(where(is.numeric), ~round(.x, 3)))

# Showing table
rhat_survival_table

```

```

# A tibble: 5 x 4
  Variable                `p-tau181`   NfL Multivariate
  <chr>                   <dbl> <dbl>         <dbl>
1 PTGENDERMale           1.00  1.00           1
2 APOE4_status1 allele   1.00  1.01          1.00
3 APOE4_status2 alleles  1.00  1.00          1.00
4 value(log_PLASMAPTAU181) 1.00 NA           1.01
5 value(log_PLASMA_NFL)   NA    1.01          1.00

```

```

# Extracting Rhat for longitudinal outcome
extract_rhat_long <- function(model) {

  s <- summary(model)$Outcome1

  data.frame(
    Variable = rownames(s),

```

```

    Rhat = s[, "Rhat"],
    row.names = NULL
  )
}

# Applying to each model
ptau_long <- extract_rhat_long(joint_model_ptau)
nfl_long <- extract_rhat_long(joint_model_nfl)
multi_long <- extract_rhat_long(multi_joint_model)

# Combining with model labels to form a table
rhat_longitudinal_table <- bind_rows(
  ptau_long %>% mutate(Model = "p-tau181"),
  nfl_long %>% mutate(Model = "NfL"),
  multi_long %>% mutate(Model = "Multivariate")
) %>%
  pivot_wider(
    names_from = Model,
    values_from = Rhat
  ) %>%
  mutate(across(where(is.numeric), ~round(.x, 3)))

# Showing table
rhat_longitudinal_table

```

```

# A tibble: 6 x 4
  Variable      `p-tau181`   NfL Multivariate
  <chr>          <dbl> <dbl>         <dbl>
1 (Intercept)      1    1.00         1.00
2 time_years      1.02  1.02         1.09
3 PTGENDERMale     1     1          1.00
4 APOE4_status1 allele 1.00  1.00         1
5 APOE4_status2 alleles 1     1          1
6 sigma           1.01  1.02         1.02

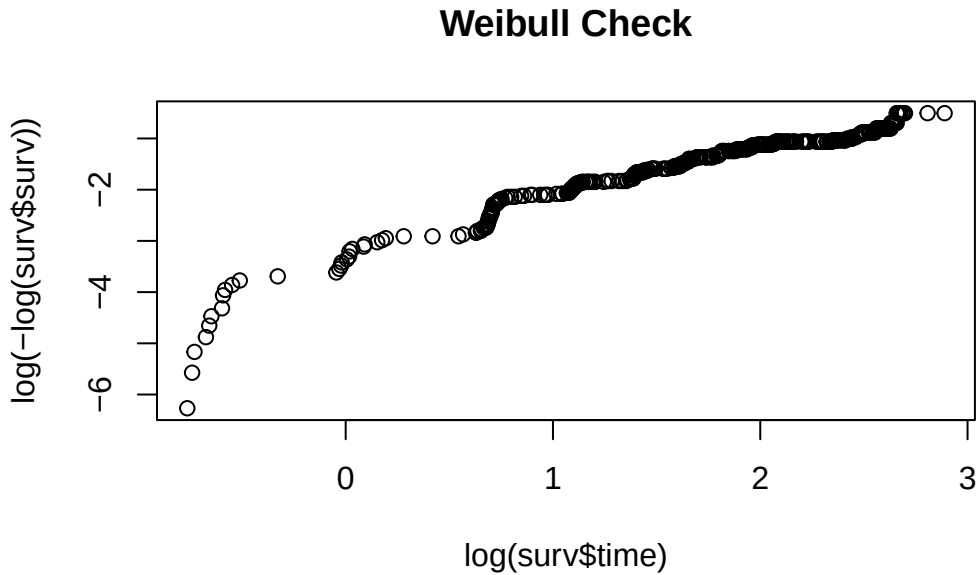
```

7 Latent Class Mixed Models

Choice of p-value <https://pmc.ncbi.nlm.nih.gov/articles/PMC5059270/>

8 JLCMM: NFL

```
surv <- survfit(Surv(surv_data_in$time_to_event, surv_data_in$event) ~ 1)
plot(log(surv$time), log(-log(surv$surv)), main="Weibull Check")
```



```
# merging the longitudinal and survival data for jlcmm
jlcmm_data <- surv_data_in %>%
  dplyr::select(RID, event, time_to_event) %>%
  merge(long_data_in, by = "RID")
```

8.1 M0 (1 latent class)

```
# Fitting a one class joint latent class mixed model with a random intercept
jlcmm_m0_1 <- jlcmm(
  fixed = log_PLASMA_NFL ~ time_years,
  random = ~ 1,
  subject = "RID",
  survival = Surv(time_to_event, event) ~ 1,
```

```
data = jlcmm_data
)
```

Warning in Surv(time = c(7.11506849315068, 7.11506849315068, 7.11506849315068,
: type= 'mstate' is deprecated, use a factor variable as status

```
# Fitting a one class joint latent class mixed model with flexible hazard
jlcmm_m0_1_piecewise <- jlcmm(
  fixed = log_PLASMA_NFL ~ time_years,
  random = ~ 1,
  subject = "RID",
  survival = Surv(time_to_event, event) ~ 1,
  hazard = "3-quant-piecewise",
  data = jlcmm_data
)
```

Warning in Surv(time = c(7.11506849315068, 7.11506849315068, 7.11506849315068,
: type= 'mstate' is deprecated, use a factor variable as status

```
summary(jlcmm_m0_1)
```

Joint latent class model for quantitative outcome and competing risks
fitted by maximum likelihood method

```
jlcmm(fixed = log_PLASMA_NFL ~ time_years, random = ~1, subject = "RID",
      survival = Surv(time_to_event, event) ~ 1, data = jlcmm_data)
```

Statistical Model:

```
Dataset: jlcmm_data
Number of subjects: 528
Number of observations: 2074
Number of latent classes: 1
Number of parameters: 6
Event 1:
  Number of events: 128
  Weibull baseline risk function
```

Iteration process:

```
Convergence criteria satisfied
Number of iterations: 16
```

```
Convergence criteria: parameters= 1.3e-12
                      : likelihood= 3e-09
                      : second derivatives= 4.9e-16
```

Goodness-of-fit statistics:

```
maximum log-likelihood: -1060.31
AIC: 2132.63
BIC: 2158.24
Score test statistic for CI assumption: 33.409 (p-value=0)
```

Maximum Likelihood Estimates:

Parameters in the proportional hazard model:

	coef	Se	Wald	p-value
event1 +/-sqrt(Weibull1)	0.19672	0.01272	15.465	0.00000
event1 +/-sqrt(Weibull2)	0.99403	0.03788	26.241	0.00000

Fixed effects in the longitudinal model:

	coef	Se	Wald	p-value
intercept	3.55322	0.02046	173.668	0.00000
time_years	0.04454	0.00372	11.977	0.00000

Variance-covariance matrix of the random-effects:

```
intercept
intercept 0.19035
```

	coef	Se
Residual standard error	0.21728	0.00391

```
summary(jlcmm_m0_1_piecewise)
```

Joint latent class model for quantitative outcome and competing risks
fitted by maximum likelihood method

```
jlcmm(fixed = log_PLASMA_NFL ~ time_years, random = ~1, subject = "RID",
      survival = Surv(time_to_event, event) ~ 1, hazard = "3-quant-piecewise",
      data = jlcmm_data)
```

Statistical Model:

Dataset: jlcmm_data
 Number of subjects: 528
 Number of observations: 2074
 Number of latent classes: 1
 Number of parameters: 6
 Event 1:
 Number of events: 128
 Piecewise constant baseline risk function with nodes
 0 5.020548 17.981

Iteration process:
 Convergence criteria satisfied
 Number of iterations: 14
 Convergence criteria: parameters= 1.8e-09
 : likelihood= 4.9e-05
 : second derivatives= 2.5e-11

Goodness-of-fit statistics:
 maximum log-likelihood: -1059.04
 AIC: 2130.08
 BIC: 2155.69
 Score test statistic for CI assumption: 32.364 (p-value=0)

Maximum Likelihood Estimates:

Parameters in the proportional hazard model:

	coef	Se	Wald	p-value
event1 +/-sqrt(piecewise1)	0.20818	0.01091	19.078	0.00000
event1 +/-sqrt(piecewise2)	0.17859	0.01468	12.165	0.00000

Fixed effects in the longitudinal model:

	coef	Se	Wald	p-value
intercept	3.55322	0.02046	173.668	0.00000
time_years	0.04454	0.00372	11.977	0.00000

Variance-covariance matrix of the random-effects:

	intercept
intercept	0.19035

	coef	Se
--	------	----

Residual standard error 0.21728 0.00391

```
# Fitting a one class joint latent class mixed model with a random intercept and slope
jlcmm_m0_2 <- jlcmm(
  fixed = log_PLASMA_NFL ~ time_years,
  random = ~ 1 + time_years,
  subject = "RID",
  survival = Surv(time_to_event, event) ~ 1,
  data = jlcmm_data
)
```

Warning in Surv(time = c(7.11506849315068, 7.11506849315068, 7.11506849315068,
: type= 'mstate' is deprecated, use a factor variable as status

```
summary(jlcmm_m0_2)
```

Joint latent class model for quantitative outcome and competing risks
fitted by maximum likelihood method

```
jlcmm(fixed = log_PLASMA_NFL ~ time_years, random = ~1 + time_years,
      subject = "RID", survival = Surv(time_to_event, event) ~
        1, data = jlcmm_data)
```

Statistical Model:

Dataset: jlcmm_data
Number of subjects: 528
Number of observations: 2074
Number of latent classes: 1
Number of parameters: 8
Event 1:
 Number of events: 128
 Weibull baseline risk function

Iteration process:

Convergence criteria satisfied
Number of iterations: 15
Convergence criteria: parameters= 1.1e-09
 : likelihood= 3.3e-06
 : second derivatives= 2.3e-12

Goodness-of-fit statistics:

```

maximum log-likelihood: -1055.06
AIC: 2126.12
BIC: 2160.28
Score test statistic for CI assumption: 36.199 (p-value=0)

```

Maximum Likelihood Estimates:

Parameters in the proportional hazard model:

	coef	Se	Wald	p-value
event1 +/-sqrt(Weibull1)	0.19672	0.01272	15.465	0.00000
event1 +/-sqrt(Weibull2)	0.99403	0.03788	26.240	0.00000

Fixed effects in the longitudinal model:

	coef	Se	Wald	p-value
intercept	3.55302	0.01975	179.909	0.00000
time_years	0.04473	0.00389	11.488	0.00000

Variance-covariance matrix of the random-effects:

	intercept	time_years
intercept	0.17593	
time_years	0.00427	0.00063

	coef	Se
Residual standard error	0.21407	0.00463

8.1.1 Determining the random effects

```

# Comparing BIC and log-likelihood for the two one class models with different random effects
summarytable(jlcmm_m0_1, jlcmm_m0_2)

```

	G	loglik	npm	BIC	%class1
jlcmm_m0_1	1	-1060.314	6	2158.243	100
jlcmm_m0_2	1	-1055.061	8	2160.275	100

```

# log likelihood for the full model
l = jlcmm_m0_2$loglik

```

```
# log likelihood for the reduced model
l_0 = jlcmm_m0_1$loglik

# computing the LR test statistic
LR = - 2 * (l_0 - l)

# computing the p-value given the null hypothesis is true (the reduced model is the true model)
pchisq(LR, 2, lower.tail = FALSE)
```

```
[1] 0.005231058
```

```
LR
```

```
[1] 10.50628
```

8.2 M1 (2 latent classes)

```
# Running gridsearch() for a two class model
# jlcmm_m1_nfl_search <- gridsearch(m = jlcmm(fixed = log_PLASMA_NFL ~ time_years,
#      random = ~ 1,
#      mixture = ~ 1,
#      classmb = ~ 1,
#      ng = 2,
#      subject = "RID",
#      survival = Surv(time_to_event, event) ~ 1,
#      data = jlcmm_data),
# minit = jlcmm_m0_1,
# rep = 50,
# maxiter = 50)

# Estimating two class model using the best result from gridsearch() as a starting value
# jlcmm_m1_nfl <- jlcmm(
#   fixed = log_PLASMA_NFL ~ time_years,
#   random = ~ 1,
#   mixture = ~ 1,
#   classmb = ~ 1,
#   ng = 2,
#   subject = "RID",
#   survival = Surv(time_to_event, event) ~ 1,
```

```
# data = jlcmm_data,
# B = jlcmm_m1_nfl_search$best
# )
```

```
# Fitting a two class joint latent class mixed model
# Results from gridsearch() and B = random(jlcmm_m0_1) are very similar, global maximum found
jlcmm_m1_nfl <- jlcmm(
  fixed = log_PLASMA_NFL ~ time_years,
  random = ~ 1,
  mixture = ~ 1,
  classmb = ~ 1,
  ng = 2,
  subject = "RID",
  survival = Surv(time_to_event, event) ~ 1,
  data = jlcmm_data,
  B = random(jlcmm_m0_1)
)
```

Warning in Surv(time = c(7.11506849315068, 7.11506849315068, 7.11506849315068,
: type= 'mstate' is deprecated, use a factor variable as status

```
summary(jlcmm_m1_nfl)
```

Joint latent class model for quantitative outcome and competing risks
fitted by maximum likelihood method

```
jlcmm(fixed = log_PLASMA_NFL ~ time_years, mixture = ~1, random = ~1,
  subject = "RID", classmb = ~1, ng = 2, survival = Surv(time_to_event,
    event) ~ 1, data = jlcmm_data)
```

Statistical Model:

```
Dataset: jlcmm_data
Number of subjects: 528
Number of observations: 2074
Number of latent classes: 2
Number of parameters: 10
Event 1:
  Number of events: 128
  Class-specific hazards and
  Weibull baseline risk function
```


Iteration process:

Convergence criteria satisfied
Number of iterations: 18
Convergence criteria: parameters= 1.5e-05
: likelihood= 4.3e-05
: second derivatives= 1.2e-10

Goodness-of-fit statistics:

maximum log-likelihood: -1032.38
AIC: 2084.77
BIC: 2127.46
Score test statistic for CI assumption: 0.368 (p-value=0.5444)

Maximum Likelihood Estimates:

Fixed effects in the class-membership model:
(the class of reference is the last class)

	coef	Se	Wald	p-value
intercept class1	-0.80937	0.14485	-5.588	0.00000

Parameters in the proportional hazard model:

	coef	Se	Wald	p-value
event1 +/-sqrt(Weibull1) class 1	0.47076	0.02606	18.062	0.00000
event1 +/-sqrt(Weibull2) class 1	1.23898	0.05811	21.319	0.00000
event1 +/-sqrt(Weibull1) class 2	0.23156	0.01211	19.117	0.00000
event1 +/-sqrt(Weibull2) class 2	2.68119	0.47553	5.638	0.00000

Fixed effects in the longitudinal model:

	coef	Se	Wald	p-value
intercept class1	3.76627	0.03897	96.638	0.00000
intercept class2	3.45835	0.02643	130.853	0.00000
time_years	0.04451	0.00372	11.971	0.00000

Variance-covariance matrix of the random-effects:

intercept
intercept 0.17008

	coef	Se
Residual standard error	0.21730	0.00391

```
# Checking flexible hazard
jlcmm_m1_nfl_piecewise <- jlcmm(
  fixed = log_PLASMA_NFL ~ time_years,
  random = ~ 1,
  mixture = ~ 1,
  classmb = ~ 1,
  ng = 2,
  subject = "RID",
  survival = Surv(time_to_event, event) ~ 1,
  data = jlcmm_data,
  B = random(jlcmm_m0_1_piecewise),
)
```

Warning in Surv(time = c(7.11506849315068, 7.11506849315068, 7.11506849315068,
: type= 'mstate' is deprecated, use a factor variable as status

```
# Does not converge
summary(jlcmm_m1_nfl_piecewise)
```

Joint latent class model for quantitative outcome and competing risks
fitted by maximum likelihood method

```
jlcmm(fixed = log_PLASMA_NFL ~ time_years, mixture = ~1, random = ~1,
  subject = "RID", classmb = ~1, ng = 2, survival = Surv(time_to_event,
    event) ~ 1, data = jlcmm_data)
```

Statistical Model:

```
Dataset: jlcmm_data
Number of subjects: 528
Number of observations: 2074
Number of latent classes: 2
Number of parameters: 10
Event 1:
  Number of events: 128
  Class-specific hazards and
  Weibull baseline risk function
```

Iteration process:

```
Maximum number of iteration reached without convergence
Number of iterations: 100
Convergence criteria: parameters= 2.6e-13
```

```
: likelihood= 2.6e-05
: second derivatives= 1
```

Goodness-of-fit statistics:

```
maximum log-likelihood: -1044.21
AIC: 2108.43
BIC: 2151.12
```

Maximum Likelihood Estimates:

Fixed effects in the class-membership model:
(the class of reference is the last class)

```
              coef Se Wald p-value
intercept class1 0.29890
```

Parameters in the proportional hazard model:

```
              coef Se Wald p-value
event1 +/-sqrt(Weibull1) class 1 0.28275
event1 +/-sqrt(Weibull2) class 1 0.99221
event1 +/-sqrt(Weibull1) class 2 0.00000
event1 +/-sqrt(Weibull2) class 2 0.59359
```

Fixed effects in the longitudinal model:

```
              coef Se Wald p-value
intercept class1 3.69363
intercept class2 3.36430
time_years      0.04457
```

Variance-covariance matrix of the random-effects:

```
      intercept
intercept 0.16274
```

```
              coef Se
Residual standard error 0.21728
```

```
# Defining a function to extract the posterior probability metrics as presenting in postprob
get_pprob_table <- function(model) {
```

```

G <- model$ng
pprobs <- model$pprob

# Mean posterior probabilities per class
mean_pprob <- do.call(rbind, lapply(1:G, function(g) {
  # Individuals assigned to class g
  members <- pprobs[pprobs$class == g, 3:(2 + G)]
  colMeans(members)
}))

rownames(mean_pprob) <- paste0("Class", 1:G)
colnames(mean_pprob) <- paste0("Prob", 1:G)

# Percentage of individuals above posterior probability thresholds per class
thresholds <- c(0.7, 0.8, 0.9)

threshold_table <- do.call(rbind, lapply(1:G, function(g) {
#   Extracting individuals whose highest posterior probability is for class g
  pprob_g <- pprobs[pprobs$class == g, 2 + g]
  sapply(thresholds, function(t)
    round(mean(pprob_g > t) * 100, 2))
}))

rownames(threshold_table) <- paste0("Class", 1:G)
colnames(threshold_table) <- paste0("Prob > ", thresholds, " (%)")

# Combine into one table
out <- cbind(
  as.data.frame(round(mean_pprob, 4)),
  as.data.frame(threshold_table)
)

return(out)
}

```

```

# defining a function to extract the model parameters
get_jlcmm_params <- function(model) {

  p_full <- length(model$best)
#   model$V is a vector of the upper triangle of a matrix, not the var-covar matrix itself
  V_mat <- matrix(0, nrow = p_full, ncol = p_full)
  V_mat[upper.tri(V_mat, diag = TRUE)] <- model$V

```

```

# Fill the matrix by adding its transpose, then removing the doubled values on the diagonal
V_mat <- V_mat + t(V_mat) - diag(diag(V_mat))

# Dropping unwanted values stored in model$best
drop <- "varcov|stderr"
# Creates a logical vector in order to drop values from estimates and var-covar matrix
keep <- !grepl(drop, names(model$best))
estimates <- model$best[keep]
V_mat <- V_mat[keep, keep]
parameters <- names(estimates)

se <- sqrt(diag(V_mat))
z <- estimates / se
p_values <- 2 * (1 - pnorm(abs(z)))

submodel <- c("Class Membership",
              "Survival", "Survival", "Survival", "Survival",
              "Longitudinal", "Longitudinal", "Longitudinal")

out <- data.frame(
  "Submodel" = submodel,
  "Parameter" = parameters,
  "Estimate" = round(estimates, 3),
  "SE" = round(se, 3),
  "Z" = round(z, 3),
  "P value" = round(p_values, 3),
  check.names = FALSE
)

return(out)
}

```

8.3 M2 (3 latent classes)

```

# Running gridsearch() for a three class model
# jlcmm_m2_nfl_search <- gridsearch(m = jlcmm(fixed = log_PLASMA_NFL ~ time_years,
#                                           random = ~ 1,
#                                           mixture = ~ 1,
#                                           classmb = ~ 1,
#                                           ng = 3,

```

```
#
#                                     subject = "RID",
#                                     survival = Surv(time_to_event, event) ~ 1,
#                                     data = jlcmm_data),
#                                     minit = jlcmm_m0_1,
#                                     rep = 50,
#                                     maxiter = 50)
```

```
# Estimating two class model using the best result from gridsearch() as a starting value
# jlcmm_m2_nfl <- jlcmm(
#   fixed = log_PLASMA_NFL ~ time_years,
#   random = ~ 1,
#   mixture = ~ 1,
#   classmb = ~ 1,
#   ng = 3,
#   subject = "RID",
#   survival = Surv(time_to_event, event) ~ 1,
#   data = jlcmm_data,
#   B = jlcmm_m2_nfl_search$best
# )
```

```
# Fitting a three class joint latent class mixed model
# Results from gridsearch() and B = random(jlcmm_m0_1) are very similar, global maximum found
jlcmm_m2_nfl <- jlcmm(
  fixed = log_PLASMA_NFL ~ time_years,
  random = ~ 1,
  mixture = ~ 1,
  classmb = ~ 1,
  ng = 3,
  subject = "RID",
  survival = Surv(time_to_event, event) ~ 1,
  data = jlcmm_data,
  B = random(jlcmm_m0_1)
)
```

Warning in Surv(time = c(7.11506849315068, 7.11506849315068, 7.11506849315068,
: type= 'mstate' is deprecated, use a factor variable as status

```
summary(jlcmm_m2_nfl)
```

Joint latent class model for quantitative outcome and competing risks
fitted by maximum likelihood method

```
jlcmm(fixed = log_PLASMA_NFL ~ time_years, mixture = ~1, random = ~1,
      subject = "RID", classmb = ~1, ng = 3, survival = Surv(time_to_event,
        event) ~ 1, data = jlcmm_data)
```

Statistical Model:

```
Dataset: jlcmm_data
Number of subjects: 528
Number of observations: 2074
Number of latent classes: 3
Number of parameters: 14
Event 1:
  Number of events: 128
  Class-specific hazards and
  Weibull baseline risk function
```

Iteration process:

```
Convergence criteria satisfied
Number of iterations: 32
Convergence criteria: parameters= 8.1e-07
                     : likelihood= 2.6e-06
                     : second derivatives= 1.7e-07
```

Goodness-of-fit statistics:

```
maximum log-likelihood: -1031.04
AIC: 2090.08
BIC: 2149.85
Score test statistic for CI assumption: 0.314 (p-value=0.575)
```

Maximum Likelihood Estimates:

Fixed effects in the class-membership model:
(the class of reference is the last class)

	coef	Se	Wald	p-value
intercept class1	-1.33463	0.63662	-2.096	0.03604
intercept class2	-0.56564	0.18627	-3.037	0.00239

Parameters in the proportional hazard model:

	coef	Se	Wald	p-value
event1 +/-sqrt(Weibull1) class 1	0.26871	0.00837	32.093	0.00000
event1 +/-sqrt(Weibull2) class 1	3.53167	0.57422	6.150	0.00000

event1 +/-sqrt(Weibull1)	class 2	0.46946	0.02379	19.732	0.00000
event1 +/-sqrt(Weibull2)	class 2	1.24035	0.05640	21.993	0.00000
event1 +/-sqrt(Weibull1)	class 3	-0.00079	0.43239	-0.002	0.99854
event1 +/-sqrt(Weibull2)	class 3	1.15317	241.86777	0.005	0.99620

Fixed effects in the longitudinal model:

	coef	Se	Wald p-value
intercept class1	3.54283	0.12829	27.615 0.00000
intercept class2	3.76305	0.03869	97.250 0.00000
intercept class3	3.43675	0.03988	86.172 0.00000
time_years	0.04450	0.00372	11.971 0.00000

Variance-covariance matrix of the random-effects:

	intercept
intercept	0.1692

	coef	Se
Residual standard error	0.21730	0.00391

```
postprob(jlcmm_m2_nfl)
```

Posterior classification based on longitudinal and time-to-event data:

	class1	class2	class3
N	10.00	130.00	388.00
%	1.89	24.62	73.48

Posterior classification table:

	--> mean of posterior probabilities in each class		
	prob1	prob2	prob3
class1	0.8920	0.1080	0.0000
class2	0.0133	0.9584	0.0283
class3	0.1682	0.0982	0.7336

Posterior probabilities above a threshold (%):

	class1	class2	class3
prob>0.7	90	92.31	62.63
prob>0.8	80	91.54	31.19
prob>0.9	70	90.77	7.99

Posterior classification based only on longitudinal data:

	class1	class2	class3
N	0	103.00	425.00
%	0	19.51	80.49

8.4 Comparing M0, M1, M2

```
# Function for extracting the class proportions from each model
get_class_proportions <- function(model) {
#   Getting the number of classes
  G <- model$ng
#   Calculating the class proportions (number of individuals assigned to a class / number of
  props <- table(model$pprob$class) / nrow(model$pprob) * 100

#   Creating a vector to store the proportions in, with NA for extra classes
  out <- rep(NA, G)
  out[1:G] <- as.numeric(props)
  names(out) <- paste0("Class", 1:G)

#   Returning results
  return(round(out, 2))
}

# Function for extracting entropy
get_entropy <- function(model) {
  G <- model$ng
  N <- nrow(model$pprob)

  if (G == 1) return(1.00)

  pprobs <- model$pprob[, 3:(2 + G)]
#   computing entropy
  entropy <- -sum(pprobs * log(pprobs + 0.0001))

#   scaling to between 0-1
  entropy_r2 <- 1 - (entropy / (N * log(G)))

  return(round(entropy_r2, 3))
}
```

```

models <- list(jlcmm_m0_1, jlcmm_m1_nfl, jlcmm_m2_nfl)

# Find max number of classes across all models
G_max <- max(sapply(models, function(m) m$ng))

# Extract proportions for each model, with NA for cells where the model has fewer classes than G_max
props <- do.call(rbind, lapply(models, function(m) {
  p <- get_class_proportions(m)
  padded <- c(p, rep(NA, G_max - length(p)))
  return(padded)
}))

colnames(props) <- paste0("Class", 1:G_max, " (%)")

summary_jlcmm_nfl <- data.frame(
  "Candidate model" = paste0("Model", 1:length(models)),
  "G" = sapply(models, function(m) m$ng),
  "Log Likelihood" = sapply(models, function(m) m$loglik),
  "BIC" = sapply(models, function(m) m$BIC),
  "Entropy" = sapply(models, get_entropy),
  props,
  check.names = FALSE
)

```

```
summary(jlcmm_m1_nfl)
```

Joint latent class model for quantitative outcome and competing risks
fitted by maximum likelihood method

```

jlcmm(fixed = log_PLASMA_NFL ~ time_years, mixture = ~1, random = ~1,
      subject = "RID", classmb = ~1, ng = 2, survival = Surv(time_to_event,
        event) ~ 1, data = jlcmm_data)

```

Statistical Model:

```

Dataset: jlcmm_data
Number of subjects: 528
Number of observations: 2074
Number of latent classes: 2
Number of parameters: 10
Event 1:
  Number of events: 128
  Class-specific hazards and

```

Weibull baseline risk function

Iteration process:

Convergence criteria satisfied
Number of iterations: 18
Convergence criteria: parameters= 1.5e-05
: likelihood= 4.3e-05
: second derivatives= 1.2e-10

Goodness-of-fit statistics:

maximum log-likelihood: -1032.38
AIC: 2084.77
BIC: 2127.46
Score test statistic for CI assumption: 0.368 (p-value=0.5444)

Maximum Likelihood Estimates:

Fixed effects in the class-membership model:
(the class of reference is the last class)

	coef	Se	Wald p-value
intercept class1	-0.80937	0.14485	-5.588 0.00000

Parameters in the proportional hazard model:

	coef	Se	Wald p-value
event1 +/-sqrt(Weibull1) class 1	0.47076	0.02606	18.062 0.00000
event1 +/-sqrt(Weibull2) class 1	1.23898	0.05811	21.319 0.00000
event1 +/-sqrt(Weibull1) class 2	0.23156	0.01211	19.117 0.00000
event1 +/-sqrt(Weibull2) class 2	2.68119	0.47553	5.638 0.00000

Fixed effects in the longitudinal model:

	coef	Se	Wald p-value
intercept class1	3.76627	0.03897	96.638 0.00000
intercept class2	3.45835	0.02643	130.853 0.00000
time_years	0.04451	0.00372	11.971 0.00000

Variance-covariance matrix of the random-effects:

	intercept
intercept	0.17008

	coef	Se
Residual standard error	0.21730	0.00391

8.5 Class Descriptions

```
# Saving which class each individual was predicted to be in
class_preds <- jlcmm_m1_nfl$pprobY

# Merging with model variables
data_preds <- class_preds %>% merge(
  jlcmm_data, by = "RID") %>% merge(
  surv_data %>% dplyr::select(age_at_baseline, RID), by = "RID"
)

# Merging with baseline biomarker values
data_preds <- data_preds %>% merge(
  baseline, by = "RID"
)

# Summarising age and baseline nfl & ptau by class
continuous_summary <- data_preds %>%
  pivot_longer(
    cols = c(age_at_baseline, baseline_value_nfl, baseline_value_ptau),
    names_to = "variable",
    values_to = "value"
  ) %>%
  group_by(variable, class) %>%
  summarise(
    stat = sprintf("%.1f (%.1f, %.1f)", median(value, na.rm = TRUE),
      quantile(value, 0.25, na.rm = TRUE),
      quantile(value, 0.75, na.rm = TRUE)),
    .groups = "drop"
  )

# Summarising APOE status and gender by class
categorical_summary <- data_preds %>%
  pivot_longer(
    cols = c(APOE4_status, PTGENDER),
    names_to = "variable",
    values_to = "value"
  ) %>%
```

```

group_by(variable, class, value) %>%
summarise(n = n(), .groups = "drop") %>%
group_by(variable, class) %>%
mutate(pct = n / sum(n) * 100,
       stat = sprintf("%d (%.1f%%)", n, pct),
       variable = paste0(variable, ": ", value)) %>%
ungroup() %>%
dplyr::select(variable, class, stat)

# Combining the summary statistics and pivot_wider makes each class a column
class_summary_nfl <- bind_rows(continuous_summary, categorical_summary) %>%
  pivot_wider(names_from = class, values_from = stat)

names(class_summary_nfl) <- c("Variable", "Class1", "Class2")

class_summary_nfl$Variable <- c("Age at baseline",
                               "Baseline NfL", "Baseline p-tau181",
                               r"(Genotype: 0 alleles)",
                               r"(Genotype: 1 alleles)",
                               r"(Genotype: 2 alleles)",
                               "Gender: Female", "Gender: Male")

saveRDS(class_summary_nfl, "tables/class_summary_nfl.rds")

```

8.5.1 Survival curves by class

```

# Saving the plot
png("figures/class_specific_nfl.png",
    width = 6, height = 4, units = "in", res = 300)
plot(
  jlcmm_m1_nfl,
  which = "survival",
  col = c("cornflowerblue", "firebrick"),
  lty = c(1, 2, 3),
  lwd = 2,
  legend.loc = "bottomleft",
  legend = c("Class 1 (high risk)", "Class 2 (low risk)"),
  cex = 0.6,
  bty = "n",
  xlab = "Time (years)",

```

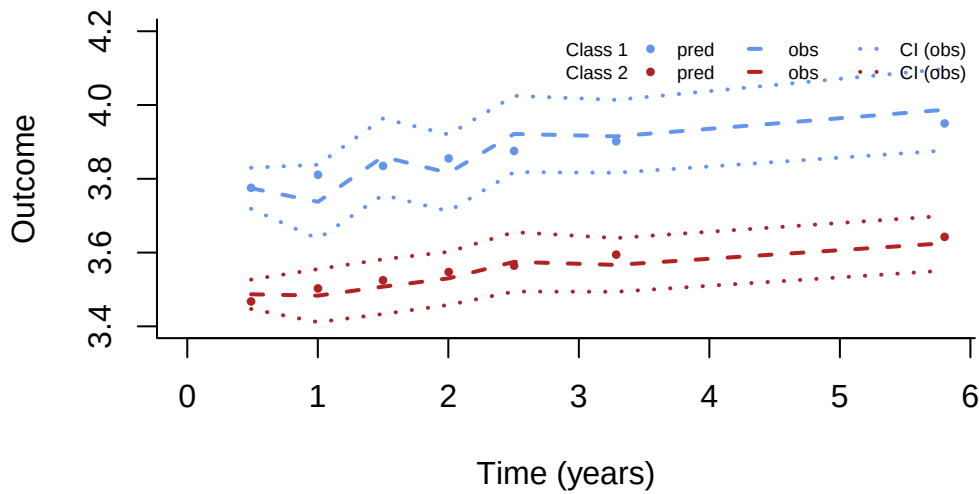
```
    ylab = "Event-free probability",  
    main = ""  
  )  
  dev.off()
```

pdf
2

8.5.2 Longitudinal trajectory by class

```
# longitudinal trajectories by class  
plot(  
  jlcmm_m1_nfl,  
  which = "fit",  
  var.time = "time_years",  
  marg = TRUE,  
  col = c("cornflowerblue", "firebrick"),  
  lty = c(1, 2, 3),  
  lwd = 2,  
  pch = c(16, 17),  
  legend.loc = "topright",  
  cex = 0.6,  
  legend = c("Class 1", "Class 2"),  
  bty = "n",  
  xlab = "Time (years)",  
  ylab = "Outcome",  
  ylim = c(3.4, 4.2)  
)
```

Weighted marginal predictions



9 JLCMM: PTAU

9.1 M0 (1 latent class)

```
# Fitting a one class joint latent class mixed model with random intercept
jlcmm_m0_1 <- jlcmm(
  fixed = log_PLASMAPTAU181 ~ time_years,
  random = ~ 1,
  subject = "RID",
  survival = Surv(time_to_event, event) ~ 1,
  data = jlcmm_data
)
```

Warning in Surv(time = c(7.11506849315068, 7.11506849315068, 7.11506849315068,
: type= 'mstate' is deprecated, use a factor variable as status

```
# Fitting a one class joint latent class mixed model with flexible baseline hazard to check p
jlcmm_m0_1_pieewise <- jlcmm(
  fixed = log_PLASMAPTAU181 ~ time_years,
  random = ~ 1,
```

```

subject = "RID",
survival = Surv(time_to_event, event) ~ 1,
hazard = "3-quant-piecewise",
data = jlcmm_data
)

```

Warning in Surv(time = c(7.11506849315068, 7.11506849315068, 7.11506849315068,
: type= 'mstate' is deprecated, use a factor variable as status

```
summary(jlcmm_m0_1)
```

Joint latent class model for quantitative outcome and competing risks
fitted by maximum likelihood method

```
jlcmm(fixed = log_PLASMAPTAU181 ~ time_years, random = ~1, subject = "RID",
      survival = Surv(time_to_event, event) ~ 1, data = jlcmm_data)
```

Statistical Model:

```

Dataset: jlcmm_data
Number of subjects: 528
Number of observations: 2074
Number of latent classes: 1
Number of parameters: 6
Event 1:
  Number of events: 128
  Weibull baseline risk function

```

Iteration process:

```

Convergence criteria satisfied
Number of iterations: 11
Convergence criteria: parameters= 2.7e-09
                     : likelihood= 2e-05
                     : second derivatives= 7.7e-12

```

Goodness-of-fit statistics:

```

maximum log-likelihood: -2019.69
AIC: 4051.39
BIC: 4077
Score test statistic for CI assumption: 50.336 (p-value=0)

```

Maximum Likelihood Estimates:

Parameters in the proportional hazard model:

	coef	Se	Wald	p-value
event1 +/-sqrt(Weibull1)	0.19672	0.01272	15.465	0.00000
event1 +/-sqrt(Weibull2)	0.99403	0.03788	26.241	0.00000

Fixed effects in the longitudinal model:

	coef	Se	Wald	p-value
intercept	2.67306	0.02439	109.575	0.00000
time_years	0.02794	0.00661	4.229	0.00002

Variance-covariance matrix of the random-effects:

	intercept
intercept	0.21662

	coef	Se
Residual standard error	0.38829	0.00698

```
summary(jlcmm_m0_1_piecewise)
```

Joint latent class model for quantitative outcome and competing risks
fitted by maximum likelihood method

```
jlcmm(fixed = log_PLASMAPTAU181 ~ time_years, random = ~1, subject = "RID",  
      survival = Surv(time_to_event, event) ~ 1, hazard = "3-quant-piecewise",  
      data = jlcmm_data)
```

Statistical Model:

Dataset: jlcmm_data
Number of subjects: 528
Number of observations: 2074
Number of latent classes: 1
Number of parameters: 6
Event 1:
Number of events: 128
Piecewise constant baseline risk function with nodes
0 5.020548 17.981

Iteration process:

Convergence criteria satisfied
Number of iterations: 12
Convergence criteria: parameters= 2.5e-08
: likelihood= 4.3e-05
: second derivatives= 1.9e-12

Goodness-of-fit statistics:
maximum log-likelihood: -2018.42
AIC: 4048.84
BIC: 4074.45
Score test statistic for CI assumption: 48.67 (p-value=0)

Maximum Likelihood Estimates:

Parameters in the proportional hazard model:

	coef	Se	Wald	p-value
event1 +/-sqrt(piecewise1)	0.20818	0.01091	19.078	0.00000
event1 +/-sqrt(piecewise2)	0.17859	0.01468	12.165	0.00000

Fixed effects in the longitudinal model:

	coef	Se	Wald	p-value
intercept	2.67306	0.02439	109.575	0.00000
time_years	0.02794	0.00661	4.229	0.00002

Variance-covariance matrix of the random-effects:

	intercept
intercept	0.21662

	coef	Se
Residual standard error	0.38829	0.00698

```
# Fitting a one class joint latent class mixed model with a random intercept and slope
jlcmm_m0_2 <- jlcmm(
  fixed = log_PLASMAPTAU181 ~ time_years,
  random = ~ time_years,
  subject = "RID",
  survival = Surv(time_to_event, event) ~ 1,
  data = jlcmm_data
)
```

Warning in Surv(time = c(7.11506849315068, 7.11506849315068, 7.11506849315068,
: type= 'mstate' is deprecated, use a factor variable as status

```
summary(jlcmm_m0_2)
```

Joint latent class model for quantitative outcome and competing risks
fitted by maximum likelihood method

```
jlcmm(fixed = log_PLASMAPTAU181 ~ time_years, random = ~time_years,  
      subject = "RID", survival = Surv(time_to_event, event) ~  
      1, data = jlcmm_data)
```

Statistical Model:

Dataset: jlcmm_data
Number of subjects: 528
Number of observations: 2074
Number of latent classes: 1
Number of parameters: 8
Event 1:
 Number of events: 128
 Weibull baseline risk function

Iteration process:

Convergence criteria satisfied
Number of iterations: 26
Convergence criteria: parameters= 2.6e-08
 : likelihood= 1.1e-05
 : second derivatives= 9.3e-11

Goodness-of-fit statistics:

maximum log-likelihood: -2019
AIC: 4054.01
BIC: 4088.16
Score test statistic for CI assumption: 50.87 (p-value=0)

Maximum Likelihood Estimates:

Parameters in the proportional hazard model:

	coef	Se	Wald	p-value
event1 +/-sqrt(Weibull1)	0.19672	0.01272	15.465	0.00000
event1 +/-sqrt(Weibull2)	0.99404	0.03788	26.241	0.00000

Fixed effects in the longitudinal model:

	coef	Se	Wald	p-value
intercept	2.67338	0.02426	110.206	0.00000
time_years	0.02772	0.00680	4.078	0.00005

Variance-covariance matrix of the random-effects:

	intercept	time_years
intercept	0.21425	
time_years	-0.00031	0.00119

	coef	Se
Residual standard error	0.38507	0.00784

9.1.1 Determining the random effects

```
# Comparing BIC and log-likelihood for the two one class models with different random effects
summarytable(jlcmm_m0_1, jlcmm_m0_2)
```

	G	loglik	npm	BIC	%class1
jlcmm_m0_1	1	-2019.695	6	4077.005	100
jlcmm_m0_2	1	-2019.005	8	4088.162	100

```
# log likelihood for the full model
l = jlcmm_m0_2$loglik

# log likelihood for the reduced model
l_0 = jlcmm_m0_1$loglik

# computing the LR test statistic
LR = - 2 * (l_0 - l)

# computing the p-value given the null hypothesis is true (the reduced model is the true model)
pchisq(LR, 2, lower.tail = FALSE)
```

```
[1] 0.5013604
```

LR

[1] 1.38086

9.2 M1 (2 latent classes)

```
# Running gridsearch() for a two class model
# jlcmm_m1_ptau_search <- gridsearch(m = jlcmm(fixed = log_PLASMAPTAU181 ~ time_years,
#                                           random = ~ 1,
#                                           mixture = ~ 1,
#                                           classmb = ~ 1,
#                                           ng = 2,
#                                           subject = "RID",
#                                           survival = Surv(time_to_event, event) ~ 1,
#                                           data = jlcmm_data),
#                                   minit = jlcmm_m0_1,
#                                   rep = 50,
#                                   maxiter = 50)
```

```
# Estimating two class model using the best result from gridsearch() as a starting value
# jlcmm_m1_ptau <- jlcmm(
#   fixed = log_PLASMAPTAU181 ~ time_years,
#   random = ~ 1,
#   mixture = ~ 1,
#   classmb = ~ 1,
#   ng = 2,
#   subject = "RID",
#   survival = Surv(time_to_event, event) ~ 1,
#   data = jlcmm_data,
#   B = jlcmm_m1_ptau_search$best
# )
```

```
# Fitting a two class joint latent class mixed model
# Results from gridsearch() and B = random(jlcmm_m0_1) are very similar, global maximum found
jlcmm_m1_ptau <- jlcmm(
  fixed = log_PLASMAPTAU181 ~ time_years,
  random = ~ 1,
  mixture = ~ 1,
  classmb = ~ 1,
  ng = 2,
```

```

subject = "RID",
survival = Surv(time_to_event, event) ~ 1,
data = jlcmm_data,
B = random(jlcmm_m0_1)
)

```

Warning in Surv(time = c(7.11506849315068, 7.11506849315068, 7.11506849315068,
: type= 'mstate' is deprecated, use a factor variable as status

```
summary(jlcmm_m1_ptau)
```

Joint latent class model for quantitative outcome and competing risks
fitted by maximum likelihood method

```

jlcmm(fixed = log_PLASMAPTAU181 ~ time_years, mixture = ~1, random = ~1,
      subject = "RID", classmb = ~1, ng = 2, survival = Surv(time_to_event,
        event) ~ 1, data = jlcmm_data)

```

Statistical Model:

```

Dataset: jlcmm_data
Number of subjects: 528
Number of observations: 2074
Number of latent classes: 2
Number of parameters: 10
Event 1:
  Number of events: 128
  Class-specific hazards and
  Weibull baseline risk function

```

Iteration process:

```

Convergence criteria satisfied
Number of iterations: 16
Convergence criteria: parameters= 1.2e-05
                     : likelihood= 2.8e-05
                     : second derivatives= 4.4e-11

```

Goodness-of-fit statistics:

```

maximum log-likelihood: -1989.5
AIC: 3999
BIC: 4041.69
Score test statistic for CI assumption: 0.122 (p-value=0.7268)

```

Maximum Likelihood Estimates:

Fixed effects in the class-membership model:
(the class of reference is the last class)

	coef	Se	Wald p-value
intercept class1	0.72040	0.17626	4.087 0.00004

Parameters in the proportional hazard model:

	coef	Se	Wald p-value
event1 +/-sqrt(Weibull1) class 1	0.23144	0.01222	18.937 0.00000
event1 +/-sqrt(Weibull2) class 1	2.74571	0.47381	5.795 0.00000
event1 +/-sqrt(Weibull1) class 2	0.45317	0.03330	13.608 0.00000
event1 +/-sqrt(Weibull2) class 2	1.21551	0.06335	19.188 0.00000

Fixed effects in the longitudinal model:

	coef	Se	Wald p-value
intercept class1	2.55090	0.03527	72.324 0.00000
intercept class2	2.92477	0.04366	66.992 0.00000
time_years	0.02762	0.00660	4.183 0.00003

Variance-covariance matrix of the random-effects:

	intercept
intercept	0.18576

	coef	Se
Residual standard error	0.38831	0.00698

```
postprob(jlcmm_m1_ptau)
```

Posterior classification based on longitudinal and time-to-event data:

	class1	class2
N	399.00	129.00
%	75.57	24.43

Posterior classification table:

--> mean of posterior probabilities in each class

```

      prob1  prob2
class1 0.8796 0.1204
class2 0.0329 0.9671

```

Posterior probabilities above a threshold (%):

```

      class1 class2
prob>0.7  90.98  93.80
prob>0.8  76.19  91.47
prob>0.9  52.63  91.47

```

Posterior classification based only on longitudinal data:

```

      class1 class2
N 458.00  70.00
%  86.74  13.26

```

```

# Posterior probability table for the 2 class jlcmm for ptau
get_pprob_table(jlcmm_m1_ptau)

```

	Prob1	Prob2	Prob > 0.7 (%)	Prob > 0.8 (%)	Prob > 0.9 (%)
Class1	0.8796	0.1204	90.98	76.19	52.63
Class2	0.0329	0.9671	93.80	91.47	91.47

```

# The model estimates in a table
get_jlcmm_params(jlcmm_m1_ptau)

```

	Submodel	Parameter	Estimate	SE	Z
1	Class Membership	intercept class1	0.720	0.176	4.087
2	Survival event1	+/-sqrt(Weibull1) class 1	0.231	0.012	18.937
3	Survival event1	+/-sqrt(Weibull2) class 1	2.746	0.474	5.795
4	Survival event1	+/-sqrt(Weibull1) class 2	0.453	0.033	13.608
5	Survival event1	+/-sqrt(Weibull2) class 2	1.216	0.063	19.188
6	Longitudinal	intercept class1	2.551	0.035	72.324
7	Longitudinal	intercept class2	2.925	0.044	66.992
8	Longitudinal	time_years	0.028	0.007	4.183
P value					
1	0				
2	0				
3	0				
4	0				


```

5      0
6      0
7      0
8      0

```

9.3 M2 (3 latent classes)

```

# Running gridsearch() for a three class model
# jlcmm_m2_ptau_search <- gridsearch(m = jlcmm(fixed = log_PLASMAPTAU181 ~ time_years,
#                                           random = ~ 1,
#                                           mixture = ~ 1,
#                                           classmb = ~ 1,
#                                           ng = 3,
#                                           subject = "RID",
#                                           survival = Surv(time_to_event, event) ~ 1,
#                                           data = jlcmm_data),
#                                   minit = jlcmm_m0_1,
#                                   rep = 50,
#                                   maxiter = 50)

```

```

# Estimating two class model using the best result from gridsearch() as a starting value
# jlcmm_m2_ptau <- jlcmm(
#   fixed = log_PLASMAPTAU181 ~ time_years,
#   random = ~ 1,
#   mixture = ~ 1,
#   classmb = ~ 1,
#   ng = 3,
#   subject = "RID",
#   survival = Surv(time_to_event, event) ~ 1,
#   data = jlcmm_data,
#   B = jlcmm_m2_ptau_search$best
# )

```

```

# Fitting a three class joint latent class mixed model
# Results from gridsearch() and B = random(jlcmm_m0_1) are very similar, global maximum found
jlcmm_m2_ptau <- jlcmm(
  fixed = log_PLASMAPTAU181 ~ time_years,
  random = ~ 1,
  mixture = ~ 1,
  classmb = ~ 1,

```

```

ng = 3,
subject = "RID",
survival = Surv(time_to_event, event) ~ 1,
data = jlcmm_data,
B = random(jlcmm_m0_1)
)

```

Warning in Surv(time = c(7.11506849315068, 7.11506849315068, 7.11506849315068,
: type= 'mstate' is deprecated, use a factor variable as status

```
summary(jlcmm_m2_ptau)
```

Joint latent class model for quantitative outcome and competing risks
fitted by maximum likelihood method

```

jlcmm(fixed = log_PLASMAPTAU181 ~ time_years, mixture = ~1, random = ~1,
      subject = "RID", classmb = ~1, ng = 3, survival = Surv(time_to_event,
      event) ~ 1, data = jlcmm_data)

```

Statistical Model:

```

Dataset: jlcmm_data
Number of subjects: 528
Number of observations: 2074
Number of latent classes: 3
Number of parameters: 14
Event 1:
  Number of events: 128
  Class-specific hazards and
  Weibull baseline risk function

```

Iteration process:

```

Convergence criteria satisfied
Number of iterations: 22
Convergence criteria: parameters= 6.8e-06
                     : likelihood= 3.7e-06
                     : second derivatives= 4.9e-12

```

Goodness-of-fit statistics:

```

maximum log-likelihood: -1984.35
AIC: 3996.71
BIC: 4056.47

```

Score test statistic for CI assumption: 0.478 (p-value=0.4892)

Maximum Likelihood Estimates:

Fixed effects in the class-membership model:
(the class of reference is the last class)

	coef	Se	Wald p-value
intercept class1	1.58220	0.71371	2.217 0.02663
intercept class2	2.20295	0.73411	3.001 0.00269

Parameters in the proportional hazard model:

	coef	Se	Wald p-value
event1 +/-sqrt(Weibull1) class 1	0.44967	0.03793	11.855 0.00000
event1 +/-sqrt(Weibull2) class 1	1.20205	0.06666	18.033 0.00000
event1 +/-sqrt(Weibull1) class 2	0.23348	0.01214	19.226 0.00000
event1 +/-sqrt(Weibull2) class 2	2.83837	0.48214	5.887 0.00000
event1 +/-sqrt(Weibull1) class 3	0.18511	0.06609	2.801 0.00510
event1 +/-sqrt(Weibull2) class 3	1.38193	0.37589	3.676 0.00024

Fixed effects in the longitudinal model:

	coef	Se	Wald p-value
intercept class1	2.95776	0.04286	69.004 0.00000
intercept class2	2.61453	0.05528	47.295 0.00000
intercept class3	1.81783	0.21547	8.437 0.00000
time_years	0.02770	0.00660	4.196 0.00003

Variance-covariance matrix of the random-effects:

	intercept
intercept	0.1392

	coef	Se
Residual standard error	0.38820	0.00697

```
# Checking the posterior probabilities for the 3 class model
postprob(jlcmm_m2_ptau)
```

Posterior classification based on longitudinal and time-to-event data:

	class1	class2	class3
N	132	376.00	20.00
%	25	71.21	3.79

Posterior classification table:

--> mean of posterior probabilities in each class

	prob1	prob2	prob3
class1	0.9494	0.0433	0.0072
class2	0.1237	0.8242	0.0521
class3	0.0183	0.2388	0.7429

Posterior probabilities above a threshold (%):

	class1	class2	class3
prob>0.7	93.18	86.17	65
prob>0.8	87.88	62.77	45
prob>0.9	84.09	28.72	15

Posterior classification based only on longitudinal data:

	class1	class2	class3
N	93.00	414.00	21.00
%	17.61	78.41	3.98

```
# Posterior probability table for the 3 class jlcmm for ptau
get_pprob_table(jlcmm_m2_ptau)
```

	Prob1	Prob2	Prob3	Prob > 0.7 (%)	Prob > 0.8 (%)	Prob > 0.9 (%)
Class1	0.9494	0.0433	0.0072	93.18	87.88	84.09
Class2	0.1237	0.8242	0.0521	86.17	62.77	28.72
Class3	0.0183	0.2388	0.7429	65.00	45.00	15.00

```
# Checking flexible baseline hazard (piecewise)
jlcmm_m2_ptau_pieewise <- jlcmm(
  fixed = log_PLASMAPTAU181 ~ time_years,
  random = ~ 1,
  mixture = ~ 1,
  classmb = ~ 1,
  ng = 3,
  subject = "RID",
  survival = Surv(time_to_event, event) ~ 1,
  hazard = "3-quant-pieewise",
```

```

data = jlcmm_data,
B = random(jlcmm_m0_1_piecewise)
)

```

Warning in Surv(time = c(7.11506849315068, 7.11506849315068, 7.11506849315068,
: type= 'mstate' is deprecated, use a factor variable as status

```

# Flexible hazard with two parameters has higher BIC than Weibull hazard
summary(jlcmm_m2_ptau_piecewise)

```

Joint latent class model for quantitative outcome and competing risks
fitted by maximum likelihood method

```

jlcmm(fixed = log_PLASMAPTAU181 ~ time_years, mixture = ~1, random = ~1,
      subject = "RID", classmb = ~1, ng = 3, survival = Surv(time_to_event,
        event) ~ 1, hazard = "3-quant-piecewise", data = jlcmm_data)

```

Statistical Model:

```

Dataset: jlcmm_data
Number of subjects: 528
Number of observations: 2074
Number of latent classes: 3
Number of parameters: 14
Event 1:
  Number of events: 128
  Class-specific hazards and
  Piecewise constant baseline risk function with nodes
  0 5.020548 17.981

```

Iteration process:

```

Convergence criteria satisfied
Number of iterations: 29
Convergence criteria: parameters= 8e-08
                     : likelihood= 1.9e-05
                     : second derivatives= 5.7e-07

```

Goodness-of-fit statistics:

```

maximum log-likelihood: -1989.27
AIC: 4006.54
BIC: 4066.3
Score test statistic for CI assumption: 0.172 (p-value=0.6783)

```

Maximum Likelihood Estimates:

Fixed effects in the class-membership model:
(the class of reference is the last class)

	coef	Se	Wald	p-value
intercept class1	0.32122	0.40438	0.794	0.42700
intercept class2	-1.91950	0.66666	-2.879	0.00399

Parameters in the proportional hazard model:

	coef	Se	Wald	p-value
event1 +/-sqrt(piecewise1) class 1	0.09515	0.05238	1.816	0.06930
event1 +/-sqrt(piecewise2) class 1	0.06800	0.05507	1.235	0.21692
event1 +/-sqrt(piecewise1) class 2	0.00031	0.06057	0.005	0.99589
event1 +/-sqrt(piecewise2) class 2	0.19124	0.07536	2.538	0.01116
event1 +/-sqrt(piecewise1) class 3	0.33412	0.03654	9.143	0.00000
event1 +/-sqrt(piecewise2) class 3	0.39002	0.07985	4.884	0.00000

Fixed effects in the longitudinal model:

	coef	Se	Wald	p-value
intercept class1	2.54156	0.06430	39.526	0.00000
intercept class2	1.73822	0.17757	9.789	0.00000
intercept class3	2.99162	0.07074	42.290	0.00000
time_years	0.02766	0.00659	4.197	0.00003

Variance-covariance matrix of the random-effects:

intercept	
intercept	0.11655

	coef	Se
Residual standard error	0.38816	0.00697

9.4 M3 (4 latent classes)

```
# Running gridsearch() for a four class model
# jlcmm_m3_ptau_search <- gridsearch(m = jlcmm(fixed = log_PLASMAPTAU181 ~ time_years,
#                                             random = ~ 1,
```

```
#
#                               mixture = ~ 1,
#                               classmb = ~ 1,
#                               ng = 4,
#                               subject = "RID",
#                               survival = Surv(time_to_event, event) ~ 1,
#                               data = jlcmm_data),
#                               minit = jlcmm_m0_1,
#                               rep = 50,
#                               maxiter = 50)
```

```
# Estimating two class model using the best result from gridsearch() as a starting value
# jlcmm_m3_ptau <- jlcmm(
#   fixed = log_PLASMAPTAU181 ~ time_years,
#   random = ~ 1,
#   mixture = ~ 1,
#   classmb = ~ 1,
#   ng = 4,
#   subject = "RID",
#   survival = Surv(time_to_event, event) ~ 1,
#   data = jlcmm_data,
#   B = jlcmm_m3_ptau_search$best
# )
```

```
# Fitting a four class joint latent class mixed model
# Results from gridsearch() and B = random(jlcmm_m0_1) are very similar, global maximum found
jlcmm_m3_ptau <- jlcmm(
  fixed = log_PLASMAPTAU181 ~ time_years,
  random = ~ 1,
  mixture = ~ 1,
  classmb = ~ 1,
  ng = 4,
  subject = "RID",
  survival = Surv(time_to_event, event) ~ 1,
  data = jlcmm_data,
  B = random(jlcmm_m0_1)
)
```

Warning in Surv(time = c(7.11506849315068, 7.11506849315068, 7.11506849315068,
: type= 'mstate' is deprecated, use a factor variable as status

```
summary(jlcmm_m3_ptau)
```

Joint latent class model for quantitative outcome and competing risks
fitted by maximum likelihood method

```
jlcmm(fixed = log_PLASMAPTAU181 ~ time_years, mixture = ~1, random = ~1,  
      subject = "RID", classmb = ~1, ng = 4, survival = Surv(time_to_event,  
        event) ~ 1, data = jlcmm_data)
```

Statistical Model:

Dataset: jlcmm_data
Number of subjects: 528
Number of observations: 2074
Number of latent classes: 4
Number of parameters: 18
Event 1:
 Number of events: 128
 Class-specific hazards and
 Weibull baseline risk function

Iteration process:

Convergence criteria satisfied
Number of iterations: 47
Convergence criteria: parameters= 3.2e-10
 : likelihood= 3.4e-09
 : second derivatives= 7e-15

Goodness-of-fit statistics:

maximum log-likelihood: -1974.83
AIC: 3985.67
BIC: 4062.51
Score test statistic for CI assumption: 0.829 (p-value=0.3625)

Maximum Likelihood Estimates:

Fixed effects in the class-membership model:
(the class of reference is the last class)

	coef	Se	Wald	p-value
intercept class1	-1.11307	0.49715	-2.239	0.02516
intercept class2	-2.32398	0.48998	-4.743	0.00000
intercept class3	0.64934	0.15978	4.064	0.00005

Parameters in the proportional hazard model:

			coef	Se	Wald	p-value
event1 +/-sqrt(Weibull1)	class 1		0.18057	0.05210	3.466	0.00053
event1 +/-sqrt(Weibull2)	class 1		1.36539	0.31550	4.328	0.00002
event1 +/-sqrt(Weibull1)	class 2		0.27313	0.00670	40.767	0.00000
event1 +/-sqrt(Weibull2)	class 2		4.18365	1.16761	3.583	0.00034
event1 +/-sqrt(Weibull1)	class 3		0.22920	0.01651	13.880	0.00000
event1 +/-sqrt(Weibull2)	class 3		2.69872	0.59105	4.566	0.00000
event1 +/-sqrt(Weibull1)	class 4		0.47329	0.02571	18.412	0.00000
event1 +/-sqrt(Weibull2)	class 4		1.23379	0.05724	21.555	0.00000

Fixed effects in the longitudinal model:

			coef	Se	Wald	p-value
intercept	class1		1.85764	0.14233	13.052	0.00000
intercept	class2		3.69150	0.16968	21.755	0.00000
intercept	class3		2.62528	0.04794	54.762	0.00000
intercept	class4		2.93631	0.03755	78.198	0.00000
time_years			0.02706	0.00660	4.100	0.00004

Variance-covariance matrix of the random-effects:

```

      intercept
intercept  0.09862

```

		coef	Se
Residual standard error		0.38826	0.00698

```
postprob(jlcmm_m3_ptau)
```

Posterior classification based on longitudinal and time-to-event data:

	class1	class2	class3	class4
N	35.00	16.00	356.00	121.00
%	6.63	3.03	67.42	22.92

Posterior classification table:

```

--> mean of posterior probabilities in each class
      prob1  prob2  prob3  prob4
class1 0.7840 0.0000 0.1983 0.0176

```

```

class2 0.0000 0.6795 0.1243 0.1962
class3 0.0653 0.0103 0.8171 0.1074
class4 0.0103 0.0079 0.0226 0.9593

```

Posterior probabilities above a threshold (%):

```

      class1 class2 class3 class4
prob>0.7  68.57  43.75  82.87  93.39
prob>0.8  51.43  25.00  59.55  93.39
prob>0.9  37.14  12.50  30.34  90.91

```

Posterior classification based only on longitudinal data:

```

      class1 class2 class3 class4
N   35.00  10.00 389.00   94.0
%    6.63   1.89 73.67   17.8

```

9.5 Comparing M0, M1, M2, M3

```

models <- list(jlcmm_m0_1, jlcmm_m1_ptau, jlcmm_m2_ptau, jlcmm_m3_ptau)

# Find max number of classes across all models
G_max <- max(sapply(models, function(m) m$ng))

# Extract proportions for each model, with NA for cells where the model has fewer classes than
props <- do.call(rbind, lapply(models, function(m) {
  p <- get_class_proportions(m)
  padded <- c(p, rep(NA, G_max - length(p)))
  return(padded)
}))

colnames(props) <- paste0("Class", 1:G_max, " (%)")

summary_jlcmm_ptau <- data.frame(
  "Candidate model" = paste0("Model", 1:length(models)),
  "G" = sapply(models, function(m) m$ng),
  "Log Likelihood" = sapply(models, function(m) m$loglik),
  "BIC" = sapply(models, function(m) m$BIC),
  "Entropy" = sapply(models, get_entropy),
  props,

```

```

    check.names = FALSE
  )

```

9.6 Class Descriptions

```

# Saving which class each individual was predicted to be in
class_preds <- jlcmm_m1_ptau$pprobY

# Merging with model variables
data_preds <- class_preds %>% merge(
  jlcmm_data, by = "RID") %>% merge(
  surv_data %>% dplyr::select(age_at_baseline, RID), by = "RID"
)

# Merging with baseline biomarker values
data_preds <- data_preds %>% merge(
  baseline, by = "RID"
)

# Summarising age and baseline nfl & ptau by class
continuous_summary <- data_preds %>%
  pivot_longer(
    cols = c(age_at_baseline, baseline_value_nfl, baseline_value_ptau),
    names_to = "variable",
    values_to = "value"
  ) %>%
  group_by(variable, class) %>%
  summarise(
    stat = sprintf("%.1f (%.1f, %.1f)", median(value, na.rm = TRUE),
      quantile(value, 0.25, na.rm = TRUE),
      quantile(value, 0.75, na.rm = TRUE)),
    .groups = "drop"
  )

# Summarising APOE status and gender by class
categorical_summary <- data_preds %>%
  pivot_longer(
    cols = c(APOE4_status, PTGENDER),
    names_to = "variable",
    values_to = "value"
  )

```

```

) %>%
group_by(variable, class, value) %>%
summarise(n = n(), .groups = "drop") %>%
group_by(variable, class) %>%
mutate(pct = n / sum(n) * 100,
       stat = sprintf("%d (%.1f%%)", n, pct),
       variable = paste0(variable, ": ", value)) %>%
ungroup() %>%
dplyr::select(variable, class, stat)

# Combining the summary statistics and pivot_wider makes each class a column
class_summary_ptau <- bind_rows(continuous_summary, categorical_summary) %>%
  pivot_wider(names_from = class, values_from = stat)

names(class_summary_ptau) <- c("Variable", "Class1", "Class2")

class_summary_ptau$Variable <- c("Age at baseline",
                                "Baseline NfL", "Baseline p-tau181",
                                r"(Genotype: 0 alleles)",
                                r"(Genotype: 1 alleles)",
                                r"(Genotype: 2 alleles)",
                                "Gender: Female", "Gender: Male")

saveRDS(class_summary_ptau, "tables/class_summary_ptau.rds")

```

9.6.1 Survival curves by class

```

# Saving the plot
png("figures/class_specific_ptau.png",
    width = 6, height = 4, units = "in", res = 300)
plot(
  jlcmm_m1_ptau,
  which = "survival",
  col = c("cornflowerblue", "firebrick"),
  lty = c(1, 2, 3),
  lwd = 2,
  legend.loc = "bottomleft",
  legend = c("Class 1 (high risk)", "Class 2 (low risk)"),
  cex = 0.6,
  bty = "l",

```

```

    xlab = "Time (years)",
    ylab = "Event-free probability",
    main = ""
)
dev.off()

```

pdf
2

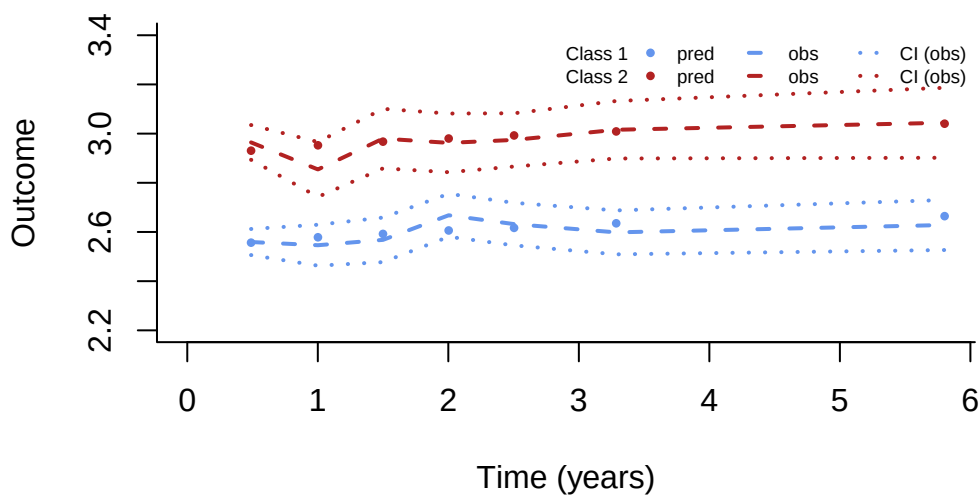
9.6.2 Longitudinal trajectories by class

```

# longitudinal trajectories by class
# longitudinal trajectories by class
plot(
  jlcmm_m1_ptau,
  which = "fit",
  var.time = "time_years",
  marg = TRUE,
  col = c("cornflowerblue", "firebrick"),
  lty = c(1, 2, 3),
  lwd = 2,
  pch = c(16, 17),
  legend.loc = "topright",
  cex = 0.6,
  legend = c("Class 1", "Class 2"),
  bty = "n",
  xlab = "Time (years)",
  ylab = "Outcome",
  ylim = c(2.2, 3.4)
)

```

Weighted marginal predictions



9.7 Defining time points for evaluation of predictive performance

```
# We define evaluation time points for time-dependent AUC comparison
times <- c(2, 3, 4, 5, 6)
```

10 Time-dependent AUC

10.1 Time-dependent AUC for fitted Cox PH models

```
# We define a function to compute time-dependent AUC for Cox models using timeROC
get_cox_auc <- function(model, times) {

  T <- surv_data_out$time_to_event
  delta <- surv_data_out$event

  lp <- predict(model, newdata = surv_data_out, type = "lp")

  timeROC::timeROC(
```

```

    T = T,
    delta = delta,
    marker = lp,
    cause = 1,
    times = times,
    iid = TRUE
  )$AUC
}

```

```

# We compute time-dependent AUC for Cox model with p-tau181
auc_ptau_bl <- get_cox_auc(surv_model_ptau_bl_final, times)

# # We compute time-dependent AUC for Cox model with NfL
auc_nfl_bl <- get_cox_auc(surv_model_nfl_bl_final, times)

# We compute time-dependent AUC for Cox model with both p-tau181 and NfL
auc_both_bl <- get_cox_auc(surv_model_both_bl_final, times)

```

10.2 Time-dependent AUC for fitted joint models

```

# We construct prediction datasets for the joint models by merging the longitudinal and survival data
newdata <- dplyr::left_join(
  long_data_out,
  dplyr::select(surv_data_out, RID, age_at_baseline, time_to_event, event),
  by = "RID"
)

```

```

# We define a function to compute time-dependent AUC for joint models using tvAUC
get_jm_auc <- function(model, times) {

  aucs <- sapply(times, function(t) {

    tmp <- tvAUC(
      model,
      newdata = newdata,
      Tstart = 1,
      Thoriz = t
    )

    tmp$auc
  })
}

```

```

}))

return(as.numeric(aucs))
}

# We compute time-dependent AUC for the joint model including p-tau181
auc_ptau_jm <- get_jm_auc(joint_model_ptau, times)

# We compute time-dependent AUC for the joint model including NfL
auc_nfl_jm <- get_jm_auc(joint_model_nfl, times)

# We compute time-dependent AUC for the multivariate joint model
auc_multi_jm <- get_jm_auc(multi_joint_model, times)

```

10.3 Time-dependent AUC for latent class joint models

10.3.0.1 Function for getting predictions from multivariate joint latent class model

```

# Defining a function to predict cumulative hazard (1 - S(t)) for multivariate joint latent class model
predict_mjlcmm <- function(model, data, ID, landmark, times) {

#   Extracting the unique ID's
ids <- unique(data[[ID]])
#   Extracting number of individuals to make predictions for
N <- length(unique(data[[ID]]))
#   Extracting the number of classes
G <- model$ng
#   Number of horizons
n_h <- length(times)
print(n_h)

#   Extracting the posterior probabilities (given a two class model)
post_prob <- model$pprob[, 3:4]

#   Extracting the Weibull hazard parameters from the model object
scale <- sapply(1:G, function(g) {
  exp(model$best[paste0("event1 +/-sqrt(Weibull2) class ", as.character(g))])
})
shape <- sapply(1:G, function(g) {
  exp(model$best[paste0("event1 +/-sqrt(Weibull1) class ", as.character(g))])
})

#   Initiating a matrix to save the predictions for each individual at each horizon

```



```

S_pred <- matrix(0, nrow = N, ncol = n_h)

# Saving the class-specific survivals at the landmark time
S_land <- sapply(1:G, function(g)
  exp(-(landmark / scale[g])^shape[g]))

# Looping over all individuals
for(i in 1:N) {

# Getting the posterior probabilities for each individual
pp_i <- post_prob[i, ]

# Looping over the horizons
for(j in 1:n_h) {
  t <- times[j]

# Getting the conditional class-specific survivals
Sg <- sapply(1:G, function(g)
  exp(-(t / scale[g])^shape[g]) / S_land[g])

# Calculating the cumulative hazard (1 - survival) for individual i based on their posterior
S_pred[i, j] <- 1 - sum(pp_i * Sg)
}
}

# Convert to long format matching dynpred() output structure
S_pred <- as.data.frame(S_pred)
colnames(S_pred) <- as.character(times)
S_pred[[ID]] <- ids

S_pred_long <- S_pred %>%
  pivot_longer(
    cols = -all_of(ID),
    names_to = "horizon",
    values_to = "pred"
  ) %>%
  mutate(horizon = as.numeric(horizon))

names(S_pred_long) <- c("RID", "horizon", "pred")

return(S_pred_long)
}

```

10.3.0.2 Function for extracting time-dependent AUC

```
# Defining a function to compute time dependent AUC for joint latent class models
jlcmm_timeAUC <- function(model, newdata, ID, time_var, surv_time_var, event_var,
                           landmark, times, cause = 1) {

#   Initiating a data frame to save the results in at each horizon time
output <- data.frame(
  times = times,
  cases = NA,
  AUC = NA
)

#   Calculating the survival predictions previously defined function
if(class(model) == "mpjlcmm") {
  dfpred <- predict_mjlcmm(
    model = model,
    data = newdata,
    ID = ID,
    landmark = landmark,
    times = times
  )
}

#   Calculating the dynamic survival predictions from joint models using dynpred() in the lcr
if(class(model) == "Jointlcmm") {
  dfpred <- dynpred(
    model = model,
    newdata = newdata,
    var.time = time_var,
    horizon = times,
    landmark = landmark
  )$pred
}

#   Fetching the true events for each ID for each time point
dftrue <- unique(newdata[c(ID, surv_time_var, event_var)])
names(dftrue) <- c(ID, "time", "event")

#   Merging the true and predicted data frames
merged <- merge(dfpred, dftrue, by = ID)

#   Looping through the horizon times
```

```

for(i in 1:length(times)) {
  h = times[i]

#   Filtering for the predictions from horizon h
  data_h <- merged %>% filter(horizon == h)

#   Identifying individuals that convert (cases) in the time between landmark and lan
  cases <- data_h$event == 1 & data_h$time > landmark & data_h$time <= h
  controls <- data_h$time > h
  data_h <- data_h[cases | controls, ]

  roc <- timeROC(
    T = data_h$time,
    delta = data_h$event,
    marker = data_h$pred,
    cause = cause,
    times = h,
    iid = TRUE)

  output$AUC[i] <- roc$AUC[2]
#   Checking the events in the time window between l and l + h
  output$cases[i] <- sum(cases)

}
return(output)
}

```

```

# Computing AUC for 2 class joint latent class mixed model for NfL
auc_jlcmm_m1_nfl <- jlcmm_timeAUC(jlcmm_m1_nfl, newdata,
                                   "RID", "time_years", "time_to_event", "event",
                                   1, times)

```

	RID	landmark	horizon	pred
[1,]	31	1	2	0.14260269
[2,]	31	1	3	0.20970694
[3,]	31	1	4	0.26566526
[4,]	31	1	5	0.30933735
[5,]	31	1	6	0.34178067
[6,]	74	1	2	0.10322711
[7,]	74	1	3	0.15180598
[8,]	74	1	4	0.19232996
[9,]	74	1	5	0.22400118

[10,]	74	1	6 0.24764733
[11,]	89	1	2 0.07932390
[12,]	89	1	3 0.11665682
[13,]	89	1	4 0.14781128
[14,]	89	1	5 0.17219728
[15,]	89	1	6 0.19050305
[16,]	120	1	2 0.07738928
[17,]	120	1	3 0.11381201
[18,]	120	1	4 0.14420813
[19,]	120	1	5 0.16800450
[20,]	120	1	6 0.18587804
[21,]	127	1	2 0.05729456
[22,]	127	1	3 0.08426314
[23,]	127	1	4 0.10678258
[24,]	127	1	5 0.12445449
[25,]	127	1	6 0.13783853
[26,]	150	1	2 0.07669308
[27,]	150	1	3 0.11278826
[28,]	150	1	4 0.14291149
[29,]	150	1	5 0.16649568
[30,]	150	1	6 0.18421367
[31,]	200	1	2 0.08095357
[32,]	200	1	3 0.11905322
[33,]	200	1	4 0.15084646
[34,]	200	1	5 0.17572916
[35,]	200	1	6 0.19439902
[36,]	259	1	2 0.09769384
[37,]	259	1	3 0.14366943
[38,]	259	1	4 0.18202449
[39,]	259	1	5 0.21200928
[40,]	259	1	6 0.23441920
[41,]	260	1	2 0.09120255
[42,]	260	1	3 0.13412413
[43,]	260	1	4 0.16993475
[44,]	260	1	5 0.19794112
[45,]	260	1	6 0.21890078
[46,]	272	1	2 0.07289131
[47,]	272	1	3 0.10719784
[48,]	272	1	4 0.13583086
[49,]	272	1	5 0.15825634
[50,]	272	1	6 0.17512496
[51,]	359	1	2 0.07875047
[52,]	359	1	3 0.11581361

[53,]	359	1	4	0.14674330
[54,]	359	1	5	0.17095453
[55,]	359	1	6	0.18913219
[56,]	377	1	2	0.13999088
[57,]	377	1	3	0.20586633
[58,]	377	1	4	0.26080088
[59,]	377	1	5	0.30367695
[60,]	377	1	6	0.33553675
[61,]	384	1	2	0.07792372
[62,]	384	1	3	0.11459790
[63,]	384	1	4	0.14520351
[64,]	384	1	5	0.16916277
[65,]	384	1	6	0.18715571
[66,]	419	1	2	0.08798050
[67,]	419	1	3	0.12938617
[68,]	419	1	4	0.16393382
[69,]	419	1	5	0.19095818
[70,]	419	1	6	0.21119798
[71,]	498	1	2	0.04776483
[72,]	498	1	3	0.07024988
[73,]	498	1	4	0.08903389
[74,]	498	1	5	0.10380133
[75,]	498	1	6	0.11505626
[76,]	519	1	2	0.07492813
[77,]	519	1	3	0.11019294
[78,]	519	1	4	0.13962435
[79,]	519	1	5	0.16267062
[80,]	519	1	6	0.17999429
[81,]	522	1	2	0.08979522
[82,]	522	1	3	0.13205467
[83,]	522	1	4	0.16731365
[84,]	522	1	5	0.19489110
[85,]	522	1	6	0.21553633
[86,]	548	1	2	0.15549435
[87,]	548	1	3	0.22866386
[88,]	548	1	4	0.28967543
[89,]	548	1	5	0.33727664
[90,]	548	1	6	0.37260018
[91,]	555	1	2	0.13286449
[92,]	555	1	3	0.19538713
[93,]	555	1	4	0.24752830
[94,]	555	1	5	0.28823239
[95,]	555	1	6	0.31850003

[96,]	637	1	2	0.15511321
[97,]	637	1	3	0.22810340
[98,]	637	1	4	0.28896557
[99,]	637	1	5	0.33645062
[100,]	637	1	6	0.37168901
[101,]	668	1	2	0.06965802
[102,]	668	1	3	0.10244336
[103,]	668	1	4	0.12980900
[104,]	668	1	5	0.15124904
[105,]	668	1	6	0.16739529
[106,]	671	1	2	0.17821268
[107,]	671	1	3	0.26207068
[108,]	671	1	4	0.33198732
[109,]	671	1	5	0.38651261
[110,]	671	1	6	0.42691182
[111,]	684	1	2	0.20290239
[112,]	684	1	3	0.29837637
[113,]	684	1	4	0.37797082
[114,]	684	1	5	0.44002104
[115,]	684	1	6	0.48593634
[116,]	685	1	2	0.16430579
[117,]	685	1	3	0.24162088
[118,]	685	1	4	0.30608634
[119,]	685	1	5	0.35637309
[120,]	685	1	6	0.39366526
[121,]	686	1	2	0.19502335
[122,]	686	1	3	0.28679041
[123,]	686	1	4	0.36329646
[124,]	686	1	5	0.42294531
[125,]	686	1	6	0.46710030
[126,]	908	1	2	0.12766646
[127,]	908	1	3	0.18774353
[128,]	908	1	4	0.23784718
[129,]	908	1	5	0.27696702
[130,]	908	1	6	0.30607333
[131,]	920	1	2	0.12295067
[132,]	920	1	3	0.18080907
[133,]	920	1	4	0.22906424
[134,]	920	1	5	0.26674680
[135,]	920	1	6	0.29479953
[136,]	926	1	2	0.15126697
[137,]	926	1	3	0.22244758
[138,]	926	1	4	0.28180210

[139,]	926	1	5	0.32811489
[140,]	926	1	6	0.36249396
[141,]	951	1	2	0.20373940
[142,]	951	1	3	0.29960717
[143,]	951	1	4	0.37952972
[144,]	951	1	5	0.44183504
[145,]	951	1	6	0.48793734
[146,]	984	1	2	0.14824988
[147,]	984	1	3	0.21801101
[148,]	984	1	4	0.27618291
[149,]	984	1	5	0.32157615
[150,]	984	1	6	0.35528115
[151,]	1023	1	2	0.13999088
[152,]	1023	1	3	0.20586633
[153,]	1023	1	4	0.26080088
[154,]	1023	1	5	0.30367695
[155,]	1023	1	6	0.33553675
[156,]	1045	1	2	0.10018909
[157,]	1045	1	3	0.14733864
[158,]	1045	1	4	0.18667178
[159,]	1045	1	5	0.21741707
[160,]	1045	1	6	0.24038447
[161,]	1072	1	2	0.03827159
[162,]	1072	1	3	0.05629028
[163,]	1072	1	4	0.07135315
[164,]	1072	1	5	0.08322724
[165,]	1072	1	6	0.09236122
[166,]	1155	1	2	0.12020017
[167,]	1155	1	3	0.17676451
[168,]	1155	1	4	0.22394154
[169,]	1155	1	5	0.26078580
[170,]	1155	1	6	0.28822402
[171,]	1186	1	2	0.17291137
[172,]	1186	1	3	0.25427521
[173,]	1186	1	4	0.32211385
[174,]	1186	1	5	0.37502342
[175,]	1186	1	6	0.41423822
[176,]	1206	1	2	0.13907285
[177,]	1206	1	3	0.20451639
[178,]	1206	1	4	0.25909110
[179,]	1206	1	5	0.30168737
[180,]	1206	1	6	0.33334207
[181,]	1222	1	2	0.08132876

[182,]	1222	1	3	0.11960492
[183,]	1222	1	4	0.15154523
[184,]	1222	1	5	0.17654228
[185,]	1222	1	6	0.19529596
[186,]	1280	1	2	0.09332179
[187,]	1280	1	3	0.13724043
[188,]	1280	1	4	0.17388174
[189,]	1280	1	5	0.20253402
[190,]	1280	1	6	0.22396715
[191,]	1286	1	2	0.07884504
[192,]	1286	1	3	0.11595267
[193,]	1286	1	4	0.14691942
[194,]	1286	1	5	0.17115948
[195,]	1286	1	6	0.18935826
[196,]	1300	1	2	0.14495193
[197,]	1300	1	3	0.21316146
[198,]	1300	1	4	0.27004063
[199,]	1300	1	5	0.31442873
[200,]	1300	1	6	0.34739691
[201,]	1352	1	2	0.12244762
[202,]	1352	1	3	0.18006934
[203,]	1352	1	4	0.22812733
[204,]	1352	1	5	0.26565657
[205,]	1352	1	6	0.29359691
[206,]	1414	1	2	0.05239736
[207,]	1414	1	3	0.07706192
[208,]	1414	1	4	0.09766178
[209,]	1414	1	5	0.11384112
[210,]	1414	1	6	0.12613104
[211,]	1418	1	2	0.14800421
[212,]	1418	1	3	0.21764977
[213,]	1418	1	4	0.27572537
[214,]	1418	1	5	0.32104374
[215,]	1418	1	6	0.35469386
[216,]	2002	1	2	0.03588540
[217,]	2002	1	3	0.05278143
[218,]	2002	1	4	0.06690896
[219,]	2002	1	5	0.07805579
[220,]	2002	1	6	0.08665665
[221,]	2022	1	2	0.03288126
[222,]	2022	1	3	0.04836390
[223,]	2022	1	4	0.06131388
[224,]	2022	1	5	0.07154511

[225,]	2022	1	6	0.07947480
[226,]	2027	1	2	0.08664232
[227,]	2027	1	3	0.12741841
[228,]	2027	1	4	0.16144152
[229,]	2027	1	5	0.18805803
[230,]	2027	1	6	0.20799886
[231,]	2031	1	2	0.08122357
[232,]	2031	1	3	0.11945024
[233,]	2031	1	4	0.15134932
[234,]	2031	1	5	0.17631431
[235,]	2031	1	6	0.19504449
[236,]	2068	1	2	0.08448473
[237,]	2068	1	3	0.12424571
[238,]	2068	1	4	0.15742310
[239,]	2068	1	5	0.18338202
[240,]	2068	1	6	0.20284080
[241,]	2072	1	2	0.08627163
[242,]	2072	1	3	0.12687332
[243,]	2072	1	4	0.16075113
[244,]	2072	1	5	0.18725466
[245,]	2072	1	6	0.20711267
[246,]	2073	1	2	0.07439559
[247,]	2073	1	3	0.10940985
[248,]	2073	1	4	0.13863252
[249,]	2073	1	5	0.16151647
[250,]	2073	1	6	0.17872117
[251,]	2093	1	2	0.05702112
[252,]	2093	1	3	0.08386106
[253,]	2093	1	4	0.10627332
[254,]	2093	1	5	0.12386189
[255,]	2093	1	6	0.13718483
[256,]	2099	1	2	0.17537432
[257,]	2099	1	3	0.25789694
[258,]	2099	1	4	0.32670101
[259,]	2099	1	5	0.38036123
[260,]	2099	1	6	0.42012630
[261,]	2130	1	2	0.09823724
[262,]	2130	1	3	0.14446849
[263,]	2130	1	4	0.18303654
[264,]	2130	1	5	0.21318695
[265,]	2130	1	6	0.23571828
[266,]	2133	1	2	0.08913755
[267,]	2133	1	3	0.13108759

[268,]	2133	1	4	0.16608877
[269,]	2133	1	5	0.19346578
[270,]	2133	1	6	0.21396408
[271,]	2151	1	2	0.04579154
[272,]	2151	1	3	0.06734820
[273,]	2151	1	4	0.08535872
[274,]	2151	1	5	0.09952474
[275,]	2151	1	6	0.11033881
[276,]	2168	1	2	0.17733262
[277,]	2168	1	3	0.26077657
[278,]	2168	1	4	0.33034824
[279,]	2168	1	5	0.38460532
[280,]	2168	1	6	0.42480790
[281,]	2187	1	2	0.02518826
[282,]	2187	1	3	0.03705151
[283,]	2187	1	4	0.04698601
[284,]	2187	1	5	0.05487257
[285,]	2187	1	6	0.06108351
[286,]	2191	1	2	0.03304860
[287,]	2191	1	3	0.04860997
[288,]	2191	1	4	0.06162554
[289,]	2191	1	5	0.07190777
[290,]	2191	1	6	0.07987485
[291,]	2195	1	2	0.10883724
[292,]	2195	1	3	0.16005556
[293,]	2195	1	4	0.20277858
[294,]	2195	1	5	0.23615965
[295,]	2195	1	6	0.26105920
[296,]	2208	1	2	0.08272344
[297,]	2208	1	3	0.12165577
[298,]	2208	1	4	0.15414277
[299,]	2208	1	5	0.17956488
[300,]	2208	1	6	0.19863016
[301,]	2210	1	2	0.10903475
[302,]	2210	1	3	0.16034600
[303,]	2210	1	4	0.20314644
[304,]	2210	1	5	0.23658771
[305,]	2210	1	6	0.26153138
[306,]	2225	1	2	0.05146975
[307,]	2225	1	3	0.07569788
[308,]	2225	1	4	0.09593413
[309,]	2225	1	5	0.11183075
[310,]	2225	1	6	0.12391343

[311,]	2233	1	2	0.02956684
[312,]	2233	1	3	0.04349012
[313,]	2233	1	4	0.05514092
[314,]	2233	1	5	0.06436198
[315,]	2233	1	6	0.07155117
[316,]	2247	1	2	0.14450705
[317,]	2247	1	3	0.21250727
[318,]	2247	1	4	0.26921206
[319,]	2247	1	5	0.31346457
[320,]	2247	1	6	0.34633336
[321,]	2301	1	2	0.08280425
[322,]	2301	1	3	0.12177460
[323,]	2301	1	4	0.15429327
[324,]	2301	1	5	0.17974001
[325,]	2301	1	6	0.19882335
[326,]	2304	1	2	0.02690468
[327,]	2304	1	3	0.03957547
[328,]	2304	1	4	0.05018276
[329,]	2304	1	5	0.05859245
[330,]	2304	1	6	0.06518687
[331,]	2308	1	2	0.16102051
[332,]	2308	1	3	0.23678995
[333,]	2308	1	4	0.29996765
[334,]	2308	1	5	0.34925312
[335,]	2308	1	6	0.38581130
[336,]	2357	1	2	0.13034443
[337,]	2357	1	3	0.19168143
[338,]	2357	1	4	0.24283478
[339,]	2357	1	5	0.28277081
[340,]	2357	1	6	0.31247543
[341,]	2360	1	2	0.05076495
[342,]	2360	1	3	0.07466150
[343,]	2360	1	4	0.09462149
[344,]	2360	1	5	0.11030330
[345,]	2360	1	6	0.12222852
[346,]	2378	1	2	0.04863611
[347,]	2378	1	3	0.07153107
[348,]	2378	1	4	0.09065660
[349,]	2378	1	5	0.10568959
[350,]	2378	1	6	0.11713918
[351,]	2380	1	2	0.08731119
[352,]	2380	1	3	0.12840196
[353,]	2380	1	4	0.16268726

[354,]	2380	1	5	0.18950762
[355,]	2380	1	6	0.20959789
[356,]	4003	1	2	0.09739637
[357,]	4003	1	3	0.14323201
[358,]	4003	1	4	0.18147047
[359,]	4003	1	5	0.21136460
[360,]	4003	1	6	0.23370806
[361,]	4012	1	2	0.07108005
[362,]	4012	1	3	0.10453442
[363,]	4012	1	4	0.13245747
[364,]	4012	1	5	0.15433092
[365,]	4012	1	6	0.17079487
[366,]	4029	1	2	0.02518826
[367,]	4029	1	3	0.03705151
[368,]	4029	1	4	0.04698601
[369,]	4029	1	5	0.05487257
[370,]	4029	1	6	0.06108351
[371,]	4030	1	2	0.08604222
[372,]	4030	1	3	0.12653596
[373,]	4030	1	4	0.16032385
[374,]	4030	1	5	0.18675746
[375,]	4030	1	6	0.20656421
[376,]	4032	1	2	0.07543180
[377,]	4032	1	3	0.11093357
[378,]	4032	1	4	0.14056241
[379,]	4032	1	5	0.16376218
[380,]	4032	1	6	0.18119839
[381,]	4035	1	2	0.17447402
[382,]	4035	1	3	0.25657305
[383,]	4035	1	4	0.32502422
[384,]	4035	1	5	0.37841005
[385,]	4035	1	6	0.41797397
[386,]	4050	1	2	0.06295017
[387,]	4050	1	3	0.09257960
[388,]	4050	1	4	0.11731592
[389,]	4050	1	5	0.13671154
[390,]	4050	1	6	0.15135914
[391,]	4057	1	2	NA
[392,]	4057	1	3	NA
[393,]	4057	1	4	NA
[394,]	4057	1	5	NA
[395,]	4057	1	6	NA
[396,]	4058	1	2	0.20689692

[397,]	4058	1	3	0.30425025
[398,]	4058	1	4	0.38541047
[399,]	4058	1	5	0.44867814
[400,]	4058	1	6	0.49548589
[401,]	4060	1	2	0.20115660
[402,]	4060	1	3	0.29580922
[403,]	4060	1	4	0.37471936
[404,]	4060	1	5	0.43623750
[405,]	4060	1	6	0.48176276
[406,]	4067	1	2	0.05239736
[407,]	4067	1	3	0.07706192
[408,]	4067	1	4	0.09766178
[409,]	4067	1	5	0.11384112
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[833,]	4893	1	4	0.27250496
[834,]	4893	1	5	0.31729633
[835,]	4893	1	6	0.35056013
[836,]	4945	1	2	0.06463032
[837,]	4945	1	3	0.09505023
[838,]	4945	1	4	0.12044513
[839,]	4945	1	5	0.14035283
[840,]	4945	1	6	0.15537580
[841,]	4985	1	2	0.14116145
[842,]	4985	1	3	0.20758763
[843,]	4985	1	4	0.26298101
[844,]	4985	1	5	0.30621385
[845,]	4985	1	6	0.33833517
[846,]	5023	1	2	0.04370424
[847,]	5023	1	3	0.06427887
[848,]	5023	1	4	0.08147121
[849,]	5023	1	5	0.09500107
[850,]	5023	1	6	0.10534879
[851,]	5082	1	2	0.05011966
[852,]	5082	1	3	0.07371260
[853,]	5082	1	4	0.09341965
[854,]	5082	1	5	0.10890479
[855,]	5082	1	6	0.12068583
[856,]	5093	1	2	0.05060135
[857,]	5093	1	3	0.07442092
[858,]	5093	1	4	0.09431678
[859,]	5093	1	5	0.10994873
[860,]	5093	1	6	0.12183739
[861,]	5097	1	2	0.08207542
[862,]	5097	1	3	0.12070287
[863,]	5097	1	4	0.15293586
[864,]	5097	1	5	0.17816047
[865,]	5097	1	6	0.19708097
[866,]	5118	1	2	0.07228828
[867,]	5118	1	3	0.10631110
[868,]	5118	1	4	0.13470775
[869,]	5118	1	5	0.15694944

[870,]	5118	1	6	0.17368333
[871,]	5132	1	2	0.05751045
[872,]	5132	1	3	0.08458061
[873,]	5132	1	4	0.10718468
[874,]	5132	1	5	0.12492239
[875,]	5132	1	6	0.13835466
[876,]	5142	1	2	0.11317662
[877,]	5142	1	3	0.16643653
[878,]	5142	1	4	0.21086049
[879,]	5142	1	5	0.24556411
[880,]	5142	1	6	0.27143315

```
# Computing AUC for 2 class joint latent class mixed model for ptau
auc_jlcmm_m1_ptau <- jlcmm_timeAUC(jlcmm_m1_ptau, newdata,
                                   "RID", "time_years", "time_to_event", "event",
                                   1, times)
```

	RID	landmark	horizon	pred
[1,]	31	1	2	0.120134396
[2,]	31	1	3	0.177164675
[3,]	31	1	4	0.226055689
[4,]	31	1	5	0.265754912
[5,]	31	1	6	0.296737669
[6,]	74	1	2	0.074887693
[7,]	74	1	3	0.110441258
[8,]	74	1	4	0.140932992
[9,]	74	1	5	0.165734052
[10,]	74	1	6	0.185206423
[11,]	89	1	2	0.098797485
[12,]	89	1	3	0.145700027
[13,]	89	1	4	0.185914524
[14,]	89	1	5	0.218588246
[15,]	89	1	6	0.244143069
[16,]	120	1	2	0.100370700
[17,]	120	1	3	0.148019981
[18,]	120	1	4	0.188874215
[19,]	120	1	5	0.222065942
[20,]	120	1	6	0.248020978
[21,]	127	1	2	0.105581621
[22,]	127	1	3	0.155704308
[23,]	127	1	4	0.198677530
[24,]	127	1	5	0.233585031

[25,]	127	1	6	0.260865683
[26,]	150	1	2	0.179969745
[27,]	150	1	3	0.265401359
[28,]	150	1	4	0.338624031
[29,]	150	1	5	0.398024952
[30,]	150	1	6	0.444229326
[31,]	200	1	2	0.028104393
[32,]	200	1	3	0.041451884
[33,]	200	1	4	0.052919492
[34,]	200	1	5	0.062316439
[35,]	200	1	6	0.069887525
[36,]	259	1	2	0.182276878
[37,]	259	1	3	0.268803591
[38,]	259	1	4	0.342964443
[39,]	259	1	5	0.403125023
[40,]	259	1	6	0.449916313
[41,]	260	1	2	0.030019424
[42,]	260	1	3	0.044275900
[43,]	260	1	4	0.056522242
[44,]	260	1	5	0.066549743
[45,]	260	1	6	0.074607997
[46,]	272	1	2	0.025164294
[47,]	272	1	3	0.037116244
[48,]	272	1	4	0.047388279
[49,]	272	1	5	0.055817154
[50,]	272	1	6	0.062640303
[51,]	359	1	2	0.003500999
[52,]	359	1	3	0.005170289
[53,]	359	1	4	0.006633085
[54,]	359	1	5	0.007928992
[55,]	359	1	6	0.009241178
[56,]	377	1	2	0.164429112
[57,]	377	1	3	0.242484238
[58,]	377	1	4	0.309387412
[59,]	377	1	5	0.363671344
[60,]	377	1	6	0.405922310
[61,]	384	1	2	0.085489089
[62,]	384	1	3	0.126074692
[63,]	384	1	4	0.160877415
[64,]	384	1	5	0.189169145
[65,]	384	1	6	0.211338423
[66,]	419	1	2	0.087565548
[67,]	419	1	3	0.129136759

[68,]	419	1	4	0.164783861
[69,]	419	1	5	0.193759297
[70,]	419	1	6	0.216456809
[71,]	498	1	2	0.059127516
[72,]	498	1	3	0.087200385
[73,]	498	1	4	0.111283345
[74,]	498	1	5	0.130895127
[75,]	498	1	6	0.146358239
[76,]	519	1	2	0.009149307
[77,]	519	1	3	0.013499614
[78,]	519	1	4	0.017259258
[79,]	519	1	5	0.020414955
[80,]	519	1	6	0.023164025
[81,]	522	1	2	0.037577255
[82,]	522	1	3	0.055421117
[83,]	522	1	4	0.070740802
[84,]	522	1	5	0.083256833
[85,]	522	1	6	0.093237737
[86,]	548	1	2	0.160062834
[87,]	548	1	3	0.236045471
[88,]	548	1	4	0.301173126
[89,]	548	1	5	0.354019395
[90,]	548	1	6	0.395159616
[91,]	555	1	2	0.170047564
[92,]	555	1	3	0.250769533
[93,]	555	1	4	0.319957414
[94,]	555	1	5	0.376091306
[95,]	555	1	6	0.419771560
[96,]	637	1	2	0.100400621
[97,]	637	1	3	0.148064103
[98,]	637	1	4	0.188930505
[99,]	637	1	5	0.222132083
[100,]	637	1	6	0.248094731
[101,]	668	1	2	0.002885537
[102,]	668	1	3	0.004262694
[103,]	668	1	4	0.005475216
[104,]	668	1	5	0.006568472
[105,]	668	1	6	0.007724090
[106,]	671	1	2	0.162292942
[107,]	671	1	3	0.239334118
[108,]	671	1	4	0.305368632
[109,]	671	1	5	0.358949198
[110,]	671	1	6	0.400656740

[111,]	684	1	2 0.023682505
[112,]	684	1	3 0.034931111
[113,]	684	1	4 0.044600586
[114,]	684	1	5 0.052541560
[115,]	684	1	6 0.058987754
[116,]	685	1	2 0.107295083
[117,]	685	1	3 0.158231078
[118,]	685	1	4 0.201901068
[119,]	685	1	5 0.237372753
[120,]	685	1	6 0.265089295
[121,]	686	1	2 0.041281839
[122,]	686	1	3 0.060884112
[123,]	686	1	4 0.077710243
[124,]	686	1	5 0.091446065
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[126,]	908	1	2 0.049906256
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[128,]	908	1	4 0.093935372
[129,]	908	1	5 0.110510914
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[131,]	920	1	2 0.108303710
[132,]	920	1	3 0.159718459
[133,]	920	1	4 0.203798600
[134,]	920	1	5 0.239602391
[135,]	920	1	6 0.267575520
[136,]	926	1	2 0.100847676
[137,]	926	1	3 0.148723357
[138,]	926	1	4 0.189771550
[139,]	926	1	5 0.223120328
[140,]	926	1	6 0.249196703
[141,]	951	1	2 0.210806144
[142,]	951	1	3 0.310874504
[143,]	951	1	4 0.396636599
[144,]	951	1	5 0.466190872
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[146,]	984	1	2 0.136157373
[147,]	984	1	3 0.200793089
[148,]	984	1	4 0.256199742
[149,]	984	1	5 0.301174774
[150,]	984	1	6 0.336233644
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[152,]	1023	1	3 0.134856545
[153,]	1023	1	4 0.172080903

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[155,]	1023	1	6	0.226017692
[156,]	1045	1	2	0.044124564
[157,]	1045	1	3	0.065076160
[158,]	1045	1	4	0.083058266
[159,]	1045	1	5	0.097730099
[160,]	1045	1	6	0.109376584
[161,]	1072	1	2	0.064716643
[162,]	1072	1	3	0.095442436
[163,]	1072	1	4	0.121798179
[164,]	1072	1	5	0.143250266
[165,]	1072	1	6	0.160135205
[166,]	1155	1	2	0.045432870
[167,]	1155	1	3	0.067005463
[168,]	1155	1	4	0.085519583
[169,]	1155	1	5	0.100622195
[170,]	1155	1	6	0.112601502
[171,]	1186	1	2	0.064993842
[172,]	1186	1	3	0.095851210
[173,]	1186	1	4	0.122319674
[174,]	1186	1	5	0.143863033
[175,]	1186	1	6	0.160818489
[176,]	1206	1	2	0.097497901
[177,]	1206	1	3	0.143783585
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[179,]	1206	1	5	0.215715428
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[196,]	1300	1	2	0.122801684

[197,]	1300	1	3	0.181098013
[198,]	1300	1	4	0.231073662
[199,]	1300	1	5	0.271651131
[200,]	1300	1	6	0.303312424
[201,]	1352	1	2	0.189424901
[202,]	1352	1	3	0.279344481
[203,]	1352	1	4	0.356412031
[204,]	1352	1	5	0.418926206
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[206,]	1414	1	2	0.029583846
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[209,]	1414	1	5	0.065586868
[210,]	1414	1	6	0.073534315
[211,]	1418	1	2	0.161392015
[212,]	1418	1	3	0.238005558
[213,]	1418	1	4	0.303673716
[214,]	1418	1	5	0.356957637
[215,]	1418	1	6	0.398435991
[216,]	2002	1	2	0.027235453
[217,]	2002	1	3	0.040170496
[218,]	2002	1	4	0.051284754
[219,]	2002	1	5	0.060395590
[220,]	2002	1	6	0.067745625
[221,]	2022	1	2	0.108131620
[222,]	2022	1	3	0.159464685
[223,]	2022	1	4	0.203474847
[224,]	2022	1	5	0.239221975
[225,]	2022	1	6	0.267151325
[226,]	2027	1	2	0.034485530
[227,]	2027	1	3	0.050861880
[228,]	2027	1	4	0.064924336
[229,]	2027	1	5	0.076422370
[230,]	2027	1	6	0.085616765
[231,]	2031	1	2	0.115852554
[232,]	2031	1	3	0.170850423
[233,]	2031	1	4	0.218000253
[234,]	2031	1	5	0.256289615
[235,]	2031	1	6	0.286183107
[236,]	2068	1	2	0.086655472
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[238,]	2068	1	4	0.163071735
[239,]	2068	1	5	0.191747514

[240,]	2068	1	6	0.214213510
[241,]	2072	1	2	0.179793630
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[243,]	2072	1	4	0.338292706
[244,]	2072	1	5	0.397635638
[245,]	2072	1	6	0.443795211
[246,]	2073	1	2	0.016926599
[247,]	2073	1	3	0.024968460
[248,]	2073	1	4	0.031890689
[249,]	2073	1	5	0.037607178
[250,]	2073	1	6	0.042334726
[251,]	2093	1	2	0.070881967
[252,]	2093	1	3	0.104534182
[253,]	2093	1	4	0.133397014
[254,]	2093	1	5	0.156879127
[255,]	2093	1	6	0.175332474
[256,]	2099	1	2	0.080305059
[257,]	2099	1	3	0.118430020
[258,]	2099	1	4	0.151124691
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[264,]	2130	1	5	0.198369642
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[268,]	2133	1	4	0.232293123
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[270,]	2133	1	6	0.304910212
[271,]	2151	1	2	0.042256876
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[273,]	2151	1	4	0.079544581
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[279,]	2168	1	5	0.115471633
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[281,]	2187	1	2	0.112300196
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[285,]	2187	1	6	0.277426693
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[291,]	2195	1	2	0.162317733
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[293,]	2195	1	4	0.305415271
[294,]	2195	1	5	0.359003999
[295,]	2195	1	6	0.400717848
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[297,]	2208	1	3	0.060623228
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[299,]	2208	1	5	0.091054989
[300,]	2208	1	6	0.101933303
[301,]	2210	1	2	0.091083376
[302,]	2210	1	3	0.134324354
[303,]	2210	1	4	0.171401958
[304,]	2210	1	5	0.201535692
[305,]	2210	1	6	0.225128110
[306,]	2225	1	2	0.126342938
[307,]	2225	1	3	0.186320153
[308,]	2225	1	4	0.237735830
[309,]	2225	1	5	0.279479310
[310,]	2225	1	6	0.312041469
[311,]	2233	1	2	0.051561530
[312,]	2233	1	3	0.076043142
[313,]	2233	1	4	0.097049443
[314,]	2233	1	5	0.114170009
[315,]	2233	1	6	0.127708396
[316,]	2247	1	2	0.089975465
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[318,]	2247	1	4	0.169317643
[319,]	2247	1	5	0.199086581
[320,]	2247	1	6	0.222397156
[321,]	2301	1	2	0.047254283
[322,]	2301	1	3	0.069691425
[323,]	2301	1	4	0.088946212
[324,]	2301	1	5	0.104648551
[325,]	2301	1	6	0.117091211

[326,]	2304	1	2	0.042828574
[327,]	2304	1	3	0.063165018
[328,]	2304	1	4	0.080620119
[329,]	2304	1	5	0.094865227
[330,]	2304	1	6	0.106182023
[331,]	2308	1	2	0.145775046
[332,]	2308	1	3	0.214975868
[333,]	2308	1	4	0.274293486
[334,]	2308	1	5	0.322435283
[335,]	2308	1	6	0.359940809
[336,]	2357	1	2	0.080695766
[337,]	2357	1	3	0.119006180
[338,]	2357	1	4	0.151859730
[339,]	2357	1	5	0.178573186
[340,]	2357	1	6	0.199523082
[341,]	2360	1	2	0.047002805
[342,]	2360	1	3	0.069320581
[343,]	2360	1	4	0.088473106
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[355,]	2380	1	6	0.164137503
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[358,]	4003	1	4	0.192398726
[359,]	4003	1	5	0.226207312
[360,]	4003	1	6	0.252638937
[361,]	4012	1	2	0.013241813
[362,]	4012	1	3	0.019534660
[363,]	4012	1	4	0.024958496
[364,]	4012	1	5	0.029461713
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[366,]	4029	1	2	0.111407965
[367,]	4029	1	3	0.164296173
[368,]	4029	1	4	0.209638640

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[785,]	4777	1	6	0.300761009
[786,]	4782	1	2	0.124135153
[787,]	4782	1	3	0.183064424
[788,]	4782	1	4	0.233582319
[789,]	4782	1	5	0.274598853
[790,]	4782	1	6	0.306599369
[791,]	4807	1	2	0.190182328
[792,]	4807	1	3	0.280461428
[793,]	4807	1	4	0.357836981
[794,]	4807	1	5	0.420600550
[795,]	4807	1	6	0.469402921
[796,]	4817	1	2	0.090118784
[797,]	4817	1	3	0.132901910
[798,]	4817	1	4	0.169587269

[799,]	4817	1	5	0.199403396
[800,]	4817	1	6	0.222750430
[801,]	4844	1	2	0.103731701
[802,]	4844	1	3	0.152976308
[803,]	4844	1	4	0.195197271
[804,]	4844	1	5	0.229495658
[805,]	4844	1	6	0.256305705
[806,]	4857	1	2	NA
[807,]	4857	1	3	NA
[808,]	4857	1	4	NA
[809,]	4857	1	5	NA
[810,]	4857	1	6	NA
[811,]	4869	1	2	0.091470546
[812,]	4869	1	3	0.134895297
[813,]	4869	1	4	0.172130341
[814,]	4869	1	5	0.202391557
[815,]	4869	1	6	0.226082468
[816,]	4883	1	2	0.014812593
[817,]	4883	1	3	0.021851024
[818,]	4883	1	4	0.027913606
[819,]	4883	1	5	0.032934027
[820,]	4883	1	6	0.037123789
[821,]	4888	1	2	NA
[822,]	4888	1	3	NA
[823,]	4888	1	4	NA
[824,]	4888	1	5	NA
[825,]	4888	1	6	NA
[826,]	4889	1	2	0.086014756
[827,]	4889	1	3	0.126849872
[828,]	4889	1	4	0.161866354
[829,]	4889	1	5	0.190331168
[830,]	4889	1	6	0.212634171
[831,]	4893	1	2	0.109104976
[832,]	4893	1	3	0.160900052
[833,]	4893	1	4	0.205306023
[834,]	4893	1	5	0.241373643
[835,]	4893	1	6	0.269550607
[836,]	4945	1	2	0.183477097
[837,]	4945	1	3	0.270573504
[838,]	4945	1	4	0.345222417
[839,]	4945	1	5	0.405778187
[840,]	4945	1	6	0.452874803
[841,]	4985	1	2	0.170416812

[842,]	4985	1	3	0.251314048
[843,]	4985	1	4	0.320652081
[844,]	4985	1	5	0.376907554
[845,]	4985	1	6	0.420681741
[846,]	5023	1	2	0.032576266
[847,]	5023	1	3	0.048046369
[848,]	5023	1	4	0.061332435
[849,]	5023	1	5	0.072201814
[850,]	5023	1	6	0.080910508
[851,]	5082	1	2	0.075272271
[852,]	5082	1	3	0.111008379
[853,]	5082	1	4	0.141656499
[854,]	5082	1	5	0.166584187
[855,]	5082	1	6	0.186154391
[856,]	5093	1	2	0.075380603
[857,]	5093	1	3	0.111168132
[858,]	5093	1	4	0.141860305
[859,]	5093	1	5	0.166823663
[860,]	5093	1	6	0.186421425
[861,]	5097	1	2	0.082965696
[862,]	5097	1	3	0.122353551
[863,]	5097	1	4	0.156130153
[864,]	5097	1	5	0.183591018
[865,]	5097	1	6	0.205118366
[866,]	5118	1	2	0.146718331
[867,]	5118	1	3	0.216366891
[868,]	5118	1	4	0.276068090
[869,]	5118	1	5	0.324520477
[870,]	5118	1	6	0.362265967
[871,]	5132	1	2	0.147262357
[872,]	5132	1	3	0.217169144
[873,]	5132	1	4	0.277091568
[874,]	5132	1	5	0.325723085
[875,]	5132	1	6	0.363606971
[876,]	5142	1	2	0.126194747
[877,]	5142	1	3	0.186101621
[878,]	5142	1	4	0.237457037
[879,]	5142	1	5	0.279151724
[880,]	5142	1	6	0.311676184

ON IN-SAMPLE DATA

Computing AUC for 2 class multivariate joint latent class mixed model for ptau & NfL

```
# auc_mult_jlcmm_m1 <- jlcmm_timeAUC(mult_jlcmm_m1, jlcmm_data,
#                                   "RID", "time_years", "time_to_event", "event",
#                                   1, times)
```

11 Time-depending Brier score

11.1 Time-dependent Brier score for fitted Cox PH models

```
# We specify Cox models for comparison
cox_models <- list(
  "Cox p-tau" = surv_model_ptau_bl_final,
  "Cox NfL" = surv_model_nfl_bl_final,
  "Cox both" = surv_model_both_bl_final
)

# We compute Brier score using pec for Cox models
brier_cox <- pec(
  object = cox_models,
  formula = Surv(time_to_event, event) ~
    age_at_baseline + PTGENDER + APOE4_status,
  data = surv_data_out,
  times = times,
  cens.model = "cox"
)
```

```
Registered S3 method overwritten by 'cmprsk':
  method      from
plot.cuminc lcmm
```

```
cox_brier <- lapply(brier_cox$AppErr, function(x)
  approx(brier_cox$time, x, xout = times)$y
)
```

```
Warning in regularize.values(x, y, ties, missing(ties), na.rm = na.rm):
collapsing to unique 'x' values
Warning in regularize.values(x, y, ties, missing(ties), na.rm = na.rm):
collapsing to unique 'x' values
Warning in regularize.values(x, y, ties, missing(ties), na.rm = na.rm):
```

```
collapsing to unique 'x' values
Warning in regularize.values(x, y, ties, missing(ties), na.rm = na.rm):
collapsing to unique 'x' values
```

11.2 Time-dependent Brier score for fitted joint models

```
# We define a function to compute time-dependent Brier score for joint models using tvBrier
get_jm_brier <- function(model, times) {

  briers <- sapply(times, function(t) {

    tmp <- tvBrier(
      model,
      newdata = newdata,
      Tstart = 1,
      Thoriz = t
    )

    return(tmp$Brier)
  })

  return(as.numeric(briers))
}

# We compute time-dependent Brier score for the joint model including p-tau181
brier_ptau_jm <- get_jm_brier(joint_model_ptau, times)
```

```
Warning in br(Thoriz): there are fewer than 5 subjects with an event in the interval [1.0000
```

```
brier_ptau_jm
```

```
[1] 0.01768942 0.06845960 0.09792387 0.12350775 0.13774459
```

```
# We compute time-dependent Brier score for the joint model including Nfl
brier_nfl_jm <- get_jm_brier(joint_model_nfl, times)
```

```
Warning in br(Thoriz): there are fewer than 5 subjects with an event in the interval [1.0000
```

```
brier_nfl_jm
```

```
[1] 0.01684773 0.06810325 0.09608422 0.11971086 0.13539816
```

```
# We compute time-dependent Brier score for the multivariate joint model  
brier_multi_jm <- get_jm_brier(multi_joint_model, times)
```

Warning in br(Thoriz): there are fewer than 5 subjects with an event in the interval [1.0000

```
brier_multi_jm
```

```
[1] 0.01681539 0.06552209 0.09200193 0.11455822 0.12657805
```

11.3 Time-dependent Brier score for latent class joint models

```
brier_jlcmm <- function(model, data, ID, time_var, surv_time_var, event_var, times, landmark, ...)  
  
# Initiating a data frame to save the results in at each horizon time  
output <- data.frame(  
  times = times,  
  Brier = NA,  
  IBS = NA  
)  
  
# Calculating the survival predictions previously defined function  
if(class(model) == "mpjlcmm") {  
  dfpred <- predict_mjlcmm(  
    model = model,  
    data = data,  
    ID = ID,  
    landmark = landmark,  
    times = times  
  )  
}  
  
# Calculating the dynamic survival predictions from joint models using dynpred() in the lcm  
if(class(model) == "Jointlcmm") {  
  dfpred <- dynpred(  
    model = model,  
    data = data,  
    ID = ID,  
    landmark = landmark,  
    times = times  
  )  
}
```

```

    model = model,
    newdata = newdata,
    var.time = time_var,
    horizon = times,
    landmark = landmark
  )$pred
}

# Fetching the true event times and events
dftrue <- unique(data[c(ID, surv_time_var, event_var)])
names(dftrue) <- c(ID, "time", "event")

# Merging prediction df with true df
merged <- merge(dfpred, dftrue, by = ID)

# Saving censoring indicator (1 - event indicator), times remain the same as event/censored
merged$cens_event <- 1 - merged$event

# Estimating the censoring distribution with the Kaplan-Meier estimate
km_cens <- survfit(Surv(merged$time, merged$cens_event) ~ 1)

# Function to evaluate estimate of censoring probability
Ghat <- function(t) {
  sapply(t, function(x) {
    idx <- max(which(km_cens$time <= x))
    if (length(idx) == 0) return(1)
    km_cens$surv[idx]
  })
}

brier_one_time <- function(t) {

# Saving all rows belonging to horizon t
sub <- merged[merged$horizon == t, ]
# Saving the observed survival status
status_t <- as.numeric(sub$time > t)
# Saving the predicted survival probability: 1 - pred since dynpred outputs cumulative ri
pred_t <- 1 - sub$pred

# Calculating the IPCW weights
w <- ifelse(sub$time > t,
            1 / Ghat(t),

```



```

        ifelse(sub$event == 1 & sub$time <= t,
              1 / Ghat(sub$time),
              0))

#    Calculating the IPCW-adjusted Brier score
sum(w * (status_t - pred_t)^2, na.rm = TRUE) / sum(w, na.rm = TRUE)

}

#    Calculating the Brier score for each horizon
brier_scores <- sapply(times, brier_one_time)
#    Saving to output
output$Brier <- brier_scores

#    Integrated Brier Score (IBS)
IBS <- sum(diff(times) *
          (brier_scores[-1] + brier_scores[-length(brier_scores)]) / 2) /
      (max(times) - landmark)
#    Saving to output
output$IBS <- IBS

return(output)
}

```

```

# Computing Brier scores for 2 class jlcmm for NfL
brier_jlcmm_m1_nfl <- brier_jlcmm(
  model = jlcmm_m1_nfl,
  data = newdata,
  ID = "RID",
  time_var = "time_years",
  surv_time_var = "time_to_event",
  event_var = "event",
  times = times,
  landmark = 1
)

```

	RID	landmark	horizon	pred
[1,]	31	1	2	0.14260269
[2,]	31	1	3	0.20970694
[3,]	31	1	4	0.26566526
[4,]	31	1	5	0.30933735
[5,]	31	1	6	0.34178067

[6,]	74	1	2 0.10322711
[7,]	74	1	3 0.15180598
[8,]	74	1	4 0.19232996
[9,]	74	1	5 0.22400118
[10,]	74	1	6 0.24764733
[11,]	89	1	2 0.07932390
[12,]	89	1	3 0.11665682
[13,]	89	1	4 0.14781128
[14,]	89	1	5 0.17219728
[15,]	89	1	6 0.19050305
[16,]	120	1	2 0.07738928
[17,]	120	1	3 0.11381201
[18,]	120	1	4 0.14420813
[19,]	120	1	5 0.16800450
[20,]	120	1	6 0.18587804
[21,]	127	1	2 0.05729456
[22,]	127	1	3 0.08426314
[23,]	127	1	4 0.10678258
[24,]	127	1	5 0.12445449
[25,]	127	1	6 0.13783853
[26,]	150	1	2 0.07669308
[27,]	150	1	3 0.11278826
[28,]	150	1	4 0.14291149
[29,]	150	1	5 0.16649568
[30,]	150	1	6 0.18421367
[31,]	200	1	2 0.08095357
[32,]	200	1	3 0.11905322
[33,]	200	1	4 0.15084646
[34,]	200	1	5 0.17572916
[35,]	200	1	6 0.19439902
[36,]	259	1	2 0.09769384
[37,]	259	1	3 0.14366943
[38,]	259	1	4 0.18202449
[39,]	259	1	5 0.21200928
[40,]	259	1	6 0.23441920
[41,]	260	1	2 0.09120255
[42,]	260	1	3 0.13412413
[43,]	260	1	4 0.16993475
[44,]	260	1	5 0.19794112
[45,]	260	1	6 0.21890078
[46,]	272	1	2 0.07289131
[47,]	272	1	3 0.10719784
[48,]	272	1	4 0.13583086

[49,]	272	1	5	0.15825634
[50,]	272	1	6	0.17512496
[51,]	359	1	2	0.07875047
[52,]	359	1	3	0.11581361
[53,]	359	1	4	0.14674330
[54,]	359	1	5	0.17095453
[55,]	359	1	6	0.18913219
[56,]	377	1	2	0.13999088
[57,]	377	1	3	0.20586633
[58,]	377	1	4	0.26080088
[59,]	377	1	5	0.30367695
[60,]	377	1	6	0.33553675
[61,]	384	1	2	0.07792372
[62,]	384	1	3	0.11459790
[63,]	384	1	4	0.14520351
[64,]	384	1	5	0.16916277
[65,]	384	1	6	0.18715571
[66,]	419	1	2	0.08798050
[67,]	419	1	3	0.12938617
[68,]	419	1	4	0.16393382
[69,]	419	1	5	0.19095818
[70,]	419	1	6	0.21119798
[71,]	498	1	2	0.04776483
[72,]	498	1	3	0.07024988
[73,]	498	1	4	0.08903389
[74,]	498	1	5	0.10380133
[75,]	498	1	6	0.11505626
[76,]	519	1	2	0.07492813
[77,]	519	1	3	0.11019294
[78,]	519	1	4	0.13962435
[79,]	519	1	5	0.16267062
[80,]	519	1	6	0.17999429
[81,]	522	1	2	0.08979522
[82,]	522	1	3	0.13205467
[83,]	522	1	4	0.16731365
[84,]	522	1	5	0.19489110
[85,]	522	1	6	0.21553633
[86,]	548	1	2	0.15549435
[87,]	548	1	3	0.22866386
[88,]	548	1	4	0.28967543
[89,]	548	1	5	0.33727664
[90,]	548	1	6	0.37260018
[91,]	555	1	2	0.13286449

[92,]	555	1	3	0.19538713
[93,]	555	1	4	0.24752830
[94,]	555	1	5	0.28823239
[95,]	555	1	6	0.31850003
[96,]	637	1	2	0.15511321
[97,]	637	1	3	0.22810340
[98,]	637	1	4	0.28896557
[99,]	637	1	5	0.33645062
[100,]	637	1	6	0.37168901
[101,]	668	1	2	0.06965802
[102,]	668	1	3	0.10244336
[103,]	668	1	4	0.12980900
[104,]	668	1	5	0.15124904
[105,]	668	1	6	0.16739529
[106,]	671	1	2	0.17821268
[107,]	671	1	3	0.26207068
[108,]	671	1	4	0.33198732
[109,]	671	1	5	0.38651261
[110,]	671	1	6	0.42691182
[111,]	684	1	2	0.20290239
[112,]	684	1	3	0.29837637
[113,]	684	1	4	0.37797082
[114,]	684	1	5	0.44002104
[115,]	684	1	6	0.48593634
[116,]	685	1	2	0.16430579
[117,]	685	1	3	0.24162088
[118,]	685	1	4	0.30608634
[119,]	685	1	5	0.35637309
[120,]	685	1	6	0.39366526
[121,]	686	1	2	0.19502335
[122,]	686	1	3	0.28679041
[123,]	686	1	4	0.36329646
[124,]	686	1	5	0.42294531
[125,]	686	1	6	0.46710030
[126,]	908	1	2	0.12766646
[127,]	908	1	3	0.18774353
[128,]	908	1	4	0.23784718
[129,]	908	1	5	0.27696702
[130,]	908	1	6	0.30607333
[131,]	920	1	2	0.12295067
[132,]	920	1	3	0.18080907
[133,]	920	1	4	0.22906424
[134,]	920	1	5	0.26674680

[135,]	920	1	6	0.29479953
[136,]	926	1	2	0.15126697
[137,]	926	1	3	0.22244758
[138,]	926	1	4	0.28180210
[139,]	926	1	5	0.32811489
[140,]	926	1	6	0.36249396
[141,]	951	1	2	0.20373940
[142,]	951	1	3	0.29960717
[143,]	951	1	4	0.37952972
[144,]	951	1	5	0.44183504
[145,]	951	1	6	0.48793734
[146,]	984	1	2	0.14824988
[147,]	984	1	3	0.21801101
[148,]	984	1	4	0.27618291
[149,]	984	1	5	0.32157615
[150,]	984	1	6	0.35528115
[151,]	1023	1	2	0.13999088
[152,]	1023	1	3	0.20586633
[153,]	1023	1	4	0.26080088
[154,]	1023	1	5	0.30367695
[155,]	1023	1	6	0.33553675
[156,]	1045	1	2	0.10018909
[157,]	1045	1	3	0.14733864
[158,]	1045	1	4	0.18667178
[159,]	1045	1	5	0.21741707
[160,]	1045	1	6	0.24038447
[161,]	1072	1	2	0.03827159
[162,]	1072	1	3	0.05629028
[163,]	1072	1	4	0.07135315
[164,]	1072	1	5	0.08322724
[165,]	1072	1	6	0.09236122
[166,]	1155	1	2	0.12020017
[167,]	1155	1	3	0.17676451
[168,]	1155	1	4	0.22394154
[169,]	1155	1	5	0.26078580
[170,]	1155	1	6	0.28822402
[171,]	1186	1	2	0.17291137
[172,]	1186	1	3	0.25427521
[173,]	1186	1	4	0.32211385
[174,]	1186	1	5	0.37502342
[175,]	1186	1	6	0.41423822
[176,]	1206	1	2	0.13907285
[177,]	1206	1	3	0.20451639

[178,]	1206	1	4	0.25909110
[179,]	1206	1	5	0.30168737
[180,]	1206	1	6	0.33334207
[181,]	1222	1	2	0.08132876
[182,]	1222	1	3	0.11960492
[183,]	1222	1	4	0.15154523
[184,]	1222	1	5	0.17654228
[185,]	1222	1	6	0.19529596
[186,]	1280	1	2	0.09332179
[187,]	1280	1	3	0.13724043
[188,]	1280	1	4	0.17388174
[189,]	1280	1	5	0.20253402
[190,]	1280	1	6	0.22396715
[191,]	1286	1	2	0.07884504
[192,]	1286	1	3	0.11595267
[193,]	1286	1	4	0.14691942
[194,]	1286	1	5	0.17115948
[195,]	1286	1	6	0.18935826
[196,]	1300	1	2	0.14495193
[197,]	1300	1	3	0.21316146
[198,]	1300	1	4	0.27004063
[199,]	1300	1	5	0.31442873
[200,]	1300	1	6	0.34739691
[201,]	1352	1	2	0.12244762
[202,]	1352	1	3	0.18006934
[203,]	1352	1	4	0.22812733
[204,]	1352	1	5	0.26565657
[205,]	1352	1	6	0.29359691
[206,]	1414	1	2	0.05239736
[207,]	1414	1	3	0.07706192
[208,]	1414	1	4	0.09766178
[209,]	1414	1	5	0.11384112
[210,]	1414	1	6	0.12613104
[211,]	1418	1	2	0.14800421
[212,]	1418	1	3	0.21764977
[213,]	1418	1	4	0.27572537
[214,]	1418	1	5	0.32104374
[215,]	1418	1	6	0.35469386
[216,]	2002	1	2	0.03588540
[217,]	2002	1	3	0.05278143
[218,]	2002	1	4	0.06690896
[219,]	2002	1	5	0.07805579
[220,]	2002	1	6	0.08665665

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[224,]	2022	1	5	0.07154511
[225,]	2022	1	6	0.07947480
[226,]	2027	1	2	0.08664232
[227,]	2027	1	3	0.12741841
[228,]	2027	1	4	0.16144152
[229,]	2027	1	5	0.18805803
[230,]	2027	1	6	0.20799886
[231,]	2031	1	2	0.08122357
[232,]	2031	1	3	0.11945024
[233,]	2031	1	4	0.15134932
[234,]	2031	1	5	0.17631431
[235,]	2031	1	6	0.19504449
[236,]	2068	1	2	0.08448473
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[239,]	2068	1	5	0.18338202
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[250,]	2073	1	6	0.17872117
[251,]	2093	1	2	0.05702112
[252,]	2093	1	3	0.08386106
[253,]	2093	1	4	0.10627332
[254,]	2093	1	5	0.12386189
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[257,]	2099	1	3	0.25789694
[258,]	2099	1	4	0.32670101
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[263,]	2130	1	4	0.18303654

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[647,]	4539	1	3	0.12039002
[648,]	4539	1	4	0.15253961
[649,]	4539	1	5	0.17769938
[650,]	4539	1	6	0.19657234

[651,]	4545	1	2	0.08862386
[652,]	4545	1	3	0.13033221
[653,]	4545	1	4	0.16513204
[654,]	4545	1	5	0.19235248
[655,]	4545	1	6	0.21273602
[656,]	4547	1	2	0.11501127
[657,]	4547	1	3	0.16913434
[658,]	4547	1	4	0.21427744
[659,]	4547	1	5	0.24954023
[660,]	4547	1	6	0.27581917
[661,]	4553	1	2	0.07529583
[662,]	4553	1	3	0.11073363
[663,]	4553	1	4	0.14030917
[664,]	4553	1	5	0.16346750
[665,]	4553	1	6	0.18087332
[666,]	4556	1	2	0.09086519
[667,]	4556	1	3	0.13362805
[668,]	4556	1	4	0.16930643
[669,]	4556	1	5	0.19720999
[670,]	4556	1	6	0.21809427
[671,]	4566	1	2	0.09172539
[672,]	4566	1	3	0.13489295
[673,]	4566	1	4	0.17090851
[674,]	4566	1	5	0.19907424
[675,]	4566	1	6	0.22015071
[676,]	4577	1	2	0.11343965
[677,]	4577	1	3	0.16682331
[678,]	4577	1	4	0.21135037
[679,]	4577	1	5	0.24613417
[680,]	4577	1	6	0.27206197
[681,]	4578	1	2	0.05208894
[682,]	4578	1	3	0.07660839
[683,]	4578	1	4	0.09708735
[684,]	4578	1	5	0.11317269
[685,]	4578	1	6	0.12539370
[686,]	4586	1	2	0.05054686
[687,]	4586	1	3	0.07434079
[688,]	4586	1	4	0.09421529
[689,]	4586	1	5	0.10983063
[690,]	4586	1	6	0.12170712
[691,]	4587	1	2	0.09337747
[692,]	4587	1	3	0.13732230
[693,]	4587	1	4	0.17398544

[694,]	4587	1	5	0.20265468
[695,]	4587	1	6	0.22410025
[696,]	4607	1	2	0.04715653
[697,]	4607	1	3	0.06935539
[698,]	4607	1	4	0.08790095
[699,]	4607	1	5	0.10248299
[700,]	4607	1	6	0.11360202
[701,]	4611	1	2	0.08892604
[702,]	4611	1	3	0.13077657
[703,]	4611	1	4	0.16569485
[704,]	4611	1	5	0.19300739
[705,]	4611	1	6	0.21345844
[706,]	4621	1	2	0.12668316
[707,]	4621	1	3	0.18629761
[708,]	4621	1	4	0.23601584
[709,]	4621	1	5	0.27483599
[710,]	4621	1	6	0.30372262
[711,]	4630	1	2	0.04590275
[712,]	4630	1	3	0.06751172
[713,]	4630	1	4	0.08556583
[714,]	4630	1	5	0.09976575
[715,]	4630	1	6	0.11060466
[716,]	4635	1	2	0.09963288
[717,]	4635	1	3	0.14652074
[718,]	4635	1	4	0.18563586
[719,]	4635	1	5	0.21621163
[720,]	4635	1	6	0.23905477
[721,]	4649	1	2	0.04712597
[722,]	4649	1	3	0.06931045
[723,]	4649	1	4	0.08784404
[724,]	4649	1	5	0.10241677
[725,]	4649	1	6	0.11352897
[726,]	4674	1	2	0.03745489
[727,]	4674	1	3	0.05508934
[728,]	4674	1	4	0.06983208
[729,]	4674	1	5	0.08145726
[730,]	4674	1	6	0.09040878
[731,]	4711	1	2	0.12428132
[732,]	4711	1	3	0.18276576
[733,]	4711	1	4	0.23154251
[734,]	4711	1	5	0.26963063
[735,]	4711	1	6	0.29798065
[736,]	4712	1	2	NA

[737,]	4712	1	3	NA
[738,]	4712	1	4	NA
[739,]	4712	1	5	NA
[740,]	4712	1	6	NA
[741,]	4713	1	2	0.08902540
[742,]	4713	1	3	0.13092267
[743,]	4713	1	4	0.16587989
[744,]	4713	1	5	0.19322271
[745,]	4713	1	6	0.21369596
[746,]	4720	1	2	0.11687218
[747,]	4720	1	3	0.17187078
[748,]	4720	1	4	0.21774331
[749,]	4720	1	5	0.25357327
[750,]	4720	1	6	0.28026796
[751,]	4736	1	2	0.06772504
[752,]	4736	1	3	0.09960094
[753,]	4736	1	4	0.12620890
[754,]	4736	1	5	0.14705981
[755,]	4736	1	6	0.16277419
[756,]	4739	1	2	0.05925870
[757,]	4739	1	3	0.08715138
[758,]	4739	1	4	0.11044072
[759,]	4739	1	5	0.12871126
[760,]	4739	1	6	0.14253412
[761,]	4746	1	2	0.21125031
[762,]	4746	1	3	0.31065182
[763,]	4746	1	4	0.39351847
[764,]	4746	1	5	0.45811297
[765,]	4746	1	6	0.50589334
[766,]	4750	1	2	0.20089406
[767,]	4750	1	3	0.29542317
[768,]	4750	1	4	0.37423041
[769,]	4750	1	5	0.43566853
[770,]	4750	1	6	0.48113514
[771,]	4764	1	2	0.10835030
[772,]	4764	1	3	0.15933953
[773,]	4764	1	4	0.20187168
[774,]	4764	1	5	0.23510435
[775,]	4764	1	6	0.25989511
[776,]	4765	1	2	0.16530428
[777,]	4765	1	3	0.24308915
[778,]	4765	1	4	0.30794599
[779,]	4765	1	5	0.35853707

[780,]	4765	1	6	0.39605231
[781,]	4777	1	2	0.20251836
[782,]	4777	1	3	0.29781167
[783,]	4777	1	4	0.37725559
[784,]	4777	1	5	0.43918877
[785,]	4777	1	6	0.48501827
[786,]	4782	1	2	0.05763689
[787,]	4782	1	3	0.08476654
[788,]	4782	1	4	0.10742017
[789,]	4782	1	5	0.12519642
[790,]	4782	1	6	0.13865694
[791,]	4807	1	2	0.10897891
[792,]	4807	1	3	0.16026388
[793,]	4807	1	4	0.20304243
[794,]	4807	1	5	0.23646668
[795,]	4807	1	6	0.26139788
[796,]	4817	1	2	0.05363098
[797,]	4817	1	3	0.07887593
[798,]	4817	1	4	0.09995933
[799,]	4817	1	5	0.11651466
[800,]	4817	1	6	0.12908019
[801,]	4844	1	2	0.11448865
[802,]	4844	1	3	0.16836584
[803,]	4844	1	4	0.21330409
[804,]	4844	1	5	0.24840759
[805,]	4844	1	6	0.27456976
[806,]	4857	1	2	NA
[807,]	4857	1	3	NA
[808,]	4857	1	4	NA
[809,]	4857	1	5	NA
[810,]	4857	1	6	NA
[811,]	4869	1	2	0.07828332
[812,]	4869	1	3	0.11512668
[813,]	4869	1	4	0.14587325
[814,]	4869	1	5	0.16994210
[815,]	4869	1	6	0.18801539
[816,]	4883	1	2	0.07887827
[817,]	4883	1	3	0.11600153
[818,]	4883	1	4	0.14698131
[819,]	4883	1	5	0.17123149
[820,]	4883	1	6	0.18943770
[821,]	4888	1	2	NA
[822,]	4888	1	3	NA

[823,]	4888	1	4	NA
[824,]	4888	1	5	NA
[825,]	4888	1	6	NA
[826,]	4889	1	2	0.14317349
[827,]	4889	1	3	0.21054630
[828,]	4889	1	4	0.26672836
[829,]	4889	1	5	0.31057443
[830,]	4889	1	6	0.34314528
[831,]	4893	1	2	0.14627509
[832,]	4893	1	3	0.21510713
[833,]	4893	1	4	0.27250496
[834,]	4893	1	5	0.31729633
[835,]	4893	1	6	0.35056013
[836,]	4945	1	2	0.06463032
[837,]	4945	1	3	0.09505023
[838,]	4945	1	4	0.12044513
[839,]	4945	1	5	0.14035283
[840,]	4945	1	6	0.15537580
[841,]	4985	1	2	0.14116145
[842,]	4985	1	3	0.20758763
[843,]	4985	1	4	0.26298101
[844,]	4985	1	5	0.30621385
[845,]	4985	1	6	0.33833517
[846,]	5023	1	2	0.04370424
[847,]	5023	1	3	0.06427887
[848,]	5023	1	4	0.08147121
[849,]	5023	1	5	0.09500107
[850,]	5023	1	6	0.10534879
[851,]	5082	1	2	0.05011966
[852,]	5082	1	3	0.07371260
[853,]	5082	1	4	0.09341965
[854,]	5082	1	5	0.10890479
[855,]	5082	1	6	0.12068583
[856,]	5093	1	2	0.05060135
[857,]	5093	1	3	0.07442092
[858,]	5093	1	4	0.09431678
[859,]	5093	1	5	0.10994873
[860,]	5093	1	6	0.12183739
[861,]	5097	1	2	0.08207542
[862,]	5097	1	3	0.12070287
[863,]	5097	1	4	0.15293586
[864,]	5097	1	5	0.17816047
[865,]	5097	1	6	0.19708097

[866,]	5118	1	2	0.07228828
[867,]	5118	1	3	0.10631110
[868,]	5118	1	4	0.13470775
[869,]	5118	1	5	0.15694944
[870,]	5118	1	6	0.17368333
[871,]	5132	1	2	0.05751045
[872,]	5132	1	3	0.08458061
[873,]	5132	1	4	0.10718468
[874,]	5132	1	5	0.12492239
[875,]	5132	1	6	0.13835466
[876,]	5142	1	2	0.11317662
[877,]	5142	1	3	0.16643653
[878,]	5142	1	4	0.21086049
[879,]	5142	1	5	0.24556411
[880,]	5142	1	6	0.27143315

```
# Computing Brier scores for 2 class jlcmm for ptau
brier_jlcmm_m1_ptau <- brier_jlcmm(
  model = jlcmm_m1_ptau,
  data = newdata,
  ID = "RID",
  time_var = "time_years",
  surv_time_var = "time_to_event",
  event_var = "event",
  times = times,
  landmark = 1
)
```

	RID	landmark	horizon	pred
[1,]	31	1	2	0.120134396
[2,]	31	1	3	0.177164675
[3,]	31	1	4	0.226055689
[4,]	31	1	5	0.265754912
[5,]	31	1	6	0.296737669
[6,]	74	1	2	0.074887693
[7,]	74	1	3	0.110441258
[8,]	74	1	4	0.140932992
[9,]	74	1	5	0.165734052
[10,]	74	1	6	0.185206423
[11,]	89	1	2	0.098797485
[12,]	89	1	3	0.145700027
[13,]	89	1	4	0.185914524

[14,]	89	1	5	0.218588246
[15,]	89	1	6	0.244143069
[16,]	120	1	2	0.100370700
[17,]	120	1	3	0.148019981
[18,]	120	1	4	0.188874215
[19,]	120	1	5	0.222065942
[20,]	120	1	6	0.248020978
[21,]	127	1	2	0.105581621
[22,]	127	1	3	0.155704308
[23,]	127	1	4	0.198677530
[24,]	127	1	5	0.233585031
[25,]	127	1	6	0.260865683
[26,]	150	1	2	0.179969745
[27,]	150	1	3	0.265401359
[28,]	150	1	4	0.338624031
[29,]	150	1	5	0.398024952
[30,]	150	1	6	0.444229326
[31,]	200	1	2	0.028104393
[32,]	200	1	3	0.041451884
[33,]	200	1	4	0.052919492
[34,]	200	1	5	0.062316439
[35,]	200	1	6	0.069887525
[36,]	259	1	2	0.182276878
[37,]	259	1	3	0.268803591
[38,]	259	1	4	0.342964443
[39,]	259	1	5	0.403125023
[40,]	259	1	6	0.449916313
[41,]	260	1	2	0.030019424
[42,]	260	1	3	0.044275900
[43,]	260	1	4	0.056522242
[44,]	260	1	5	0.066549743
[45,]	260	1	6	0.074607997
[46,]	272	1	2	0.025164294
[47,]	272	1	3	0.037116244
[48,]	272	1	4	0.047388279
[49,]	272	1	5	0.055817154
[50,]	272	1	6	0.062640303
[51,]	359	1	2	0.003500999
[52,]	359	1	3	0.005170289
[53,]	359	1	4	0.006633085
[54,]	359	1	5	0.007928992
[55,]	359	1	6	0.009241178
[56,]	377	1	2	0.164429112

[57,]	377	1	3	0.242484238
[58,]	377	1	4	0.309387412
[59,]	377	1	5	0.363671344
[60,]	377	1	6	0.405922310
[61,]	384	1	2	0.085489089
[62,]	384	1	3	0.126074692
[63,]	384	1	4	0.160877415
[64,]	384	1	5	0.189169145
[65,]	384	1	6	0.211338423
[66,]	419	1	2	0.087565548
[67,]	419	1	3	0.129136759
[68,]	419	1	4	0.164783861
[69,]	419	1	5	0.193759297
[70,]	419	1	6	0.216456809
[71,]	498	1	2	0.059127516
[72,]	498	1	3	0.087200385
[73,]	498	1	4	0.111283345
[74,]	498	1	5	0.130895127
[75,]	498	1	6	0.146358239
[76,]	519	1	2	0.009149307
[77,]	519	1	3	0.013499614
[78,]	519	1	4	0.017259258
[79,]	519	1	5	0.020414955
[80,]	519	1	6	0.023164025
[81,]	522	1	2	0.037577255
[82,]	522	1	3	0.055421117
[83,]	522	1	4	0.070740802
[84,]	522	1	5	0.083256833
[85,]	522	1	6	0.093237737
[86,]	548	1	2	0.160062834
[87,]	548	1	3	0.236045471
[88,]	548	1	4	0.301173126
[89,]	548	1	5	0.354019395
[90,]	548	1	6	0.395159616
[91,]	555	1	2	0.170047564
[92,]	555	1	3	0.250769533
[93,]	555	1	4	0.319957414
[94,]	555	1	5	0.376091306
[95,]	555	1	6	0.419771560
[96,]	637	1	2	0.100400621
[97,]	637	1	3	0.148064103
[98,]	637	1	4	0.188930505
[99,]	637	1	5	0.222132083

[100,]	637	1	6 0.248094731
[101,]	668	1	2 0.002885537
[102,]	668	1	3 0.004262694
[103,]	668	1	4 0.005475216
[104,]	668	1	5 0.006568472
[105,]	668	1	6 0.007724090
[106,]	671	1	2 0.162292942
[107,]	671	1	3 0.239334118
[108,]	671	1	4 0.305368632
[109,]	671	1	5 0.358949198
[110,]	671	1	6 0.400656740
[111,]	684	1	2 0.023682505
[112,]	684	1	3 0.034931111
[113,]	684	1	4 0.044600586
[114,]	684	1	5 0.052541560
[115,]	684	1	6 0.058987754
[116,]	685	1	2 0.107295083
[117,]	685	1	3 0.158231078
[118,]	685	1	4 0.201901068
[119,]	685	1	5 0.237372753
[120,]	685	1	6 0.265089295
[121,]	686	1	2 0.041281839
[122,]	686	1	3 0.060884112
[123,]	686	1	4 0.077710243
[124,]	686	1	5 0.091446065
[125,]	686	1	6 0.102369384
[126,]	908	1	2 0.049906256
[127,]	908	1	3 0.073602178
[128,]	908	1	4 0.093935372
[129,]	908	1	5 0.110510914
[130,]	908	1	6 0.123628214
[131,]	920	1	2 0.108303710
[132,]	920	1	3 0.159718459
[133,]	920	1	4 0.203798600
[134,]	920	1	5 0.239602391
[135,]	920	1	6 0.267575520
[136,]	926	1	2 0.100847676
[137,]	926	1	3 0.148723357
[138,]	926	1	4 0.189771550
[139,]	926	1	5 0.223120328
[140,]	926	1	6 0.249196703
[141,]	951	1	2 0.210806144
[142,]	951	1	3 0.310874504

[143,]	951	1	4	0.396636599
[144,]	951	1	5	0.466190872
[145,]	951	1	6	0.520239771
[146,]	984	1	2	0.136157373
[147,]	984	1	3	0.200793089
[148,]	984	1	4	0.256199742
[149,]	984	1	5	0.301174774
[150,]	984	1	6	0.336233644
[151,]	1023	1	2	0.091444268
[152,]	1023	1	3	0.134856545
[153,]	1023	1	4	0.172080903
[154,]	1023	1	5	0.202333466
[155,]	1023	1	6	0.226017692
[156,]	1045	1	2	0.044124564
[157,]	1045	1	3	0.065076160
[158,]	1045	1	4	0.083058266
[159,]	1045	1	5	0.097730099
[160,]	1045	1	6	0.109376584
[161,]	1072	1	2	0.064716643
[162,]	1072	1	3	0.095442436
[163,]	1072	1	4	0.121798179
[164,]	1072	1	5	0.143250266
[165,]	1072	1	6	0.160135205
[166,]	1155	1	2	0.045432870
[167,]	1155	1	3	0.067005463
[168,]	1155	1	4	0.085519583
[169,]	1155	1	5	0.100622195
[170,]	1155	1	6	0.112601502
[171,]	1186	1	2	0.064993842
[172,]	1186	1	3	0.095851210
[173,]	1186	1	4	0.122319674
[174,]	1186	1	5	0.143863033
[175,]	1186	1	6	0.160818489
[176,]	1206	1	2	0.097497901
[177,]	1206	1	3	0.143783585
[178,]	1206	1	4	0.183469614
[179,]	1206	1	5	0.215715428
[180,]	1206	1	6	0.240939649
[181,]	1222	1	2	0.118198409
[182,]	1222	1	3	0.174309756
[183,]	1222	1	4	0.222413513
[184,]	1222	1	5	0.261475283
[185,]	1222	1	6	0.291965541

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[188,]	1280	1	4	0.097448529
[189,]	1280	1	5	0.114638943
[190,]	1280	1	6	0.128231296
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[192,]	1286	1	3	0.093440684
[193,]	1286	1	4	0.119244435
[194,]	1286	1	5	0.140249566
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[197,]	1300	1	3	0.181098013
[198,]	1300	1	4	0.231073662
[199,]	1300	1	5	0.271651131
[200,]	1300	1	6	0.303312424
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[202,]	1352	1	3	0.279344481
[203,]	1352	1	4	0.356412031
[204,]	1352	1	5	0.418926206
[205,]	1352	1	6	0.467535894
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[211,]	1418	1	2	0.161392015
[212,]	1418	1	3	0.238005558
[213,]	1418	1	4	0.303673716
[214,]	1418	1	5	0.356957637
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[599,]	4422	1	5	0.199614431
[600,]	4422	1	6	0.222985750
[601,]	4430	1	2	NA
[602,]	4430	1	3	NA
[603,]	4430	1	4	NA
[604,]	4430	1	5	NA
[605,]	4430	1	6	NA
[606,]	4443	1	2	0.084056941
[607,]	4443	1	3	0.123962764
[608,]	4443	1	4	0.158183113
[609,]	4443	1	5	0.186003287
[610,]	4443	1	6	0.207808238
[611,]	4447	1	2	0.136401611
[612,]	4447	1	3	0.201153256
[613,]	4447	1	4	0.256659227
[614,]	4447	1	5	0.301714678
[615,]	4447	1	6	0.336835680

[616,]	4458	1	2	0.151638798
[617,]	4458	1	3	0.223622898
[618,]	4458	1	4	0.285324973
[619,]	4458	1	5	0.335397500
[620,]	4458	1	6	0.374394716
[621,]	4464	1	2	0.055744868
[622,]	4464	1	3	0.082212135
[623,]	4464	1	4	0.104919563
[624,]	4464	1	5	0.123417557
[625,]	4464	1	6	0.138020151
[626,]	4476	1	2	0.147194678
[627,]	4476	1	3	0.217069340
[628,]	4476	1	4	0.276964243
[629,]	4476	1	5	0.325573475
[630,]	4476	1	6	0.363440144
[631,]	4499	1	2	0.106713217
[632,]	4499	1	3	0.157373025
[633,]	4499	1	4	0.200806403
[634,]	4499	1	5	0.236086500
[635,]	4499	1	6	0.263655020
[636,]	4513	1	2	0.110992380
[637,]	4513	1	3	0.163683327
[638,]	4513	1	4	0.208856799
[639,]	4513	1	5	0.245545875
[640,]	4513	1	6	0.274202979
[641,]	4531	1	2	0.139105062
[642,]	4531	1	3	0.205139923
[643,]	4531	1	4	0.261745235
[644,]	4531	1	5	0.307690839
[645,]	4531	1	6	0.343499576
[646,]	4539	1	2	0.061544631
[647,]	4539	1	3	0.090764803
[648,]	4539	1	4	0.115830667
[649,]	4539	1	5	0.136238321
[650,]	4539	1	6	0.152316327
[651,]	4545	1	2	0.060504357
[652,]	4545	1	3	0.089230754
[653,]	4545	1	4	0.113873598
[654,]	4545	1	5	0.133938725
[655,]	4545	1	6	0.149752094
[656,]	4547	1	2	0.191848172
[657,]	4547	1	3	0.282917977
[658,]	4547	1	4	0.360970935

[659,]	4547	1	5	0.424283008
[660,]	4547	1	6	0.473509156
[661,]	4553	1	2	0.066177056
[662,]	4553	1	3	0.097596045
[663,]	4553	1	4	0.124545655
[664,]	4553	1	5	0.146478605
[665,]	4553	1	6	0.163735061
[666,]	4556	1	2	0.090086948
[667,]	4556	1	3	0.132854962
[668,]	4556	1	4	0.169527375
[669,]	4556	1	5	0.199333020
[670,]	4556	1	6	0.222671956
[671,]	4566	1	2	0.076911567
[672,]	4566	1	3	0.113425780
[673,]	4566	1	4	0.144740509
[674,]	4566	1	5	0.170207960
[675,]	4566	1	6	0.190195187
[676,]	4577	1	2	0.109319391
[677,]	4577	1	3	0.161216240
[678,]	4577	1	4	0.205709402
[679,]	4577	1	5	0.241847621
[680,]	4577	1	6	0.270079130
[681,]	4578	1	2	0.021230396
[682,]	4578	1	3	0.031315089
[683,]	4578	1	4	0.039987429
[684,]	4578	1	5	0.047121009
[685,]	4578	1	6	0.052943407
[686,]	4586	1	2	0.018949275
[687,]	4586	1	3	0.027951216
[688,]	4586	1	4	0.035695954
[689,]	4586	1	5	0.042078439
[690,]	4586	1	6	0.047320539
[691,]	4587	1	2	0.060553769
[692,]	4587	1	3	0.089303620
[693,]	4587	1	4	0.113966557
[694,]	4587	1	5	0.134047954
[695,]	4587	1	6	0.149873893
[696,]	4607	1	2	0.071308619
[697,]	4607	1	3	0.105163347
[698,]	4607	1	4	0.134199673
[699,]	4607	1	5	0.157822268
[700,]	4607	1	6	0.176384152
[701,]	4611	1	2	0.090701396

[702,]	4611	1	3	0.133761063
[703,]	4611	1	4	0.170683337
[704,]	4611	1	5	0.200691298
[705,]	4611	1	6	0.224186544
[706,]	4621	1	2	0.072039151
[707,]	4621	1	3	0.106240632
[708,]	4621	1	4	0.135574025
[709,]	4621	1	5	0.159437159
[710,]	4621	1	6	0.178184884
[711,]	4630	1	2	0.012880814
[712,]	4630	1	3	0.019002311
[713,]	4630	1	4	0.024279348
[714,]	4630	1	5	0.028663701
[715,]	4630	1	6	0.032362035
[716,]	4635	1	2	0.086192850
[717,]	4635	1	3	0.127112499
[718,]	4635	1	4	0.162201402
[719,]	4635	1	5	0.190724857
[720,]	4635	1	6	0.213073165
[721,]	4649	1	2	0.087757533
[722,]	4649	1	3	0.129419872
[723,]	4649	1	4	0.165145043
[724,]	4649	1	5	0.194183693
[725,]	4649	1	6	0.216930044
[726,]	4674	1	2	0.025751303
[727,]	4674	1	3	0.037981881
[728,]	4674	1	4	0.048492619
[729,]	4674	1	5	0.057114776
[730,]	4674	1	6	0.064087255
[731,]	4711	1	2	0.221085497
[732,]	4711	1	3	0.326033036
[733,]	4711	1	4	0.415975163
[734,]	4711	1	5	0.488914069
[735,]	4711	1	6	0.545577951
[736,]	4712	1	2	NA
[737,]	4712	1	3	NA
[738,]	4712	1	4	NA
[739,]	4712	1	5	NA
[740,]	4712	1	6	NA
[741,]	4713	1	2	0.125517414
[742,]	4713	1	3	0.185102787
[743,]	4713	1	4	0.236182770
[744,]	4713	1	5	0.277654435

[745,]	4713	1	6	0.310006587
[746,]	4720	1	2	0.142114892
[747,]	4720	1	3	0.209578393
[748,]	4720	1	4	0.267407633
[749,]	4720	1	5	0.314344269
[750,]	4720	1	6	0.350918682
[751,]	4736	1	2	0.124426093
[752,]	4736	1	3	0.183493461
[753,]	4736	1	4	0.234129665
[754,]	4736	1	5	0.275241995
[755,]	4736	1	6	0.307316525
[756,]	4739	1	2	0.013488984
[757,]	4739	1	3	0.019899153
[758,]	4739	1	4	0.025423499
[759,]	4739	1	5	0.030008101
[760,]	4739	1	6	0.033861149
[761,]	4746	1	2	0.079611709
[762,]	4746	1	3	0.117407567
[763,]	4746	1	4	0.149820291
[764,]	4746	1	5	0.176176805
[765,]	4746	1	6	0.196850926
[766,]	4750	1	2	0.116916535
[767,]	4750	1	3	0.172419431
[768,]	4750	1	4	0.220001922
[769,]	4750	1	5	0.258641616
[770,]	4750	1	6	0.288805776
[771,]	4764	1	2	0.120739355
[772,]	4764	1	3	0.178056784
[773,]	4764	1	4	0.227193800
[774,]	4764	1	5	0.267092216
[775,]	4764	1	6	0.298228869
[776,]	4765	1	2	0.133534496
[777,]	4765	1	3	0.196925241
[778,]	4765	1	4	0.251265319
[779,]	4765	1	5	0.295376729
[780,]	4765	1	6	0.329768360
[781,]	4777	1	2	0.121766610
[782,]	4777	1	3	0.179571633
[783,]	4777	1	4	0.229126376
[784,]	4777	1	5	0.269363030
[785,]	4777	1	6	0.300761009
[786,]	4782	1	2	0.124135153
[787,]	4782	1	3	0.183064424

[788,]	4782	1	4	0.233582319
[789,]	4782	1	5	0.274598853
[790,]	4782	1	6	0.306599369
[791,]	4807	1	2	0.190182328
[792,]	4807	1	3	0.280461428
[793,]	4807	1	4	0.357836981
[794,]	4807	1	5	0.420600550
[795,]	4807	1	6	0.469402921
[796,]	4817	1	2	0.090118784
[797,]	4817	1	3	0.132901910
[798,]	4817	1	4	0.169587269
[799,]	4817	1	5	0.199403396
[800,]	4817	1	6	0.222750430
[801,]	4844	1	2	0.103731701
[802,]	4844	1	3	0.152976308
[803,]	4844	1	4	0.195197271
[804,]	4844	1	5	0.229495658
[805,]	4844	1	6	0.256305705
[806,]	4857	1	2	NA
[807,]	4857	1	3	NA
[808,]	4857	1	4	NA
[809,]	4857	1	5	NA
[810,]	4857	1	6	NA
[811,]	4869	1	2	0.091470546
[812,]	4869	1	3	0.134895297
[813,]	4869	1	4	0.172130341
[814,]	4869	1	5	0.202391557
[815,]	4869	1	6	0.226082468
[816,]	4883	1	2	0.014812593
[817,]	4883	1	3	0.021851024
[818,]	4883	1	4	0.027913606
[819,]	4883	1	5	0.032934027
[820,]	4883	1	6	0.037123789
[821,]	4888	1	2	NA
[822,]	4888	1	3	NA
[823,]	4888	1	4	NA
[824,]	4888	1	5	NA
[825,]	4888	1	6	NA
[826,]	4889	1	2	0.086014756
[827,]	4889	1	3	0.126849872
[828,]	4889	1	4	0.161866354
[829,]	4889	1	5	0.190331168
[830,]	4889	1	6	0.212634171

[831,]	4893	1	2	0.109104976
[832,]	4893	1	3	0.160900052
[833,]	4893	1	4	0.205306023
[834,]	4893	1	5	0.241373643
[835,]	4893	1	6	0.269550607
[836,]	4945	1	2	0.183477097
[837,]	4945	1	3	0.270573504
[838,]	4945	1	4	0.345222417
[839,]	4945	1	5	0.405778187
[840,]	4945	1	6	0.452874803
[841,]	4985	1	2	0.170416812
[842,]	4985	1	3	0.251314048
[843,]	4985	1	4	0.320652081
[844,]	4985	1	5	0.376907554
[845,]	4985	1	6	0.420681741
[846,]	5023	1	2	0.032576266
[847,]	5023	1	3	0.048046369
[848,]	5023	1	4	0.061332435
[849,]	5023	1	5	0.072201814
[850,]	5023	1	6	0.080910508
[851,]	5082	1	2	0.075272271
[852,]	5082	1	3	0.111008379
[853,]	5082	1	4	0.141656499
[854,]	5082	1	5	0.166584187
[855,]	5082	1	6	0.186154391
[856,]	5093	1	2	0.075380603
[857,]	5093	1	3	0.111168132
[858,]	5093	1	4	0.141860305
[859,]	5093	1	5	0.166823663
[860,]	5093	1	6	0.186421425
[861,]	5097	1	2	0.082965696
[862,]	5097	1	3	0.122353551
[863,]	5097	1	4	0.156130153
[864,]	5097	1	5	0.183591018
[865,]	5097	1	6	0.205118366
[866,]	5118	1	2	0.146718331
[867,]	5118	1	3	0.216366891
[868,]	5118	1	4	0.276068090
[869,]	5118	1	5	0.324520477
[870,]	5118	1	6	0.362265967
[871,]	5132	1	2	0.147262357
[872,]	5132	1	3	0.217169144
[873,]	5132	1	4	0.277091568

```
[874,] 5132      1      5 0.325723085
[875,] 5132      1      6 0.363606971
[876,] 5142      1      2 0.126194747
[877,] 5142      1      3 0.186101621
[878,] 5142      1      4 0.237457037
[879,] 5142      1      5 0.279151724
[880,] 5142      1      6 0.311676184
```

```
# # ON IN-SAMPLE DATA
# # Computing Brier scores for 2 class multivariate jlcmm for ptau & NfL
# brier_mult_jlcmm_m1 <- brier_jlcmm(
#   model = mult_jlcmm_m1,
#   data = jlcmm_data,
#   ID = "RID",
#   time_var = "time_years",
#   surv_time_var = "time_to_event",
#   event_var = "event",
#   times = times,
#   landmark = 1
# )
```

12 Tables of the predictive performance results

12.1 Time-depending AUC

```
# Time-dependent AUC for Cox and joint models summarized in a table
results_table_AUC <- data.frame(
  Model = c("Cox p-tau181", "Cox NfL", "Cox both",
            "JM p-tau181", "JM NfL", "JM both", "JLCMM p-tau181", "JLCMM NfL"),

  "Year 2" = c(auc_ptau_bl[1],
               auc_nfl_bl[1],
               auc_both_bl[1],
               auc_ptau_jm[1],
               auc_nfl_jm[1],
               auc_multi_jm[1],
               auc_jlcmm_m1_ptau[1,3],
               auc_jlcmm_m1_nfl[1,3]),

  "Year 3" = c(auc_ptau_bl[2],
```

```

        auc_nfl_bl[2],
        auc_both_bl[2],
        auc_ptau_jm[2],
        auc_nfl_jm[2],
        auc_multi_jm[2],
        auc_jlcmm_m1_ptau[2,3],
        auc_jlcmm_m1_nfl[2,3]),

"Year 4" = c(auc_ptau_bl[3],
        auc_nfl_bl[3],
        auc_both_bl[3],
        auc_ptau_jm[3],
        auc_nfl_jm[3],
        auc_multi_jm[3],
        auc_jlcmm_m1_ptau[3,3],
        auc_jlcmm_m1_nfl[3,3]),

"Year 5" = c(auc_ptau_bl[4],
        auc_nfl_bl[4],
        auc_both_bl[4],
        auc_ptau_jm[4],
        auc_nfl_jm[4],
        auc_multi_jm[4],
        auc_jlcmm_m1_ptau[4,3],
        auc_jlcmm_m1_nfl[4,3]),

"Year 6" = c(auc_ptau_bl[5],
        auc_nfl_bl[5],
        auc_both_bl[5],
        auc_ptau_jm[5],
        auc_nfl_jm[5],
        auc_multi_jm[5],
        auc_jlcmm_m1_ptau[5,3],
        auc_jlcmm_m1_nfl[5,3]),

    check.names = FALSE
)

results_table_AUC[, -1] <- round(results_table_AUC[, -1], 3)
results_table_AUC

```

Model Year 2 Year 3 Year 4 Year 5 Year 6

1	Cox p-tau181	0.823	0.807	0.802	0.771	0.780
2	Cox NfL	0.846	0.787	0.817	0.792	0.797
3	Cox both	0.870	0.847	0.856	0.837	0.855
4	JM p-tau181	0.902	0.812	0.806	0.770	0.786
5	JM NfL	0.953	0.799	0.823	0.808	0.804
6	JM both	0.967	0.859	0.863	0.841	0.846
7	JLCMM p-tau181	0.862	0.843	0.811	0.787	0.815
8	JLCMM NfL	0.907	0.800	0.803	0.809	0.837

12.2 Time-dependent Brier score

```
# Time-dependent Brier score for Cox and joint models summarized in a table
results_table_Brier <- data.frame(
  Model = c("Cox p-tau181", "Cox NfL", "Cox both",
            "JM p-tau181", "JM NfL", "JM both",
            "JLCMM p-tau181", "JLCMM NfL"),

  "Year 2" = c(cox_brier$`Cox p-tau`[1],
               cox_brier$`Cox NfL`[1],
               cox_brier$`Cox both`[1],
               brier_ptau_jm[1],
               brier_nfl_jm[1],
               brier_multi_jm[1],
               brier_jlcmm_m1_ptau$Brier[1],
               brier_jlcmm_m1_nfl$Brier[1]),

  "Year 3" = c(cox_brier$`Cox p-tau`[2],
               cox_brier$`Cox NfL`[2],
               cox_brier$`Cox both`[2],
               brier_ptau_jm[2],
               brier_nfl_jm[2],
               brier_multi_jm[2],
               brier_jlcmm_m1_ptau$Brier[2],
               brier_jlcmm_m1_nfl$Brier[2]),

  "Year 4" = c(cox_brier$`Cox p-tau`[3],
               cox_brier$`Cox NfL`[3],
               cox_brier$`Cox both`[3],
               brier_ptau_jm[3],
               brier_nfl_jm[3],
               brier_multi_jm[3],
```

```

        brier_jlcmm_m1_ptau$Brier[3],
        brier_jlcmm_m1_nfl$Brier[3]),

"Year 5" = c(cox_brier$`Cox p-tau`[4],
             cox_brier$`Cox NfL`[4],
             cox_brier$`Cox both`[4],
             brier_ptau_jm[4],
             brier_nfl_jm[4],
             brier_multi_jm[4],
             brier_jlcmm_m1_ptau$Brier[4],
             brier_jlcmm_m1_nfl$Brier[4]),

"Year 6" = c(cox_brier$`Cox p-tau`[5],
             cox_brier$`Cox NfL`[5],
             cox_brier$`Cox both`[5],
             brier_ptau_jm[5],
             brier_nfl_jm[5],
             brier_multi_jm[5],
             brier_jlcmm_m1_ptau$Brier[5],
             brier_jlcmm_m1_nfl$Brier[5]),

check.names = FALSE
)

results_table_Brier[, -1] <- round(results_table_Brier[, -1], 3)
results_table_Brier

```

	Model	Year 2	Year 3	Year 4	Year 5	Year 6
1	Cox p-tau181	0.054	0.094	0.118	0.142	0.158
2	Cox NfL	0.054	0.098	0.122	0.145	0.159
3	Cox both	0.052	0.090	0.113	0.133	0.142
4	JM p-tau181	0.018	0.068	0.098	0.124	0.138
5	JM NfL	0.017	0.068	0.096	0.120	0.135
6	JM both	0.017	0.066	0.092	0.115	0.127
7	JLCMM p-tau181	0.023	0.069	0.099	0.125	0.137
8	JLCMM NfL	0.023	0.071	0.101	0.124	0.137

12.3 Saving the tables

12.3.1 Original tables

```
# Creating a directory to save tables into
dir.create("tables", showWarnings = FALSE)

# # # Saving the AUC results table
# saveRDS(results_table_AUC, "tables/results_table_AUC.rds")
# # Saving the table of the Brier score results
# saveRDS(results_table_Brier, "tables/results_table_Brier.rds")
```

12.3.2 Sensitivity analysis tables

12.3.2.1 The duplicate biomarker measurements are replaced by the first measurement

```
#saveRDS(results_table_AUC, "tables/results_table_AUC_first_duplicate.rds")
#saveRDS(results_table_Brier, "tables/results_table_Brier_first_duplicate.rds")
```

12.3.2.2 The duplicate biomarker measurements are replaced by the last measurement

```
#saveRDS(results_table_AUC, "tables/results_table_AUC_last_duplicate.rds")
# saveRDS(results_table_Brier, "tables/results_table_Brier_last_duplicate.rds")
```

12.3.2.3 Baseline date is the first examination date across biomarkers

```
#saveRDS(results_table_AUC, "tables/results_table_AUC_first_biomarker_bl.rds")
#saveRDS(results_table_Brier, "tables/results_table_Brier_first_biomarker_bl.rds")
```

12.3.2.4 Including participants with at least one biomarker measurement

```
#saveRDS(results_table_AUC, "tables/results_table_AUC_full.rds")
#saveRDS(results_table_Brier, "tables/results_table_Brier_full.rds")
```

12.3.2.5 Including time-to-event as midpoint

```
# saveRDS(results_table_AUC, "tables/results_table_AUC_midpoint.rds")
# saveRDS(results_table_Brier, "tables/results_table_Brier_midpoint.rds")
```

13 Plots of the predictive performance results

```
# We reshape all AUC results into long format for plotting
plot_data <- data.frame(
  Time = rep(times, 8),
  Model = rep(c("Cox p-tau181", "Cox NfL", "Cox both",
                "JM p-tau181", "JM NfL", "JM multi", "JLCMM p-tau181", "JLCMM NfL"),
              each = length(times)),
  AUC = c(
    auc_ptau_bl,
    auc_nfl_bl,
    auc_both_bl,
    auc_ptau_jm,
    auc_nfl_jm,
    auc_multi_jm,
    auc_jlcmm_m1_ptau$AUC,
    auc_jlcmm_m1_nfl$AUC
  )
)
```

```
# We reshape all Brier score results into long format for plotting
plot_data_Brier <- data.frame(
  Time = rep(times, 8),
  Model = rep(c("Cox p-tau181", "Cox NfL", "Cox both",
                "JM p-tau181", "JM NfL", "JM multi",
                "JLCMM p-tau181", "JLCMM NfL"),
              each = length(times)),
  Brier = c(
    cox_brier$`Cox p-tau`,
    cox_brier$`Cox NfL`,
    cox_brier$`Cox both`,

    brier_ptau_jm,
    brier_nfl_jm,
  )
)
```



```

    brier_multi_jm,

    brier_jlcmm_m1_ptau$Brier,
    brier_jlcmm_m1_nfl$Brier
  )
)

```

13.1 For comparing modelling approaches

13.1.1 Time-dependent AUC

```

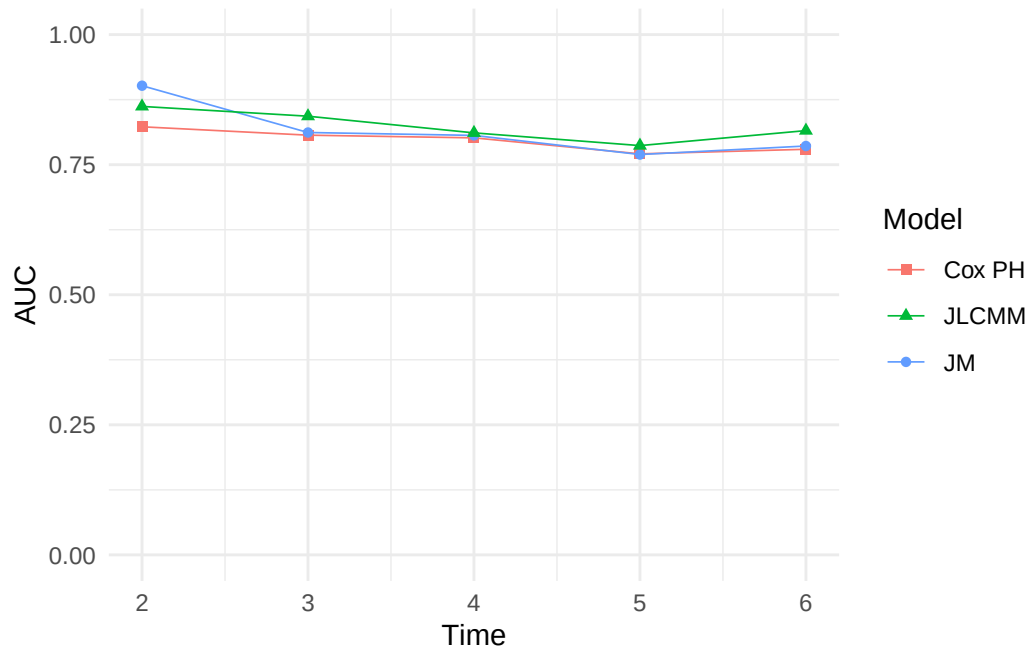
# Q1: Biomarker fixed (p-tau181) - comparing models (time-dependent AUC)
ggplot(data.frame(
  Time = times,
  Cox = auc_ptau_bl,
  JM = auc_ptau_jm,
  JLCMM = auc_jlcmm_m1_ptau$AUC
)) +
  geom_line(aes(Time, Cox, color = "Cox PH"), linewidth = 0.3) +
  geom_point(aes(Time, Cox, color = "Cox PH", shape = "Cox PH"), size = 1.5) +

  geom_line(aes(Time, JM, color = "JM"), linewidth = 0.3) +
  geom_point(aes(Time, JM, color = "JM", shape = "JM"), size = 1.5) +

  geom_line(aes(Time, JLCMM, color = "JLCMM"), linewidth = 0.3) +
  geom_point(aes(Time, JLCMM, color = "JLCMM", shape = "JLCMM"), size = 1.5) +

  scale_shape_manual(values = c("Cox PH" = 15, "JM" = 16, "JLCMM" = 17)) +
  ylim(0,1) +
  labs(
    x = "Time",
    y = "AUC",
    color = "Model",
    shape = "Model"
  ) +
  theme_minimal()

```



```
# Creating a folder to save plots
dir.create("figures", showWarnings = FALSE)

# Saving the plot
png("figures/q1_auc_ptau.png",
    width = 6, height = 4, units = "in", res = 300)
ggplot(data.frame(
  Time = times,
  Cox = auc_ptau_bl,
  JM = auc_ptau_jm,
  JLCMM = auc_jlcmm_m1_ptau$AUC
)) +
  geom_line(aes(Time, Cox, color = "Cox PH"), linewidth = 0.3) +
  geom_point(aes(Time, Cox, color = "Cox PH", shape = "Cox PH"), size = 1.5) +

  geom_line(aes(Time, JM, color = "JM"), linewidth = 0.3) +
  geom_point(aes(Time, JM, color = "JM", shape = "JM"), size = 1.5) +

  geom_line(aes(Time, JLCMM, color = "JLCMM"), linewidth = 0.3) +
  geom_point(aes(Time, JLCMM, color = "JLCMM", shape = "JLCMM"), size = 1.5) +

  scale_shape_manual(values = c("Cox PH" = 15, "JM" = 16, "JLCMM" = 17)) +
  ylim(0,1) +
```

```

labs(
  x = "Time",
  y = "AUC",
  color = "Model",
  shape = "Model"
) +
theme_minimal()
dev.off()

```

pdf
2

```

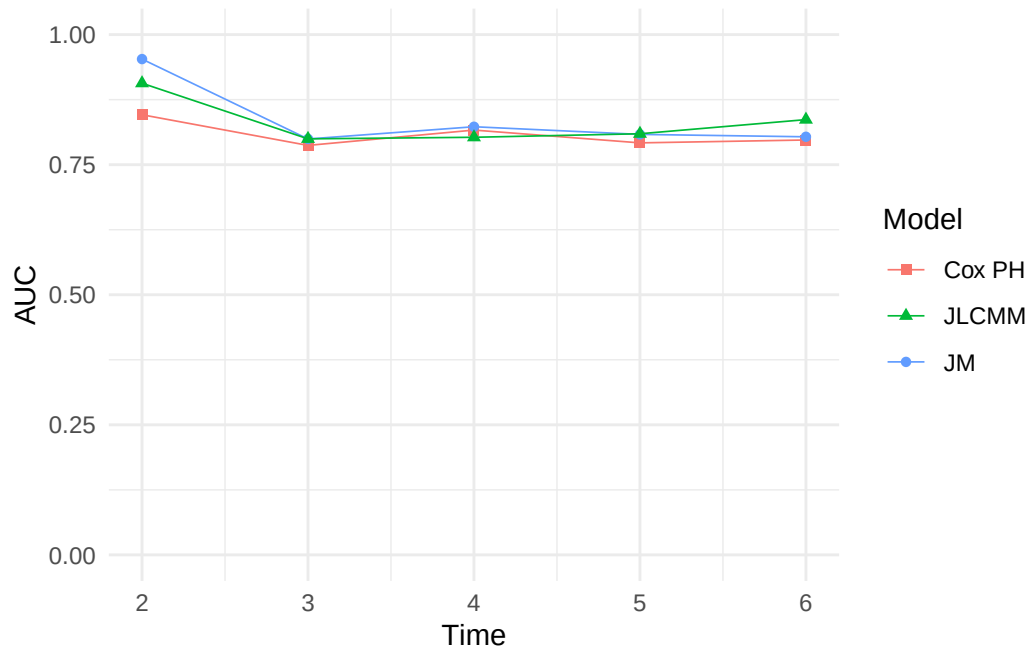
# Q1: NfL - comparing models (time-dependent AUC)
ggplot(data.frame(
  Time = times,
  Cox = auc_nfl_bl,
  JM = auc_nfl_jm,
  JLCMM = auc_jlcmm_m1_nfl$AUC
)) +
  geom_line(aes(Time, Cox, color = "Cox PH"), linewidth = 0.3) +
  geom_point(aes(Time, Cox, color = "Cox PH", shape = "Cox PH"), size = 1.5) +

  geom_line(aes(Time, JM, color = "JM"), linewidth = 0.3) +
  geom_point(aes(Time, JM, color = "JM", shape = "JM"), size = 1.5) +

  geom_line(aes(Time, JLCMM, color = "JLCMM"), linewidth = 0.3) +
  geom_point(aes(Time, JLCMM, color = "JLCMM", shape = "JLCMM"), size = 1.5) +

  scale_shape_manual(values = c("Cox PH" = 15, "JM" = 16, "JLCMM" = 17)) +
  ylim(0,1) +
  labs(
    x = "Time",
    y = "AUC",
    color = "Model",
    shape = "Model"
  ) +
  theme_minimal()

```



```
# Saving the plot
png("figures/q1_auc_nfl.png",
     width = 6, height = 4, units = "in", res = 300)
ggplot(data.frame(
  Time = times,
  Cox = auc_nfl_bl,
  JM = auc_nfl_jm,
  JLCMM = auc_jlcmm_m1_nfl$AUC
)) +
  geom_line(aes(Time, Cox, color = "Cox PH"), linewidth = 0.3) +
  geom_point(aes(Time, Cox, color = "Cox PH", shape = "Cox PH"), size = 1.5) +

  geom_line(aes(Time, JM, color = "JM"), linewidth = 0.3) +
  geom_point(aes(Time, JM, color = "JM", shape = "JM"), size = 1.5) +

  geom_line(aes(Time, JLCMM, color = "JLCMM"), linewidth = 0.3) +
  geom_point(aes(Time, JLCMM, color = "JLCMM", shape = "JLCMM"), size = 1.5) +

  scale_shape_manual(values = c("Cox PH" = 15, "JM" = 16, "JLCMM" = 17)) +
  ylim(0,1) +
  labs(
    x = "Time",
    y = "AUC",
```

```

    color = "Model",
    shape = "Model"
  ) +
  theme_minimal()
dev.off()

```

pdf
2

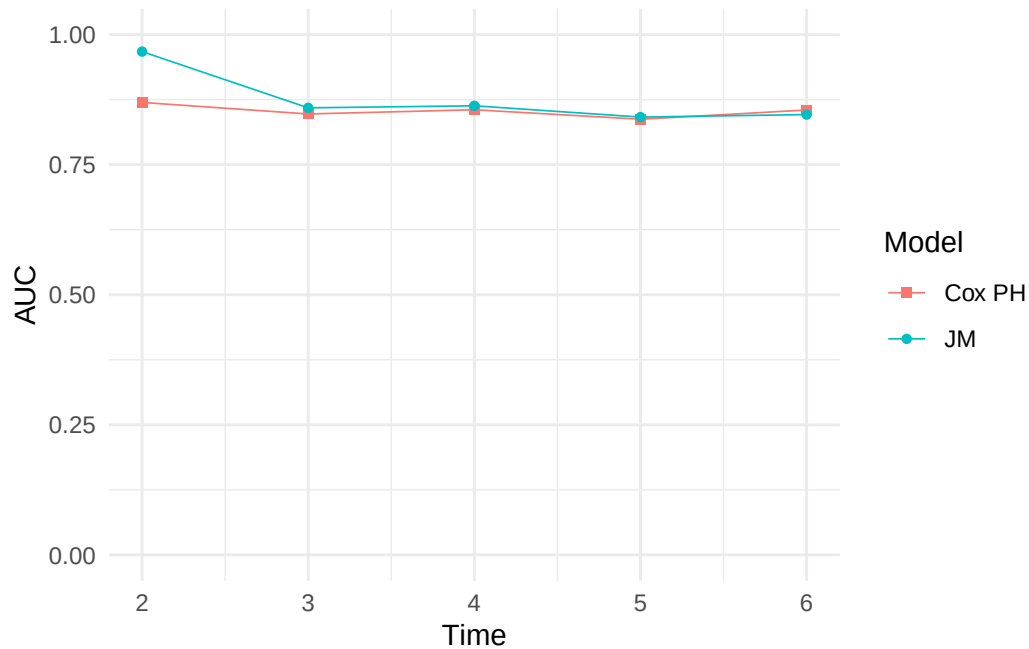
```

# Q1: Both biomarkers - comparing models (time-dependent AUC)
ggplot(data.frame(
  Time = times,
  Cox = auc_both_bl,
  JM = auc_multi_jm
)) +
  geom_line(aes(Time, Cox, color = "Cox PH"), linewidth = 0.3) +
  geom_point(aes(Time, Cox, color = "Cox PH", shape = "Cox PH"), size = 1.5) +

  geom_line(aes(Time, JM, color = "JM"), linewidth = 0.3) +
  geom_point(aes(Time, JM, color = "JM", shape = "JM"), size = 1.5) +

  scale_shape_manual(values = c("Cox PH" = 15, "JM" = 16)) +
  ylim(0,1) +
  labs(
    x = "Time",
    y = "AUC",
    color = "Model",
    shape = "Model"
  ) +
  theme_minimal()

```



```
# Saving the plot
png("figures/q1_auc_multi.png",
     width = 6, height = 4, units = "in", res = 300)
ggplot(data.frame(
  Time = times,
  Cox = auc_both_bl,
  JM = auc_multi_jm
)) +
  geom_line(aes(Time, Cox, color = "Cox PH"), linewidth = 0.3) +
  geom_point(aes(Time, Cox, color = "Cox PH", shape = "Cox PH"), size = 1.5) +

  geom_line(aes(Time, JM, color = "JM"), linewidth = 0.3) +
  geom_point(aes(Time, JM, color = "JM", shape = "JM"), size = 1.5) +

  scale_shape_manual(values = c("Cox PH" = 15, "JM" = 16)) +
  ylim(0,1) +
  labs(
    x = "Time",
    y = "AUC",
    color = "Model",
    shape = "Model"
  ) +
  theme_minimal()
```

```
dev.off()
```

pdf
2

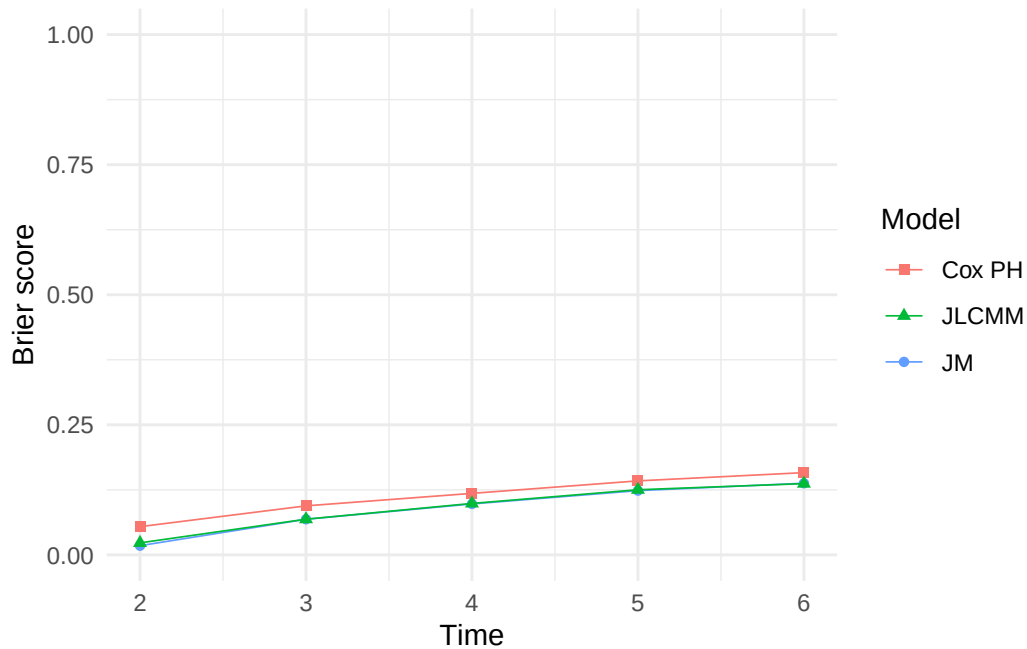
13.1.2 Time-dependent Brier score

```
# Q1: Biomarker fixed (p-tau181) - comparing models (time-dependent Brier score)
ggplot(data.frame(
  Time = times,
  Cox = cox_brier$`Cox p-tau`,
  JM = brier_ptau_jm,
  JLCMM = brier_jlcmm_m1_ptau$Brier
)) +
  geom_line(aes(Time, Cox, color = "Cox PH"), linewidth = 0.3) +
  geom_point(aes(Time, Cox, color = "Cox PH", shape = "Cox PH"), size = 1.5) +

  geom_line(aes(Time, JM, color = "JM"), linewidth = 0.3) +
  geom_point(aes(Time, JM, color = "JM", shape = "JM"), size = 1.5) +

  geom_line(aes(Time, JLCMM, color = "JLCMM"), linewidth = 0.3) +
  geom_point(aes(Time, JLCMM, color = "JLCMM", shape = "JLCMM"), size = 1.5) +

  scale_shape_manual(values = c("Cox PH" = 15, "JM" = 16, "JLCMM" = 17)) +
  ylim(0,1) +
  labs(
    x = "Time",
    y = "Brier score",
    color = "Model",
    shape = "Model"
  ) +
  theme_minimal()
```



```
# Saving the plot
png("figures/q1_brier_ptau.png",
    width = 6, height = 4, units = "in", res = 300)
ggplot(data.frame(
  Time = times,
  Cox = cox_brier$`Cox p-tau`,
  JM = brier_ptau_jm,
  JLCMM = brier_jlcmm_m1_ptau$Brier
)) +
  geom_line(aes(Time, Cox, color = "Cox PH"), linewidth = 0.3) +
  geom_point(aes(Time, Cox, color = "Cox PH", shape = "Cox PH"), size = 1.5) +

  geom_line(aes(Time, JM, color = "JM"), linewidth = 0.3) +
  geom_point(aes(Time, JM, color = "JM", shape = "JM"), size = 1.5) +

  geom_line(aes(Time, JLCMM, color = "JLCMM"), linewidth = 0.3) +
  geom_point(aes(Time, JLCMM, color = "JLCMM", shape = "JLCMM"), size = 1.5) +

  scale_shape_manual(values = c("Cox PH" = 15, "JM" = 16, "JLCMM" = 17)) +
  ylim(0,1) +
  labs(
    x = "Time",
    y = "Brier score",
```



```

    color = "Model",
    shape = "Model"
  ) +
  theme_minimal()
dev.off()

```

pdf
2

```

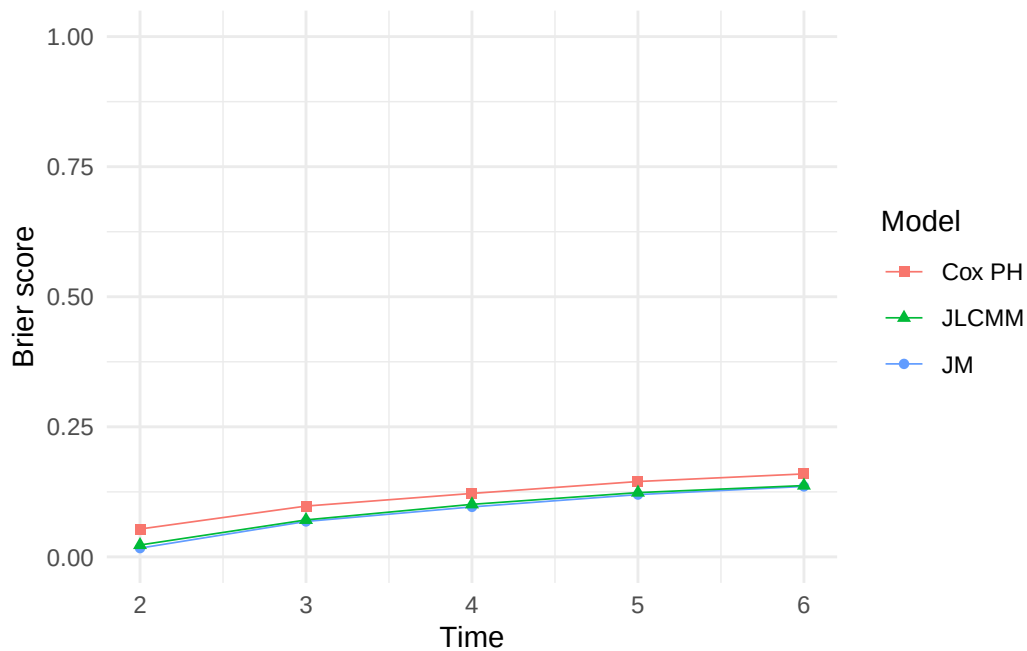
# Q1: NfL - comparing models (time-dependent Brier score)
ggplot(data.frame(
  Time = times,
  Cox = cox_brier$`Cox NfL`,
  JM = brier_nfl_jm,
  JLCMM = brier_jlcmm_m1_nfl$Brier
)) +
  geom_line(aes(Time, Cox, color = "Cox PH"), linewidth = 0.3) +
  geom_point(aes(Time, Cox, color = "Cox PH", shape = "Cox PH"), size = 1.5) +

  geom_line(aes(Time, JM, color = "JM"), linewidth = 0.3) +
  geom_point(aes(Time, JM, color = "JM", shape = "JM"), size = 1.5) +

  geom_line(aes(Time, JLCMM, color = "JLCMM"), linewidth = 0.3) +
  geom_point(aes(Time, JLCMM, color = "JLCMM", shape = "JLCMM"), size = 1.5) +

  scale_shape_manual(values = c("Cox PH" = 15, "JM" = 16, "JLCMM" = 17)) +
  ylim(0,1) +
  labs(
    x = "Time",
    y = "Brier score",
    color = "Model",
    shape = "Model"
  ) +
  theme_minimal()

```



```
# Saving the plot
png("figures/q1_brier_nfl.png",
    width = 6, height = 4, units = "in", res = 300)
ggplot(data.frame(
  Time = times,
  Cox = cox_brier$`Cox NfL`,
  JM = brier_nfl_jm,
  JLCMM = brier_jlcmm_m1_nfl$Brier
)) +
  geom_line(aes(Time, Cox, color = "Cox PH"), linewidth = 0.3) +
  geom_point(aes(Time, Cox, color = "Cox PH", shape = "Cox PH"), size = 1.5) +

  geom_line(aes(Time, JM, color = "JM"), linewidth = 0.3) +
  geom_point(aes(Time, JM, color = "JM", shape = "JM"), size = 1.5) +

  geom_line(aes(Time, JLCMM, color = "JLCMM"), linewidth = 0.3) +
  geom_point(aes(Time, JLCMM, color = "JLCMM", shape = "JLCMM"), size = 1.5) +

  scale_shape_manual(values = c("Cox PH" = 15, "JM" = 16, "JLCMM" = 17)) +
  ylim(0,1) +
  labs(
    x = "Time",
    y = "Brier score",
```

```

    color = "Model",
    shape = "Model"
  ) +
  theme_minimal()
dev.off()

```

pdf
2

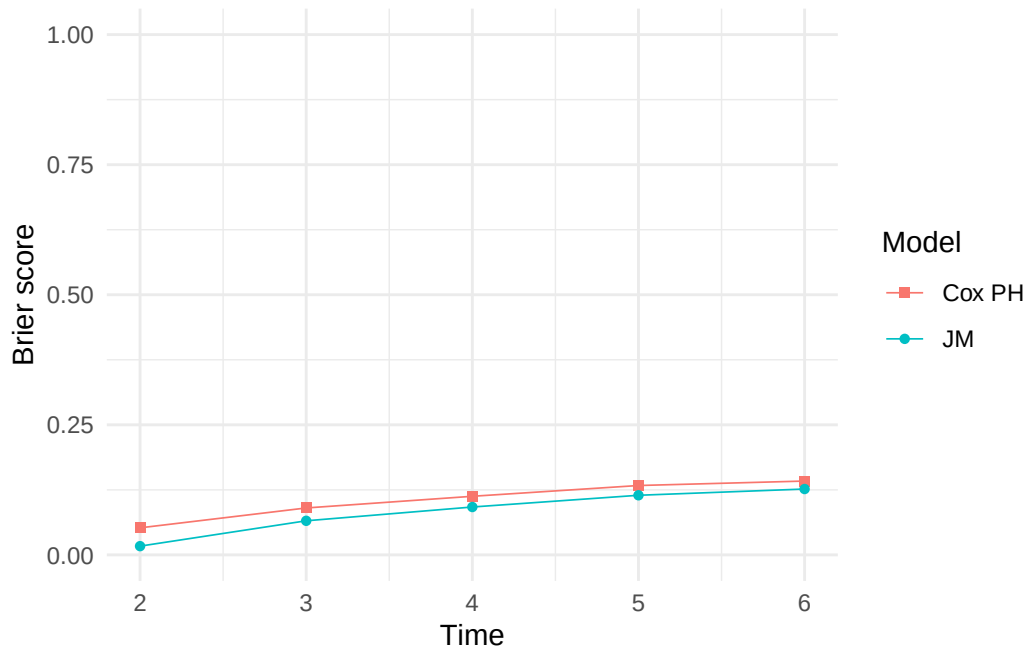
```

# Q1: Both biomarkers - comparing models (time-dependent Brier score)
ggplot(data.frame(
  Time = times,
  Cox = cox_brier$`Cox both`,
  JM = brier_multi_jm
)) +
  geom_line(aes(Time, Cox, color = "Cox PH"), linewidth = 0.3) +
  geom_point(aes(Time, Cox, color = "Cox PH", shape = "Cox PH"), size = 1.5) +

  geom_line(aes(Time, JM, color = "JM"), linewidth = 0.3) +
  geom_point(aes(Time, JM, color = "JM", shape = "JM"), size = 1.5) +

  scale_shape_manual(values = c("Cox PH" = 15, "JM" = 16)) +
  ylim(0,1) +
  labs(
    x = "Time",
    y = "Brier score",
    color = "Model",
    shape = "Model"
  ) +
  theme_minimal()

```



```
# Saving the plot
png("figures/q1_brier_multi.png",
    width = 6, height = 4, units = "in", res = 300)
ggplot(data.frame(
  Time = times,
  Cox = cox_brier$`Cox both`,
  JM = brier_multi_jm
)) +
  geom_line(aes(Time, Cox, color = "Cox PH"), linewidth = 0.3) +
  geom_point(aes(Time, Cox, color = "Cox PH", shape = "Cox PH"), size = 1.5) +

  geom_line(aes(Time, JM, color = "JM"), linewidth = 0.3) +
  geom_point(aes(Time, JM, color = "JM", shape = "JM"), size = 1.5) +

  scale_shape_manual(values = c("Cox PH" = 15, "JM" = 16)) +
  ylim(0,1) +
  labs(
    x = "Time",
    y = "Brier score",
    color = "Model",
    shape = "Model"
  ) +
  theme_minimal()
```

```
dev.off()
```

```
pdf  
  2
```

13.2 For comparing biomarker specifications

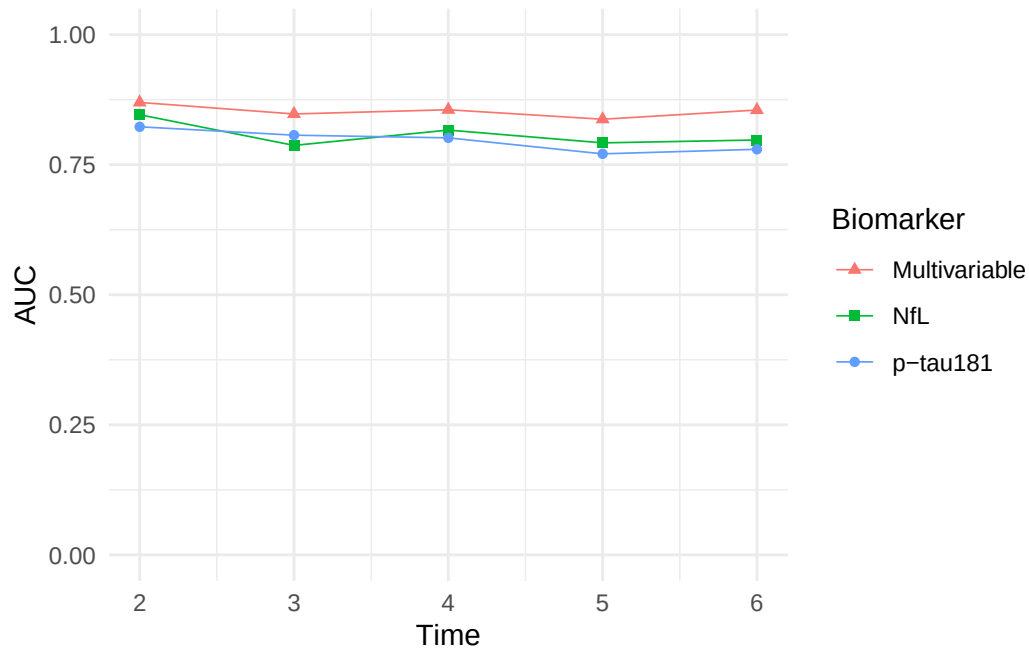
13.2.1 Time-dependent AUC

```
# Q2: Cox PH - comparing biomarkers (time-dependent AUC)
ggplot(data.frame(
  Time = times,
  NfL = auc_nfl_bl,
  pTau = auc_ptau_bl,
  Multi = auc_both_bl
)) +
  geom_line(aes(Time, NfL, color = "NfL"), linewidth = 0.3) +
  geom_point(aes(Time, NfL, color = "NfL", shape = "NfL"), size = 1.5) +

  geom_line(aes(Time, pTau, color = "p-tau181"), linewidth = 0.3) +
  geom_point(aes(Time, pTau, color = "p-tau181", shape = "p-tau181"), size = 1.5) +

  geom_line(aes(Time, Multi, color = "Multivariable"), linewidth = 0.3) +
  geom_point(aes(Time, Multi, color = "Multivariable", shape = "Multivariable"), size = 1.5)

  scale_shape_manual(values = c("NfL" = 15, "p-tau181" = 16, "Multivariable" = 17)) +
  ylim(0,1) +
  labs(
    x = "Time",
    y = "AUC",
    color = "Biomarker",
    shape = "Biomarker"
  ) +
  theme_minimal()
```



```
# Saving the plot
png("figures/q2_auc_cox.png",
    width = 6, height = 4, units = "in", res = 300)
ggplot(data.frame(
  Time = times,
  NfL = auc_nfl_bl,
  pTau = auc_ptau_bl,
  Multi = auc_both_bl
)) +
  geom_line(aes(Time, NfL, color = "NfL"), linewidth = 0.3) +
  geom_point(aes(Time, NfL, color = "NfL", shape = "NfL"), size = 1.5) +

  geom_line(aes(Time, pTau, color = "p-tau181"), linewidth = 0.3) +
  geom_point(aes(Time, pTau, color = "p-tau181", shape = "p-tau181"), size = 1.5) +

  geom_line(aes(Time, Multi, color = "Multivariable"), linewidth = 0.3) +
  geom_point(aes(Time, Multi, color = "Multivariable", shape = "Multivariable"), size = 1.5)

  scale_shape_manual(values = c("NfL" = 15, "p-tau181" = 16, "Multivariable" = 17)) +
  ylim(0,1) +
  labs(
    x = "Time",
    y = "AUC",
```

```

    color = "Biomarker",
    shape = "Biomarker"
  ) +
  theme_minimal()
dev.off()

```

pdf
2

```

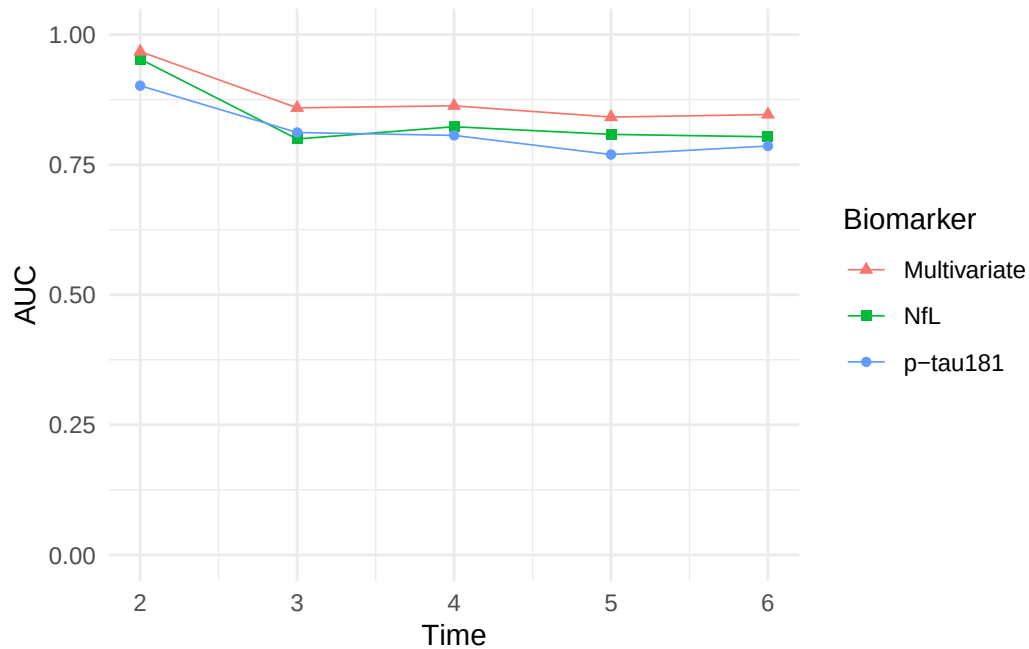
# Q2: JM - comparing biomarkers (time-dependent AUC)
ggplot(data.frame(
  Time = times,
  NfL = auc_nfl_jm,
  pTau = auc_ptau_jm,
  Multi = auc_multi_jm
)) +
  geom_line(aes(Time, NfL, color = "NfL"), linewidth = 0.3) +
  geom_point(aes(Time, NfL, color = "NfL", shape = "NfL"), size = 1.5) +

  geom_line(aes(Time, pTau, color = "p-tau181"), linewidth = 0.3) +
  geom_point(aes(Time, pTau, color = "p-tau181", shape = "p-tau181"), size = 1.5) +

  geom_line(aes(Time, Multi, color = "Multivariate"), linewidth = 0.3) +
  geom_point(aes(Time, Multi, color = "Multivariate", shape = "Multivariate"), size = 1.5) +

  scale_shape_manual(values = c("NfL" = 15, "p-tau181" = 16, "Multivariate" = 17)) +
  ylim(0,1) +
  labs(
    x = "Time",
    y = "AUC",
    color = "Biomarker",
    shape = "Biomarker"
  ) +
  theme_minimal()

```



```
# Saving the plot
png("figures/q2_auc_jm.png",
    width = 6, height = 4, units = "in", res = 300)
ggplot(data.frame(
  Time = times,
  NfL = auc_nfl_jm,
  pTau = auc_ptau_jm,
  Multi = auc_multi_jm
)) +
  geom_line(aes(Time, NfL, color = "NfL"), linewidth = 0.3) +
  geom_point(aes(Time, NfL, color = "NfL", shape = "NfL"), size = 1.5) +

  geom_line(aes(Time, pTau, color = "p-tau181"), linewidth = 0.3) +
  geom_point(aes(Time, pTau, color = "p-tau181", shape = "p-tau181"), size = 1.5) +

  geom_line(aes(Time, Multi, color = "Multivariate"), linewidth = 0.3) +
  geom_point(aes(Time, Multi, color = "Multivariate", shape = "Multivariate"), size = 1.5) +

  scale_shape_manual(values = c("NfL" = 15, "p-tau181" = 16, "Multivariate" = 17)) +
  ylim(0,1) +
  labs(
    x = "Time",
    y = "AUC",
```



```

    color = "Biomarker",
    shape = "Biomarker"
  ) +
  theme_minimal()
dev.off()

```

pdf
2

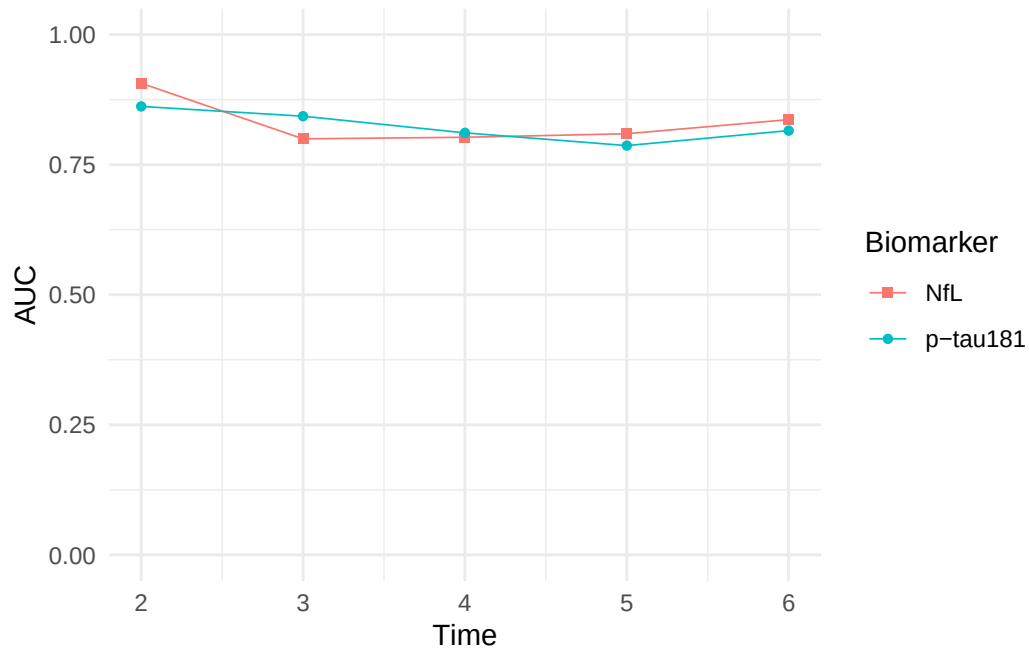
```

# Q2: LCJM - comparing biomarkers (time-dependent AUC)
ggplot(data.frame(
  Time = times,
  NfL = auc_jlcmm_m1_nfl$AUC,
  pTau = auc_jlcmm_m1_ptau$AUC
)) +
  geom_line(aes(Time, NfL, color = "NfL"), linewidth = 0.3) +
  geom_point(aes(Time, NfL, color = "NfL", shape = "NfL"), size = 1.5) +

  geom_line(aes(Time, pTau, color = "p-tau181"), linewidth = 0.3) +
  geom_point(aes(Time, pTau, color = "p-tau181", shape = "p-tau181"), size = 1.5) +

  scale_shape_manual(values = c("NfL" = 15, "p-tau181" = 16)) +
  ylim(0,1) +
  labs(
    x = "Time",
    y = "AUC",
    color = "Biomarker",
    shape = "Biomarker"
  ) +
  theme_minimal()

```



```
# Saving the plot
png("figures/q2_auc_lcjm.png",
    width = 6, height = 4, units = "in", res = 300)
ggplot(data.frame(
  Time = times,
  NfL = auc_jlcmm_m1_nfl$AUC,
  pTau = auc_jlcmm_m1_ptau$AUC
)) +
  geom_line(aes(Time, NfL, color = "NfL"), linewidth = 0.3) +
  geom_point(aes(Time, NfL, color = "NfL", shape = "NfL"), size = 1.5) +

  geom_line(aes(Time, pTau, color = "p-tau181"), linewidth = 0.3) +
  geom_point(aes(Time, pTau, color = "p-tau181", shape = "p-tau181"), size = 1.5) +

  scale_shape_manual(values = c("NfL" = 15, "p-tau181" = 16)) +
  ylim(0,1) +
  labs(
    x = "Time",
    y = "AUC",
    color = "Biomarker",
    shape = "Biomarker"
  ) +
  theme_minimal()
```

```
dev.off()
```

```
pdf  
  2
```

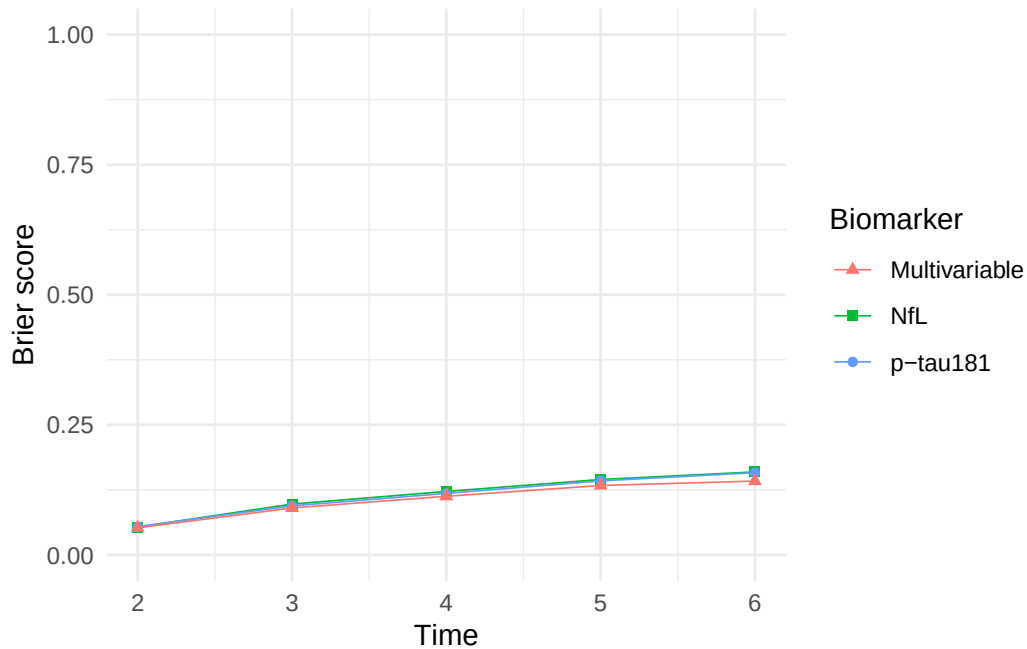
13.2.2 Time-dependent Brier score

```
# Q2: Cox PH - comparing biomarkers (time-dependent Brier score)
ggplot(data.frame(
  Time = times,
  NfL = cox_brier$`Cox NfL`,
  pTau = cox_brier$`Cox p-tau`,
  Multi = cox_brier$`Cox both`
)) +
  geom_line(aes(Time, NfL, color = "NfL"), linewidth = 0.3) +
  geom_point(aes(Time, NfL, color = "NfL", shape = "NfL"), size = 1.5) +

  geom_line(aes(Time, pTau, color = "p-tau181"), linewidth = 0.3) +
  geom_point(aes(Time, pTau, color = "p-tau181", shape = "p-tau181"), size = 1.5) +

  geom_line(aes(Time, Multi, color = "Multivariable"), linewidth = 0.3) +
  geom_point(aes(Time, Multi, color = "Multivariable", shape = "Multivariable"), size = 1.5)

  scale_shape_manual(values = c("NfL" = 15, "p-tau181" = 16, "Multivariable" = 17)) +
  ylim(0,1) +
  labs(
    x = "Time",
    y = "Brier score",
    color = "Biomarker",
    shape = "Biomarker"
  ) +
  theme_minimal()
```



```
# Saving the plot
png("figures/q2_brier_cox.png",
    width = 6, height = 4, units = "in", res = 300)
ggplot(data.frame(
  Time = times,
  NfL = cox_brier$`Cox NfL`,
  pTau = cox_brier$`Cox p-tau`,
  Multi = cox_brier$`Cox both`
)) +
  geom_line(aes(Time, NfL, color = "NfL"), linewidth = 0.3) +
  geom_point(aes(Time, NfL, color = "NfL", shape = "NfL"), size = 1.5) +

  geom_line(aes(Time, pTau, color = "p-tau181"), linewidth = 0.3) +
  geom_point(aes(Time, pTau, color = "p-tau181", shape = "p-tau181"), size = 1.5) +

  geom_line(aes(Time, Multi, color = "Multivariable"), linewidth = 0.3) +
  geom_point(aes(Time, Multi, color = "Multivariable", shape = "Multivariable"), size = 1.5)

  scale_shape_manual(values = c("NfL" = 15, "p-tau181" = 16, "Multivariable" = 17)) +
  ylim(0,1) +
  labs(
    x = "Time",
    y = "Brier score",
```

```

    color = "Biomarker",
    shape = "Biomarker"
  ) +
  theme_minimal()
dev.off()

```

pdf
2

```

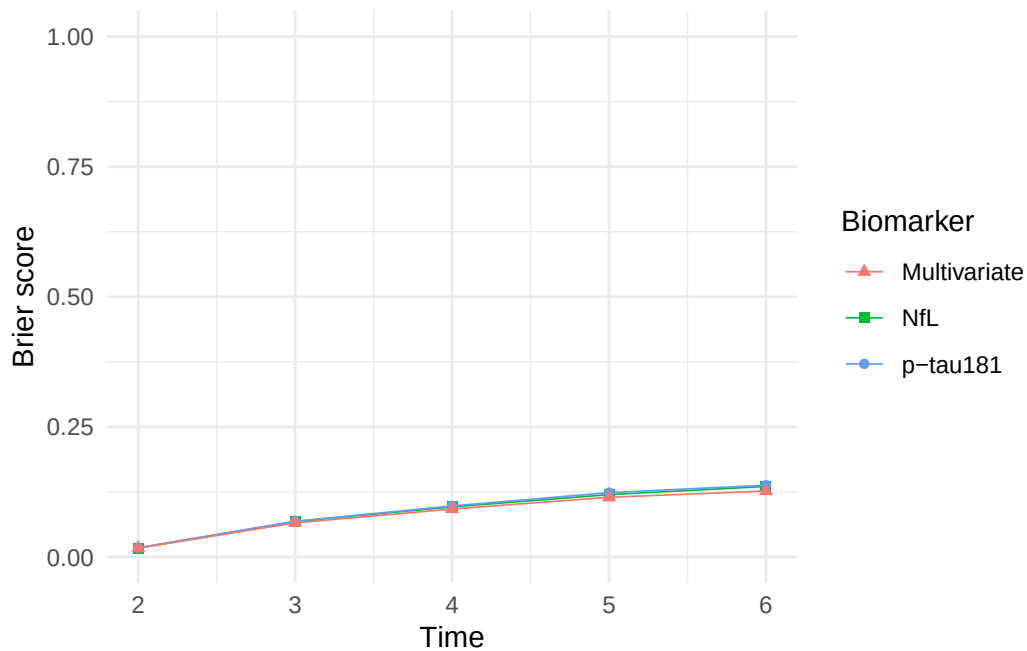
# Q2: JM - comparing biomarkers (time-dependent AUC)
ggplot(data.frame(
  Time = times,
  NfL = brier_nfl_jm,
  pTau = brier_ptau_jm,
  Multi = brier_multi_jm
)) +
  geom_line(aes(Time, NfL, color = "NfL"), linewidth = 0.3) +
  geom_point(aes(Time, NfL, color = "NfL", shape = "NfL"), size = 1.5) +

  geom_line(aes(Time, pTau, color = "p-tau181"), linewidth = 0.3) +
  geom_point(aes(Time, pTau, color = "p-tau181", shape = "p-tau181"), size = 1.5) +

  geom_line(aes(Time, Multi, color = "Multivariate"), linewidth = 0.3) +
  geom_point(aes(Time, Multi, color = "Multivariate", shape = "Multivariate"), size = 1.5) +

  scale_shape_manual(values = c("NfL" = 15, "p-tau181" = 16, "Multivariate" = 17)) +
  ylim(0,1) +
  labs(
    x = "Time",
    y = "Brier score",
    color = "Biomarker",
    shape = "Biomarker"
  ) +
  theme_minimal()

```



```
# Saving the plot
png("figures/q2_brier_jm.png",
    width = 6, height = 4, units = "in", res = 300)
ggplot(data.frame(
  Time = times,
  NfL = brier_nfl_jm,
  pTau = brier_ptau_jm,
  Multi = brier_multi_jm
)) +
  geom_line(aes(Time, NfL, color = "NfL"), linewidth = 0.3) +
  geom_point(aes(Time, NfL, color = "NfL", shape = "NfL"), size = 1.5) +

  geom_line(aes(Time, pTau, color = "p-tau181"), linewidth = 0.3) +
  geom_point(aes(Time, pTau, color = "p-tau181", shape = "p-tau181"), size = 1.5) +

  geom_line(aes(Time, Multi, color = "Multivariate"), linewidth = 0.3) +
  geom_point(aes(Time, Multi, color = "Multivariate", shape = "Multivariate"), size = 1.5) +

  scale_shape_manual(values = c("NfL" = 15, "p-tau181" = 16, "Multivariate" = 17)) +
  ylim(0,1) +
  labs(
    x = "Time",
    y = "Brier score",
```

```

    color = "Biomarker",
    shape = "Biomarker"
  ) +
  theme_minimal()
dev.off()

```

pdf
2

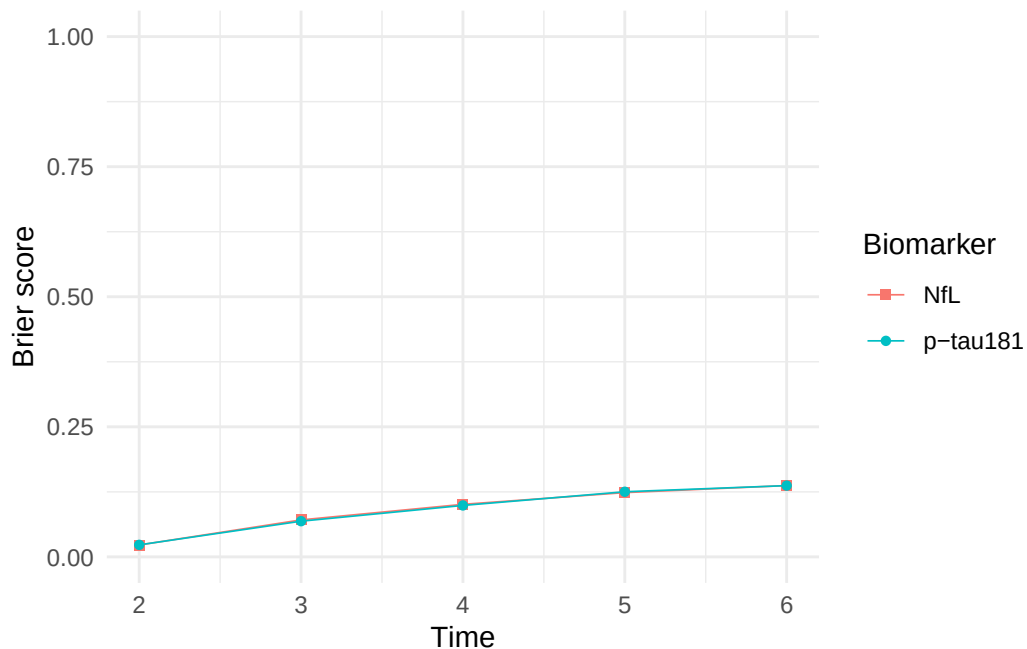
```

# Q2: LCJM - comparing biomarkers (time-dependent Brier score)
ggplot(data.frame(
  Time = times,
  NfL = brier_jlcmm_m1_nfl$Brier,
  pTau = brier_jlcmm_m1_ptau$Brier
)) +
  geom_line(aes(Time, NfL, color = "NfL"), linewidth = 0.3) +
  geom_point(aes(Time, NfL, color = "NfL", shape = "NfL"), size = 1.5) +

  geom_line(aes(Time, pTau, color = "p-tau181"), linewidth = 0.3) +
  geom_point(aes(Time, pTau, color = "p-tau181", shape = "p-tau181"), size = 1.5) +

  scale_shape_manual(values = c("NfL" = 15, "p-tau181" = 16)) +
  ylim(0,1) +
  labs(
    x = "Time",
    y = "Brier score",
    color = "Biomarker",
    shape = "Biomarker"
  ) +
  theme_minimal()

```



```
# Saving the plot
png("figures/q2_brier_lcmj.png",
    width = 6, height = 4, units = "in", res = 300)
ggplot(data.frame(
  Time = times,
  NfL = brier_jlcmm_m1_nfl$Brier,
  pTau = brier_jlcmm_m1_ptau$Brier
)) +
  geom_line(aes(Time, NfL, color = "NfL"), linewidth = 0.3) +
  geom_point(aes(Time, NfL, color = "NfL", shape = "NfL"), size = 1.5) +

  geom_line(aes(Time, pTau, color = "p-tau181"), linewidth = 0.3) +
  geom_point(aes(Time, pTau, color = "p-tau181", shape = "p-tau181"), size = 1.5) +

  scale_shape_manual(values = c("NfL" = 15, "p-tau181" = 16)) +
  ylim(0,1) +
  labs(
    x = "Time",
    y = "Brier score",
    color = "Biomarker",
    shape = "Biomarker"
  ) +
  theme_minimal()
```



```
dev.off()
```

pdf
2

14 Saving other relevant tables and figures

```
# Saving the table showing number of participants, event distribution, and event rates in in
#saveRDS(event_table, "tables/event_table.rds")
#saveRDS(event_table, "tables/event_table_full.rds")
```

```
# Saving the table showing cohort characteristics and measurement distributions
#saveRDS(cohort_summary_table, "tables/cohort_summary_table.rds")
saveRDS(cohort_summary_table, "tables/cohort_summary_table_full.rds")
```

```
# Saving log transformed ptau histogram
png("figures/hist_log_p_tau.png",
     width = 6, height = 4, units = "in", res = 300)
hist(long_data$log_PLASMAPTAU181,
      xlab = "Log-transformed p-tau181")
dev.off()
```

pdf
2

```
# Saving log transformed NfL histogram
png("figures/hist_log_nfl.png",
     width = 6, height = 4, units = "in", res = 300)
hist(long_data$log_PLASMA_NFL,
      xlab = "Log-transformed NfL"
      )
dev.off()
```

pdf
2

```
# Saving the correlation plots
png("figures/corrplot_cox.png", width = 6, height = 4, units = "in", res = 300)
corrplot(cox_cor,
          method = "color",
          type = "upper",
          addCoef.col = "black",
          tl.col = "black",
          tl.srt = 45)
dev.off()
```

pdf
2

```
png("figures/corrplot_lme.png", width = 6, height = 4, units = "in", res = 300)
corrplot(lme_cor,
          method = "color",
          type = "upper",
          addCoef.col = "black",
          tl.col = "black",
          tl.srt = 45)
dev.off()
```

pdf
2

```
# Saving the table of results from the likelihood ratio tests between models with only a random effect
saveRDS(long_re_lrt_table, "tables/long_re_lrt_table.rds")
```

```
# Saving the plot of residuals vs fitted for ptau
png("figures/residuals_vs_fitted_ptau.png",
     width = 6, height = 4, units = "in", res = 300)
plot(fitted_vals_ptau, res_ptau,
     xlab = "Fitted values",
     ylab = "Residuals",
     main = "Residuals vs Fitted for p-tau181")
abline(h = 0, col = "red", lty = 2)
dev.off()
```

pdf
2

```
# Saving the residuals vs fitted plot for nfl
png("figures/residuals_vs_fitted_nfl.png",
     width = 6, height = 4, units = "in", res = 300)
plot(fitted_vals_nfl, res_nfl,
     xlab = "Fitted values",
     ylab = "Residuals",
     main = "Residuals vs Fitted for NfL")
abline(h = 0, col = "red", lty = 2)
dev.off()
```

pdf
2

```
# Saving the QQ-plots
png("figures/qq_ptau.png",
     width = 6, height = 4, units = "in", res = 300)
qqnorm(res_ptau, main = "QQ plot: p-tau181 residuals")
qqline(res_ptau, col = "red")
dev.off()
```

pdf
2

```
png("figures/qq_nfl.png",
     width = 6, height = 4, units = "in", res = 300)
qqnorm(res_nfl, main = "QQ plot: NfL residuals")
qqline(res_nfl, col = "red")
dev.off()
```

pdf
2

```
# Saving table with PHT results
saveRDS(table_ph, "tables/table_ph.rds")
```

```
# Saving the final PH table
saveRDS(table_ph_final, "tables/table_ph_final.rds")
```

```
# Saving a table of results from the likelihood ratio tests for cox models with and without L
saveRDS(surv_lrt_table, "tables/surv_lrt_table.rds")
```

```
# Saving the cox models comparison table
saveRDS(table_cox, "tables/table_cox1.rds")
```

```
# Saving the C-index table
saveRDS(c_index_table, "tables/c_index_table.rds")
```

```
# Saving the convergence plot for the p-tau181 joint model
png("figures/convergence_plot_ptau.png",
    width = 6, height = 4, units = "in", res = 300)
ggtraceplot(joint_model_ptau, "alphas")
dev.off()
```

pdf
2

```
# Saving the convergence plot for the NfL joint model
png("figures/convergence_plot_nfl.png",
    width = 6, height = 4, units = "in", res = 300)
ggtraceplot(joint_model_nfl, "alphas")
dev.off()
```

pdf
2

```
# Saving the convergence plot for the multivariate joint model
png("figures/convergence_plot_multi.png",
    width = 6, height = 4, units = "in", res = 300)
ggtraceplot(multi_joint_model, "alphas")
dev.off()
```

pdf
2

```
# Note: Since the function does not allow direct extraction of both plots, one is saved here
```

```
# Saving the joint model comparison table
saveRDS(table_jm, "tables/table_jm1.rds")
```

```
# Saving the rhat table (survival outcome)
saveRDS(rhat_survival_table, "tables/rhat_survival_table.rds")
```

```
# Saving the rhat table (longitudinal outcome)
saveRDS(rhat_longitudinal_table, "tables/rhat_longitudinal_table.rds")
```

```
# Saving the posterior probability table for the 2 class jlcmm for nfl
pprob_m1_nfl <- get_pprob_table(jlcmm_m1_nfl)
saveRDS(pprob_m1_nfl, "tables/pprob_m1_nfl.rds")
```

```
# Saving the model estimates in a table
jlcmm_m1_nfl_output <- get_jlcmm_params(jlcmm_m1_nfl)
saveRDS(jlcmm_m1_nfl_output, "tables/jlcmm_m1_nfl_output.rds")
```

```
# Saving comparison table of fit statistics (NfL)
saveRDS(summary_jlcmm_nfl, "tables/summary_jlcmm_nfl.rds")
```

```
# Survival curves by class (NfL)
png("figures/class-specific-survival-jlcmm-nfl.png",
    width = 6, height = 4, units = "in", res = 300)
plot(jlcmm_m1_nfl, which = "survival",
     main = "",
     xlab = "Time (years)", ylab = "Survival Probability")
dev.off()
```

pdf
2

```
# Longitudinal trajectory by class (NfL)
png("figures/longitudinal-jlcmm-nfl.png",
    width = 6, height = 4, units = "in", res = 300)
plot(jlcmm_m1_nfl, var.time = "time_years", which = "fit",
     main = "",
     xlab = "Time (years)", ylab = "NfL")
dev.off()
```

pdf
2

```
# Saving the posterior probability table for the 2 class jlcmm for p-tau181
pprob_m1_ptau <- get_pprob_table(jlcmm_m1_ptau)
saveRDS(pprob_m1_ptau, "tables/pprob_m1_ptau.rds")
```

```
# Saving the model estimates in a table (p-tau181)
jlcmm_m1_ptau_output <- get_jlcmm_params(jlcmm_m1_ptau)
saveRDS(jlcmm_m1_ptau_output, "tables/jlcmm_m1_ptau_output.rds")
```

```
# Saving the posterior probability table for the 3 class jlcmm for p-tau181
pprob_m2_ptau <- get_pprob_table(jlcmm_m2_ptau)
saveRDS(pprob_m2_ptau, "tables/pprob_m2_ptau.rds")
```

```
# Comparing M0, M1, M2 (p-tau181)
saveRDS(summary_jlcmm_ptau, "tables/summary_jlcmm_ptau.rds")
```

```
# Survival curves by class (p-tau181)
png("figures/class-specific-survival-jlcmm-ptau.png",
    width = 6, height = 4, units = "in", res = 300)
plot(jlcmm_m1_nfl, which = "survival",
     main = "",
     xlab = "Time (years)", ylab = "Survival Probability")
dev.off()
```

pdf
2

```
# Longitudinal trajectories by class (p-tau181)
png("figures/longitudinal-jlcmm-ptau.png",
    width = 6, height = 4, units = "in", res = 300)
plot(jlcmm_m1_ptau, var.time = "time_years", which = "fit",
     main = "",
     xlab = "Time (years)", ylab = "ptau-181")
dev.off()
```

pdf
2

```
# Saving the posterior probability table for the 2 class jlcmm for ptau and NfL
#pprob_m1_ptau_nfl <- get_pprob_table(mult_jlcmm_m1)
#saveRDS(pprob_m1_ptau_nfl, "tables/pprob_m1_ptau_nfl.rds")
```

```
# Comparing M0, M1, M2 (NfL and p-tau181)
#saveRDS(summary_jlcmm_ptau_nfl, "tables/summary_jlcmm_ptau_nfl.rds")
```